Simulation of three-dimensional unsteady flow in hydraulic pumps

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Printing: FEBODRUK BV, Enschede.

Simulation of three-dimensional unsteady flow in hydraulic pumps /
B.P.M. van Esch.

Thesis University of Twente, Enschede. - With ref. - With summary in Dutch.
ISBN 90-365-10171
Front cover : computational grid for mixed-flow pump
(part of casing removed)

SIMULATION OF THREE-DIMENSIONAL UNSTEADY FLOW IN HYDRAULIC PUMPS

PROEFSCHRIFT

ter verkrijging van
de graad van doctor aan de Universiteit Twente,
op gezag van de rector magnificus,
Prof.dr. F.A. van Vught,
volgens het besluit van het College voor Promoties
in het openbaar te verdedigen
op donderdag 25 september 1997 te 13.15 uur

door

Bartholomeus Petrus Maria van Esch

geboren op 31 oktober 1965
te Sint-Amandsberg (België)

Dit proefschrift is goedgekeurd door de promotoren:

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Summary

The design of hydraulic turbomachines has reached the stage were improvements can only be achieved through a detailed understanding of the internal flow. The prediction of the flow in such equipment is very complicated due to the rotation and the curved three-dimensional shape of the impellers. Furthermore, the flow in turbomachines shows unsteady behaviour, especially at off-design conditions, as a result of interaction between impeller and pump casing. Considering these complexities, computer simulations will become increasingly important.

In this thesis it is shown that the flow in hydraulic pumps of the radial and mixed-flow type, operating at conditions not too far from design point, can be considered as an incompressible potential flow, where the influence of viscosity is restricted to thin boundary layers, wakes and mixing areas. A three-dimensional method for unsteady flow based on this model yields good results. In order to predict the efficiency of pumps, additional models to quantify the viscous losses can be employed successfully. Thus reads the overall conclusion which can be drawn from the investigation presented in this thesis.

The numerical method developed for solving unsteady potential flow is based on a fully three-dimensional finite-element method. The computational mesh is divided into two parts, one for the rotor and one for the pump casing, and connected by a sliding interface. In this way the impeller rotating motion with respect to the pump casing can be simulated efficiently. Some special numerical techniques are employed in order to reduce computing time. These are based on the substructuring method combined with the implicit imposition of the Kutta conditions at the trailing edges of the impeller and diffuser blades.

The losses which occur in pumps are quantified using additional models for energy dissipation in boundary layers, in mixing areas and at sudden expansions and contractions in through-flow area, as well as models for disc friction and leakage flow. Based on the velocity distribution along the rotating and stationary surfaces, as obtained from a three-dimensional potential flow computation, the state of boundary layers is determined using a one-dimensional boundary layer method.

The capability of the method is demonstrated by analyses of the flow in several laboratory and industrial pumps. For these pumps, information on the velocity and pressure distribution as well as the performance characteristics are available from experiments. In general, computational results show a good agreement with experimentally obtained values. This, and the short computing times required, make the proposed method very well suited as an analysis-tool within a pump development process.

Samenvatting

Op dit moment is het ontwerpen van hydraulische turbomachines aangeland in het stadium waarin verbeteringen slechts kunnen worden bereikt middels een gedetailleerd begrip van de interne stroming. De voorspelling daarvan is zeer gecompliceerd als gevolg van de rotatie en de drie-dimensionale, gekromde vorm van de waaiers. Bovendien vertoont de stroming een instationair gedrag, vooral buiten het ontwerpbedrijfspunt, door de interactie tussen waaier en pomphuis. Gezien de hoge graad van complexiteit zullen computersimulaties steeds belangrijker worden.

In dit proefschrift wordt aangetoond dat de stroming in pompen van het radiale- en *mixed-flow* type, werkend niet al te ver buiten het ontwerpbedrijfspunt, opgevat kan worden als een onsamendrukbare potentiaalstroming, waarbij de invloed van viscositeit beperkt blijft tot dunne grenslagen, zoggen en *mixing* gebieden. Een drie-dimensionale methode voor instationaire stromingen gebaseerd op dit model geeft goede resultaten. Aanvullende modellen voor het schatten van de viskeuze verliezen kunnen met succes worden ingezet om het rendement van pompen te voorspellen. Dit is de algemene conclusie die kan worden getrokken op basis van het in dit proefschrift gepresenteerde onderzoek.

De numerieke methode die is ontwikkeld voor de berekening van de instationaire potentiaalstroming is gebaseerd op een volledig drie-dimensionale eindige-elementen methode. Het rekenrooster wordt opgesplitst in twee delen, een voor de waaier en een voor het pomphuis. Door beide roosters met elkaar te verbinden via een *sliding interface*, kan de roterende beweging van de waaier ten opzichte van het pomphuis efficiënt worden gesimuleerd. Speciale numerieke technieken worden gebruikt om de rekentijd te verlagen. Deze zijn gebaseerd op de substructurering methode en het op een impliciete wijze voorschrijven van de Kutta condities aan de *trailing edges* van waaier- en diffusor schoepen.

De in de pomp optredende verliezen worden bepaald met aanvullende modellen. Zo zijn er modellen voor het kwantificeren van schijfwrijving en lekstroming en voor energiedissipatie in grenslagen, *mixing* gebieden en bij plotselinge verwijdingen en vernauwingen in doorstroomoppervlak. Op basis van de snelheidsverdeling lang roterende en stilstaande wanden, zoals berekend met de drie-dimensionale potentiaalmethode, wordt de aard van de grenslagen bepaald met behulp van een een-dimensionale grenslaagmethode.

De bruikbaarheid van de methode als geheel wordt aangetoond door analyse van stromingen in een aantal laboratorium- en industriële pompen. Voor deze pompen zijn zowel snelheids- en drukverdelingen als prestatiekarakteristieken bekend uit experimenten. Over het algemeen komen de resultaten van berekeningen goed overeen met experimenteel verkregen waarden. Gevoegd bij de korte rekentijden die ervoor nodig zijn, maakt dit de voorgestelde rekenmethode uitermate geschikt als analyse-tool binnen het ontwikkeltraject voor pompen.

Chapter 1
Introduction

A wide range of pumping machinery has been developed for the purpose of raising liquids, or forcing it against a resistance otherwise, and for compressing gases. When considering a pump classification based on the working principle, the centrifugal pump only forms one of many classes. The invention of the use of centrifugal force as a driving mechanism for pumps dates back to, presumably, the second half of the seventeenth century. Although nowadays centrifugal pumps are among the most successfully employed, this has not always been the case. In fact, it lasted for almost two centuries after its invention before engineers began to commercially exploit this concept. An important reason why so little progress was made is believed to be the lack of suitable gearing and prime movers of sufficiently high speed. Another explanation is the lack of a practical theory on which efficient centrifugal pumps could be designed. Even after the theoretical work of Euler in 1750, very few understood its principle. Or it may well be that existing pumping appliances satisfied most of the needs of that time. The Great Exhibition of 1851 became a turning point in the development of the centrifugal pump. Several designs were shown, of which a few appeared to be very successful. Since then the centrifugal pump has developed into a high efficiency machine which can be adapted to suit almost any working condition.

In section 1.1 an overview is given of the history of pumping machinery in general and centrifugal pumps in particular. Apart from the literature cited in this section, more information can be found in Van Esch (1997). Some of the definitions and nomenclature which will be used in the sequel of this thesis will be introduced in section 1.2. In section 1.3, a brief introduction is given on methods to predict performance. Much of the contents of sections 1.2 and 1.3 may be found in, for example, Stepanoff (1964), Csanady (1964), Shepherd (1971), Petermann (1974), Troskolañski and Lazarkiewicz (1976), Dixon (1986), Cumpsty (1989), Neumann (1991), and Turton (1994). Further references can be found in these publications. In section 1.4, an outline of this thesis is given together with comments on the usefulness of this work to pump designers.

1.1 Centrifugal pump design - historic overview

Most historians agree on the fact that Denis Papin (1647-1712) should be regarded as the originator of the centrifugal pump. Although the existence of centrifugal forces was probably known long before his time, and some machines were actually built which more or less made use of this force, he developed the first true centrifugal pump as we know it to-day: a machine in which water or air enters in axial direction near the shaft, is accelerated by revolving impeller blades and finally leaves the pump in circumferential direction.

The period before the invention of Papin in 1689 will be treated in section 1.1.1. Books written by Agricola (1556) and Ramelli (1588) are discussed. In their work they described in detail the techniques which were in use for raising water. Among these are the water wheel, the Archimedean screw, the chain of buckets, the Paternoster pump and the suction pump. Many of these devices were already in common use by the Ancients. One type of pump was however invented by Ramelli himself: the rotary displacement pump. As Denis Papin is considered the key figure in the history of centrifugal pump design, section 1.1.2 is devoted to the invention of this French engineer. Its contents is partly based on a biography by Gerland (1881). An excellent, but rather lengthy, overview of the period after Papin was presented by Harris (1953). It covers the era between the year 1689 and the Great Exhibition in 1851. An abstract is given in section 1.1.3. See also Westcott (1932) for an historic overview of pumping machinery in general.

1.1.1 Raising water in the early seventeenth century and before

A name which frequently emerges in relation to early pump design is that of the German engineer Georgius Agricola (Georg Bauer). Although a scientist in many fields, he is remembered most of all through his contribution to the mining industry in the 16th century. His book “De Re Metallica” (1556) has been regarded as a standard work for many centuries. In this book on mining and metallurgy Agricola describes, for the first time, methods and processes in mining industry in great detail. It should be stressed that what he presents are not inventions of his own, but rather the experience and knowledge of many generations before him. “De Re Metallica” consists of twelve books, of which book six describes the miners’ tools and machines. Most of its contents is devoted to a careful analysis of ventilating and pumping machinery. These machines were very important to miners as difficulty in breathing and excessive groundwater formed the limiting factor in the depth of mines.

Several different types of devices were used to remove groundwater from mining shafts. Agricola describes three types (fig. 1.1):

- A closed chain of buckets revolving over two drums, one located above the ground, the other in the shaft;
- “Rag and chain” pumps, also known as Paternoster pumps;
- Suction or piston pumps with valves.

They were driven by human force or by a water-wheel when a river could be diverted to the mine. These devices were probably known to the Ancients as well. Ventilating machines are also of three types:

- Devices that merely deflect the wind that blows in the open air into the shafts of the mine;
- A fan placed inside a drum or box-shaped casing with two square openings at its circumference, the upper one of which receives the air, while the lower one is connected to a mine shaft (fig. 1.2). Instead of a drum, a box-shaped casing was also

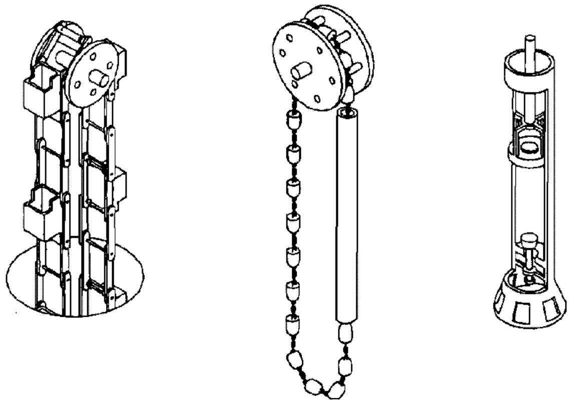

Fig. 1.1: Pumps in use around the year 1550. From left to right: chain of buckets, paternoster pump, and suction pump. Pictures from Ramelli (1588).

in use, although it was inferior to the drum. Agricola explains: “for the fans so fill the drum that they almost touch it on every side, and drive into the conduit all the air that has been accumulated; but they cannot thus fill the box-shaped casing, on account of its angles, into which the air partly retreats; therefore it cannot be as useful as the drum.” Fans were made of thin wooden boards (either fixed or flexible) or goose feathers. The fans were driven by human force, water-power or windmills;

- Bellows which could be used both for ventilation and, by suction, for the removal of toxic gasses. Bellows were compressed by human force, by the force of horses or by water-power.

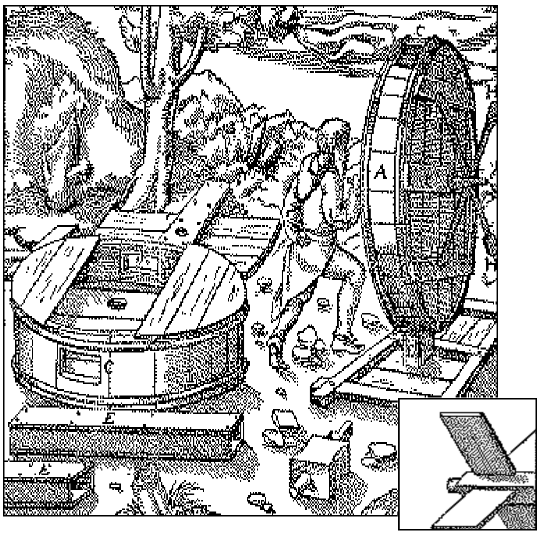

Fig. 1.2: Fan used for ventilation of mine-shafts (around 1550). Details of picture from Agricola (1556).

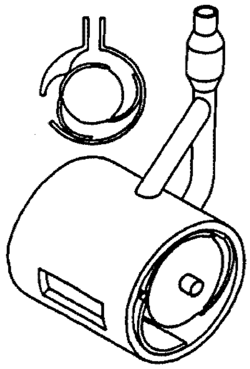

Fig. 1.3: One of the rotary displacement pumps, invented by Ramelli (picture from Ramelli, 1588).

Another book on mechanical subjects from the sixteenth century which is frequently cited by historians is that of Agostino Ramelli. In 1588 he published his book “Le Diverse et Artificiose Machine del Capitano Agostino Ramelli”. It contains no less than 195 drawings of various machines together with brief explanations, both in Italian and French. Over half of the plates illustrate water-raising devices, operated by water, wind and muscle power. Other plates include grain mills, fountains, cranes, and military equipment like bridges, hurling engines and weapons. As far as water-raising devices are concerned, most of them were well known long before Ramelli’s time; plates depicting piston pumps, water wheels, chains of buckets and Paternoster pumps resemble those of Agricola. One type of pump was however invented by Ramelli himself: the rotary displacement pump (fig. 1.3).

On January 24th 1602 the *Generale Staten* of the Netherlands granted a patent to Cornelis Cornelisz. for a device for raising water. Although its working is not clear (a drawing was missing), it is presumed that it made use of centrifugal forces. The patent mentions a number of ‘cromme oft rechte goten, oft ronde pompen’, which L.E. Harris (1953) in his paper translates to a number of ‘curved or straight tubes, or round pipes’. It certainly did not resemble the centrifugal pump as we know it today. In 1621 Pieter Sturck and Abraham Jansz. Segers patented a similar machine.

1.1.2 The invention of the centrifugal pump

One of the many inventions of Denis Papin, an engineer born in France in 1647, was the centrifugal pump. It was very different in principle from the fans mentioned by Agricola or the rotating suction pumps described by Ramelli with air or water entering in radial direction at the pump’s circumference and leaving the pump in (almost) circumferential direction. Papin’s first construction of 1689 differed from those designs in that the air or water entered the pump near the shaft, in axial direction. In the “Actis Eruditorium” (1689) he described his design (fig. 1.4). The pump consisted of only two blades running in a closely fitting circular casing with constant axial width. The discharge pipe was much too small. He found that this design did not work properly. In the “Philosophical Transactions” he described his improved design.

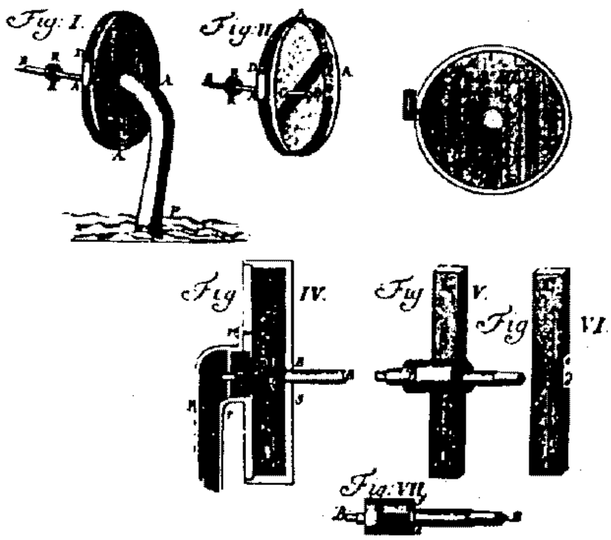

Fig. 1.4: Centrifugal pump (1689) invented by Denis Papin. Picture from Harris (1953).

Instead of changing the shape of the (radial) blades, he changed the shape of the casing to a spiral with constant width. In a letter to the Royal Society Papin wrote: “I am busie at present for a coal mine, which hath been lett off because of impurity of the air. I have therefore improved the Hessian Bellows: I don’t question but you have seen that new contrivance printed in *Lipsiae in Actis Eruditorum anno 1699* with the title *Rotatilis Suctor et Pressor Hassiacus*. And it may be applied for wind as well as for water.” He explained the improved performance as follows: “I believe that this spiral figure is a good improvement to this engine. And indeed I have made such bellows, where the radius is but 10.5 inches, the wing 2 inches broad and 9 inches high, because the tympanum [casing] is also so high, or little more; the aperture is also 9 inches, or a little more, so that it makes a square hole. When I work this engine with my foot, it makes such a wind, that it may raise up two pounds weight; and without doubt a stronger man could do much more.” And he added: “... that every wing in going round drives new air, because the air which is first in motion finds place to recede from the center towards the spiral circumference; and so it gives room to new air to come to the wing: And when the wings come near to the aperture, they drive their new air into the aperture without any friction; and the air which hath been first driven and removed from the wing, cannot lose its swiftness, because the wings which continually follow do continually drive new air, which keeps that which is before always in the same swiftness.” The pump of Papin was used in mining and during the construction of canals to supply fresh air and to remove excessive groundwater. Papin was fully aware of the drawbacks of the centrifugal pump. In a letter to Leibniz he wrote: “The machine which operates by centrifugal force is excellent in theory; but, Monsieur, I find in it a great inconvenience in practice, for where it is used to raise water to some considerable height, it is necessary to impart to it a high speed and one always uniform.” At that time, centrifugal pumps were driven by hand.

As a centrifugal pump is a high speed device, windmills could only be employed to drive the pump using gears, at the expense of huge friction losses. This is the main reason why centrifugal pumps were not used much in those years.

1.1.3 Centrifugal pump design (1689 - 1851)

It looks like the invention of Papin was already forgotten shortly after his death, as a paper entitled “Machine pour éleve de l’eau” by Le Demour in 1732 shows. In 1736, Daniel Fahrenheit, who was in Amsterdam at the time, was granted a patent for “An invention relating to a new water-machine.” Both machines in fact resemble the device made by Cornelis Cornelisz., invented more than hundred years earlier in 1602.

The first theoretical analysis was given by Euler, presented in a paper entitled “Recherche sur l’effet d’une Machine Hydraulique proposé par Mr. Segner, Professor à Göttingue” in 1750, concerning a rather primitive version of a reaction water-turbine. It formed the basis of his subsequent analysis of the centrifugal pump. These theoretical advances appear not to have led directly to improved designs until almost a century later.

The year 1818 became a turning-point in the history of the centrifugal pump. In that year the famous “Massachusetts Pump” was introduced in America. It showed a return to the original conception of Denis Papin; blades running in a circular or spiral casing. From this moment onwards the design steadily evolved into the centrifugal pump of to-day, although departures from previous designs were not always improvements. In 1831, for example, a new, supposedly improved, design of the Massachusetts pump by Blake was given a circular casing and a discharge branch at right angles to the impeller plane of rotation. A new aspect, though, was formed by the semi-shrouded impeller. It was called a centrifugal disc pump. Previous designs of both Papin and the Massachusetts pump employed open impellers (at both sides!).

Meanwhile in France, progress was mainly achieved at the theoretical plane. In 1838, M. Combes did some theoretical work, based on the work of Euler. Again its application was focussed on reaction water-turbines, but he also discussed the centrifugal ventilator and centrifugal pump. As one of the first of his time, he investigated the effect of blade curvature on hydraulic performance. Up to that time, blades were radial in shape.

In America, in 1846, W.D. Andrews patented an improved design of an older pump, explaining the novelty as follows: “I use vanes, and I enclose them within, and connect them to an additional case, which revolves with them, within the exterior or stationary case.” It was the first double-shrouded impeller. James Stuart Gwynne acquired the patent rights of Andrews’ improved design and soon built similar pumps. These pumps initially employed impellers with only one blade. Later he corrected his designs on this point. After he returned to England he was granted a patent in 1851 for the first multi-stage centrifugal pump. It was an important achievement, because up to that time centrifugal pumps were regarded as low-head machines.

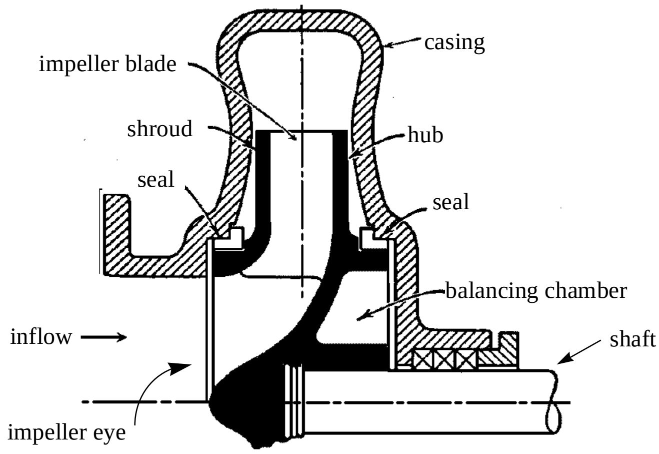

Fig. 1.5: Centrifugal impeller (from Stepanoff, 1964)

Meanwhile, in Europe, John George Appold and Bessemer were achieving great progress. Appold made his designs for centrifugal pumps after a large number of experiments. It is unknown if he ever knew about the work of Andrews and Gwynne in America, or the investigations of Combes in France. Appold employed curved blades in his pumps after he found out, by experiments, that the efficiency declined rapidly when straight blades were used. The designs made by Bessemer were in many respects more advanced than those of his competitors in America. He took out two patents, but his pumps were never brought to a commercial success.

The exhibition of 1851 marked the end of an era of centrifugal pump design which started in the year 1689 with Denis Papin. During this period, designs were mainly based on trial and error. A theoretical basis was virtually nonexistent. It is astonishing to see that progress was made even though investigators hardly relied on former achievements.

1.2 Definitions and terminology

The mechanism by which liquids are raised or gases are compressed in centrifugal pumps is based on the rotating motion of the impeller relative to the stationary pump casing (see figs. 1.5 and fig. 1.6). The name of this type of pump is taken from one of the forces which are accountable: the centrifugal force[1]. In essence its working is simple. The fluid enters the pump in axial direction through the suction nozzle into the eye of the impeller or rotor. It is subsequently accelerated into a circular motion by the rotating impeller blades and collected in the volute or diffuser. The fluid

1. Strictly speaking this is not entirely true for all types. Axial pumps are generally regarded as a subclass of centrifugal pumps, although the centrifugal force is not the driving mechanism.

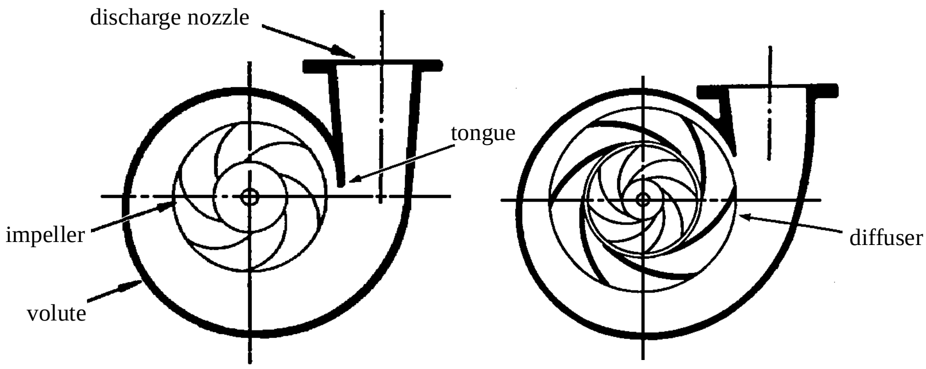

Fig. 1.6: Volute and diffusion casing pump (from Stepanoff, 1964)

finally leaves the pump through the discharge nozzle. The inlet of the pump as well as the volute (or diffuser region) are part of the pump casing. The impeller is mounted on a shaft which is connected to the driver. Seals or wearing rings are fitted on the impeller and casing to restrict flow under high pressure from leaking back to the impeller inlet or out of the pump casing.

The impeller contains several blades or vanes which are always curved backwards. They may be of single or double curvature (twisted at the suction ends). The blades are fitted to the hub (or back-shroud) of the impeller. The impeller may be equipped with a (front) shroud, or constructed as an open impeller.

The discharge region of the pump collects the fluid as it leaves the impeller. The shape of this region is such that the high velocity of the fluid is partly converted to static pressure by gradual expansion. Two main types of casings exist: the volute casing and the diffusion casing. The volute casing is formed by a single channel enclosing the impeller. Starting from the tongue of the volute, the channel through-flow area gradually increases towards the throat of the discharge nozzle to account for the increasing amount of fluid. It should be noted at this point that the ideal volute shape depends on the capacity at which the pump operates. For volute casings the major part of the conversion from kinetic energy to pressure occurs in the discharge nozzle. In a pump with diffusion casing, the impeller discharges into a channel equipped with stationary vanes. The conversion of energy now predominantly takes place between the diffusion vanes. Compared to the volute casing, the diffusion casing is compact, but its design is more complicated (see fig. 1.6).

The head H of a pump is defined as the height (in meters) against which the pump can lift a fluid. It is related to the increase in total pressure Δp_0 of the fluid by

$$H = \frac{\Delta p_0}{\rho g}, \tag{1.1}$$

where ρ is the fluid density and g is the gravitational constant. Not all of the power applied at the shaft of the pump is converted to an increase in total pressure. Several sources of losses can be identified. The most important are mechanical losses in shaft bearings, losses attributed to leakage, disc friction losses and hydraulic losses resulting from skin friction, mixing processes, boundary layer separation etc. The overall efficiency η of a pump is defined as the ratio of the pumps energy output to the power input applied at the shaft. In terms of delivered head H the overall efficiency can be written as

$$\eta = \frac{\rho g Q H}{P_{sh}}, \tag{1.2}$$

with Q the flow rate or capacity, and P_{sh} the shaft power. A convenient means of presenting the overall performance of a specific pump is the use of characteristic curves. The objective is to show the relation between quantities such as flow rate, rotational speed, delivered head, shaft power and fluid properties. The standard for centrifugal pumps is to plot the head, the efficiency and the shaft power as a function of flow rate, for a specific pump (fig. 1.7). The flow rate for which the efficiency is largest is called the best efficiency point.

A dimensional analysis (Stepanoff, 1964) reveals the existence of several scaling laws for groups of geometrically similar machines, which can be defined as follows:

$$\begin{aligned}
&\Phi = Q/(\Omega D^3) = \text{const.} && (a)\\
&\Psi = gH/(\Omega D)^2 = \text{const.} && (b)\\
&Re = \Omega D^2/\nu = \text{const.} && (c)\\
&\underset{\sim}{P} = P/\,(\rho\Omega^3 D^5) = \text{const.} && (d)
\end{aligned} \tag{1.3}$$

with Ω the rotational speed, D the diameter of the impeller, and ν the kinematic vis-

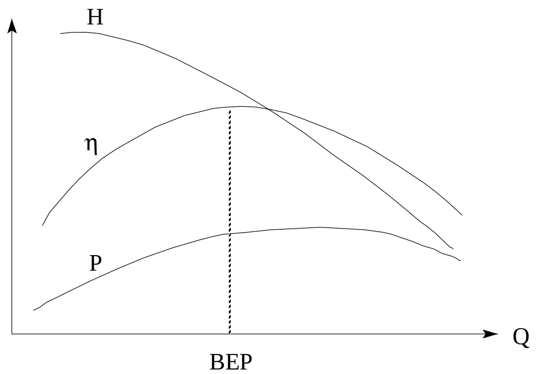

Fig. 1.7: Centrifugal pump characteristics: Head, efficiency and shaft power as a function of flow rate. Best efficiency point indicated by *BEP*.

cosity. The dimensionless quantity Φ is called the flow coefficient, Ψ the head coefficient, *Re* the Reynolds number and $\tilde{P}$ the dimensionless power coefficient. The method of similitude is based on these scaling laws and has been widely used to predict the effects of changes in operating condition (or impeller diameter) on performance. Once the characteristics for a specific pump are known from experiments, the performance of geometrically similar machines can thus be obtained fairly easily. As the dependency of flow behaviour on Reynolds number is much weaker compared to other factors, like flow rate, diameter and rotational speed, this coefficient can usually be disregarded in similarity methods. It should be noted however, that its influence cannot be neglected in case changes in Reynolds number are large, e.g. when interpreting characteristic curves of a scale model of a much larger pump.

From the flow coefficient and head coefficient (eqs. 1.3a and 1.3b) the diameter D may be eliminated to form the dimensionless specific speed coefficient n_ω:

$$n_\omega = \frac{\Phi^{1/2}}{\Psi^{3/4}} = \frac{\Omega Q^{1/2}}{(gH)^{3/4}}. \tag{1.4}$$

From its definition it follows that geometrically similar machines with similar internal flow conditions have the same specific speed value. It is commonly used as a type number to classify pumps of different design irrespective of the actual diameter, where values of the specific speed refer to a pump operating at best efficiency point. Fig. 1.8 shows the well known 'efficiency chart' where the maximum efficiency of numerous pumps is plotted against specific speed and lines of constant flow rate are drawn. It shows that there is a clear dependency of maximum attainable efficiency on specific speed. There appears to be a correlation between specific speed and the width-diameter ratio of impellers. Radial impellers are found to have low specific

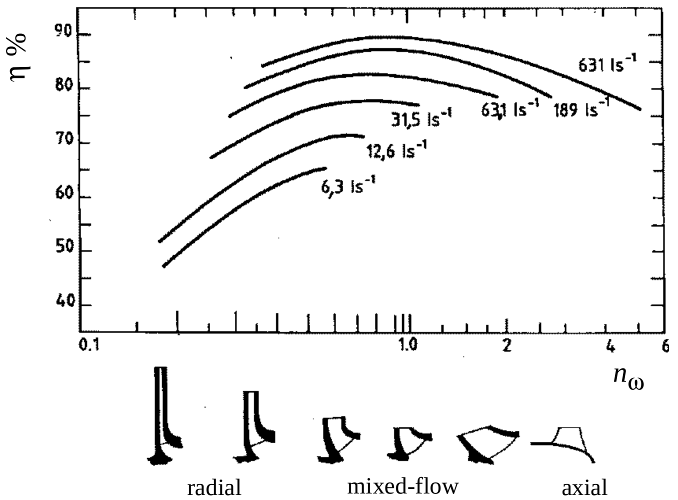

Fig. 1.8: Efficiency η_{BEP} as a function of specific speed n_ω and flow rate (liters/sec).

speed values (low capacity, high head) whereas axial impellers are associated with large specific speeds. Mixed-flow impellers form an intermediate class in terms of geometrical design and have intermediate values for the specific speed as well. The change in maximum efficiency with flow rate for a given class of similar pumps (constant specific speed in fig. 1.8) is directly related to the varying size of pumps within the class. It can be explained partly by the additional dependency of efficiency on Reynolds number (eq. 1.3c) and surface roughness (Csanady, 1964). Other causes are the varying mechanical losses with impeller diameter and the fact that seal clearances are generally not scaled according to the diameter.

1.3 Performance prediction

In this section an overview will be given of methods used for pump performance prediction. It is restricted to methods based on inviscid flow for the prediction of theoretical head, combined with additional models to quantify losses.

1.3.1 Theoretical head prediction

Consider a steady and uniform flow of an ideal fluid through a two-dimensional blade passage of an impeller revolving at constant rotational velocity Ω. Applying the principle of angular momentum leads to the well-known Euler equation for the theoretical head H_{th}

$$H_{th} = \frac{\Omega\,(r_2 v_{\theta,2} - r_1 v_{\theta,1})}{g}, \tag{1.5}$$

where v_θ is the absolute velocity in circumferential direction and subscripts *1* and *2* denote leading and trailing edge radii respectively. As radial and circumferential velocities are assumed to be uniform at impeller entrance and discharge, H_{th} can be determined from the flow rate and local blade angles. It means that no information on flow conditions within the impeller passages is required.

In practice, the flow through an impeller passage will not be uniform. Apart from the obvious deviations caused by boundary layers and wakes, the main flow will also be non-uniform due to its irrotational (absolute) velocity field. The latter is commonly referred to as ‘slip’. Many empirical methods to predict the effect of slip on the velocity field have been suggested. Busemann (1928) has set up an analytical theory for the simplified case of a potential flow through an isolated two-dimensional impeller with logarithmic spiral blades. Recent progress has been achieved by Visser et al. (1994).

Although the above mentioned methods are intended for two-dimensional flow, they have been applied to three-dimensional flows as well, in the absence of better alternatives. For this purpose, special methods were proposed to construct quasi two-dimensional stream-surfaces in passages of, for example, mixed-flow impellers.

The introduction of computers in the field of fluid dynamics led to the development of more refined flow models. One of these is the streamline-curvature method, described by, for example, Hamrick et al. (1952), Wood and Marlow (1966) and recently by Casey and Roth (1984). It is based on the assumption of constant entropy and stagnation enthalpy or rothalpy (in rotating impellers) along streamlines. Iterative procedures are adopted to satisfy mass conservation from blade to blade and to find the correct streamline shape. The method is two-dimensional, but frequently applied to stream-surfaces in three-dimensional flow. To account for truly three-dimensional flows, Wu (1952) developed the stream-surface method for incompressible flow. In this method the two-dimensional flow is solved (by means of a streamfunction formulation) on two distinct sets of stream-surfaces, i.e. oriented in hub-to-shroud and blade-to-blade directions. The exact location of the surfaces follows from an iteration procedure. As this method is complicated to implement it was soon superseded by three-dimensional methods based on the potential flow model (Daiguji, 1983a,b). Although somewhat more restrictive in terms of assumptions (entropy and stagnation enthalpy constant throughout the flow, equivalent to the flow being irrotational) its implementation is straightforward and solutions are obtained very fast.

For volute pumps, the mutual interference between impeller and volute becomes stronger at off-design operating conditions. Still, most studies of to-day concentrate on the flow through isolated impellers or volutes separately. To take full account of the interaction between impeller and volute, both regions must be considered together by means of an unsteady flow analysis. Morfiadakis et al. (1991) have performed an unsteady analysis of the potential flow in a volute pump in two dimensions, using a singularity method. Miner et al. (1992) did similar computations to obtain the velocity field using a finite element method, but neglected the unsteady shedding of trailing vortices. An improved method was presented by Badie et al. (1994) and Jonker and Van Essen (1997), which accounts for unsteady wakes behind impeller blades. It allows the computation of the unsteady pressure field. Although the method is two-dimensional, for use on radial pumps only, a variation in axial width is properly modelled.

A method to compute the unsteady three-dimensional potential flow through impeller-volute configurations of radial and mixed-flow pumps was presented by Van Esch et al. (1995) and Kruyt et al. (1996). Non-uniform blade loading and unsteady wakes behind impeller and stator blades are taken into account. It is an extension, in some respects, of the method proposed by Badie, Van Essen and Jonker, although the numerical algorithm is quite different.

1.3.2 Loss estimation

Once the theoretical head characteristic is obtained for an idealized pump by one of the methods described above, the shape of the actual head-capacity curve may be derived from an estimation of losses occurring in the pump. In addition, the efficiency-capacity curve can be constructed when the resulting head is related to shaft power. Until the present day loss estimations rely heavily on empirical models. This is partly a consequence of the lack of information on detailed flow characteristics within the pump. With three-dimensional (inviscid) flow simulations becoming available now, the accuracy of loss estimation is likely to increase.

Losses can be classified in four main groups, commonly referred to as hydraulic losses, leakage flow losses, disc friction losses and mechanical losses. The first group comprises all losses occurring within the internal fluid as it flows from the suction nozzle to the volute discharge. Hydraulic losses result in head loss as well as a decrease in efficiency. Numerous models have been put forward to account for energy dissipation in boundary layers, shock losses at impeller and diffuser vane entries, mixing processes, sudden changes in through-flow area, and gradual expansions (diffusion losses). As in general the detailed flow characteristics within the pump are unknown, models are rough and highly empirical. In pumps with shrouded impellers leakage flows will generally exist between the impeller discharge and the suction side. Many refined models (though for simple geometries) exist which are the result of both theoretical and experimental studies. One of the complicating factors is formed by the region of large clearance between the impeller discharge and the actual seal. Others are the influence of surface roughness, the distribution of circumferential velocity of the leakage flow and its effect on delivered head upon return in the main flow. Disc friction results from shear forces between the impeller external surfaces and the ambient fluid. Extensive measurements on rotating discs in closed chambers are available from which empirical models are derived. However, the influence of leakage flow and initial circumferential velocity on disc friction remains uncertain. Disc friction is a phenomenon external to the flow through the impeller and, as a consequence, does not lead to a head loss. The same is true for mechanical loss, which results from friction in bearings. For this reason, many authors (erroneously) consider disc friction and mechanical loss as belonging to one group, for which the loss in shaft power can be determined from geometrical quantities and shaft speed only.

It will be clear that some of the losses, although belonging to different groups, are closely inter-related. A correct analysis should consider these mutual influences and may require some iterative steps.

1.4 Aim and outline of this thesis

As pointed out in the previous section, pump performance prediction has largely been based (and still is, to a large extent) on experience. Characteristic curves are generally obtained from theoretical head estimations, which are subsequently corrected for viscous losses. The often unsatisfying results are only partly caused by the inadequacy of the viscous loss models used. An important contribution to the deviations should be attributed to the methods of determining the theoretical head values. Until now, theoretical head has been predicted by means of one-dimensional or (quasi-) two-dimensional methods. A method for head prediction based on truly three-dimensional computations would improve results substantially.

Anyone who is familiar with the present investigations in the field of turbomachinery, as reflected in the literature available, will know that attention is directed more and more towards methods capable of solving three-dimensional viscous flow. In principle, these Navier-Stokes models are able to predict loss implicitly. However, the present-day solvers are far from adequate, for a number of reasons. The most important reason is that turbulence is not yet fully understood (Hunt, 1985, Savill, 1987). Standard turbulence models appear to fail for flows in a rapidly rotating framework (Speziale 1985/1989). In addition, the accurate modelling of boundary layer transition and separation is a field of investigation which researchers are only beginning to explore. Tuning these methods in order to make solutions correspond to experiments before being able to use them, has little or no use. Adding to the (yet) unsolved problems, a major difficulty is posed by the extreme requirements in terms of computer resources. At present, an analysis of a single impeller channel in three dimensions, with an adequately refined mesh near the solid surfaces and in clearance gaps, exceeds the limits of most computers. Obviously, an unsteady analysis of a complete rotor-stator configuration, to account for interaction at off-design conditions, is way out of reach at this moment. As part of a design tool, current viscous methods are of limited suitability; the mere fact that they are available does not mean that it is profitable in all cases to use them.

For most pumps operating near design conditions, the influence of viscosity is restricted to thin boundary layers and wakes. The main flow can be predicted fairly accurately by means of three-dimensional inviscid methods. Using the method as presented in this thesis offers the possibility to perform an unsteady flow analysis in complete pump configurations of the radial and mixed-flow type. Based on solutions thus obtained, additional viscous loss models and a boundary layer analysis will yield good performance predictions. In practice, this means an improvement to the design tools of many engineers, and it will continue to be part of the conceptual design phase of a pump for a long time to come. It is the opinion of the author that pump designers should employ the immediate advantages offered by this method, while at the same time eagerly follow future developments.

In chapter 2 of this thesis, the inviscid flow model will be derived from the Navier-Stokes equations. Assumptions concerning the behaviour of the fluid will be explained. The heart of the thesis is formed by chapter 3, where the numerical algorithm is presented. Some special numerical techniques, which make complete pump simulations possible, are described in detail. In chapter 4, the correct implementation of some of the submodels is verified using analytical solutions to some simple problems. As a first step towards an accurate viscous analysis, some of the existing loss models will be presented in chapter 5, others will be adapted to suit the current application. Finally, in chapter 6, results of the current method will be compared with experimentally obtained values.

Nomenclature

D	Impeller diameter
H	Head
P_{sh}	Shaft power
$\underset{\sim}{P}$	Dimensionless power: $P/(\rho\Omega^3 D^5)$

Q	Flow rate
Re	Reynolds number
g	Gravitational acceleration
n_ω	Specific speed: $\Omega Q^{1/2}/(gH)^{3/4}$
p_0	Stagnation pressure
r	Radius
v	Absolute velocity

Greek symbols

Φ	Flow coefficient: $Q/(\Omega D^3)$
Ω	Angular velocity of impeller
Ψ	Head coefficient: $gH/(\Omega D)^2$
η	Efficiency
ν	Kinematic viscosity
ρ	Fluid density

Subscripts

1	Leading edge
2	Trailing edge
th	Ideal, inviscid
θ	Circumferential direction

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Chapter 2
Flow model

Fluid flow in general is described correctly by the Navier-Stokes equations. However, for most fluid flow problems these equations are far too complicated to be solved either analytically or numerically. In case of the flow through a hydraulic pump operating near design condition, the physical model can be simplified considerably without losing its overall validity. As a first assumption the fluid can be considered incompressible. Secondly, the core of the fluid can be regarded inviscid, as viscous forces are negligible when compared to inertia forces. The third assumption is that the flow enters the impeller free of vorticity. For an inviscid main flow this means that the flow remains irrotational. Combining these assumptions leads to the incompressible potential flow.

In section 2.1, the governing equations for a potential flow are derived from the Navier-Stokes equations. Lifting bodies require a special treatment in potential flow theory. This is discussed in section 2.2, where it is explained how wakes are formed in inviscid flow. In section 2.3, the boundary conditions are given for the special case of pump applications. The consequence of a multi-block approach on boundary conditions and its advantage for pumps is discussed in sections 2.4 and 2.5. The modelling of wakes behind pump impeller blades is the subject of section 2.6. Finally in section 2.7, the procedure advocated in this thesis will be introduced: an inviscid method with additional models to quantify viscous losses in confined regions.

2.1 Basic equations and assumptions

In an inertial frame of reference the flow of an isothermal Newtonian fluid is described by the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla . (\rho \underline{v}) = 0, \tag{2.1}$$

and by the Navier-Stokes equation

$$\frac{\partial \underline{v}}{\partial t} + \underline{v} . \nabla \underline{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \underline{v} + \frac{\nu}{3} \nabla \nabla . \underline{v} + \underline{F}, \tag{2.2}$$

with ρ the density, $\underline{v}$ the velocity vector, t the time, p the pressure, ν the kinematic viscosity, and $\underline{F}$ an external force. A number of assumptions can be made for the flow in hydraulic pumps:

- Mach-numbers are usually small enough to justify the assumption of incompressible flow ($Ma^2 << 1$). The continuity equation then reduces to

$$\nabla . \underline{v} = 0. \tag{2.3}$$

- Reynolds-numbers Re defined as

$$Re = \frac{\Omega D^2}{\nu}, \tag{2.4}$$

  with Ω the impeller rotational speed and D the impeller diameter, are of order 10^5 to 10^7. This means that viscous forces can be neglected when compared to inertia forces, except in boundary layers and wakes. With this assumption, the Navier-Stokes equation (2.2) for the main flow reduces to

$$\frac{\partial \underline{v}}{\partial t} + \underline{v}.\nabla \underline{v} = -\frac{1}{\rho}\nabla p + \underline{F}. \tag{2.5}$$

- Boundary layers in the impeller will generally be thin compared to the width of the impeller passage, provided that no boundary layer separation occurs. To show this, assume the boundary layer along the impeller to be turbulent from the start. If no pressure gradient acts along the stream lines, the boundary layer displacement thickness δ^* will be (Schlichting, 1979)

$$\frac{\delta^*}{x} = \frac{0.046}{Re_x^{0.2}}, \tag{2.6}$$

  with x the local coordinate along the blade and Re_x the Reynolds number based on x. Assuming the impeller passage width S at the impeller outer radius to be equal to the blade length L, the ratio of δ^* and S at the trailing edge radius will be

$$\left.\frac{\delta^*}{S}\right|_{x=L} = \frac{0.046}{Re_L^{0.2}} . \tag{2.7}$$

  For Re_L of order 10^5 to 10^7 this ratio will have values between 0.002 and 0.005.
- Flows with Reynolds number exceeding the value 10^5 will be turbulent. This means that the velocity $\underline{v}$ can be written as the sum of its time-averaged value $\underline{V}$

and a fluctuation $\underline{v}'$

$$\underline{v} = \underline{V} + \underline{v}' . \tag{2.8}$$

Inserting eq. (2.8) into eq. (2.5) and time averaging gives

$$\frac{\partial \underline{V}}{\partial t} + \underline{V} . \nabla \underline{V} = -\frac{1}{\rho} \nabla p + \underline{F} - \nabla (\overline{\underline{v}'\underline{v}'}) , \tag{2.9}$$

where the last term represents the gradient of the Reynolds stresses, defined as the ensemble averaged products of velocity fluctuations. From a dimensional analysis, the importance of Reynolds stresses relative to pressure and convective forces can be estimated as

$$Tu^2 = \frac{u'^2}{U^2} , \tag{2.10}$$

where *Tu* is called the turbulence intensity, and *U* and *u'* are characteristic values for the mean velocity and the fluctuation respectively. In practice *Tu* will be 5 percent or less for the core of the flow, which means that the relative importance of Reynold stresses is of order 10^{-3}. In case of boundary layer separation this will in general not be true.

- The final assumption which can be made in case of hydraulic pumps is the flow being irrotational

$$\nabla \times \underline{v} = 0 . \tag{2.11}$$

Vorticity is generated by viscous shear forces or non-conservative external forces. Thus, in the absence of non-conservative forces and viscous effects confined to thin (and attached) boundary layers and wakes, the core of the flow can be assumed irrotational, provided that the incoming fluid is free of rotation. A velocity potential ϕ can now be defined as

$$\underline{v} = \nabla \phi , \tag{2.12}$$

which can be substituted into eq. (2.3) to obtain the Laplace equation

$$\nabla^2 \phi = 0 . \tag{2.13}$$

Once the potential is solved from the Laplace equation with appropriate boundary conditions (given in section 2.3), the pressure can be derived from eq. (2.5). In case of irrotational flow and the external force of gravity, this equation reduces to the unsteady Bernoulli equation

$$\frac{\partial \phi}{\partial t} + \frac{1}{2} \underline{v} \cdot \underline{v} + \frac{p}{\rho} + gz = c(t) , \tag{2.14}$$

where g is the gravitational acceleration, z is the height, and c only depends on time t. It appears that the total pressure p_0, defined as

$$p_0 = p + \frac{1}{2}\rho \underline{v} \cdot \underline{v} , \tag{2.15}$$

is only changed by a flow's unsteadiness (at constant z-coordinate). Since the governing Laplace equation is steady in time, unsteadiness is introduced through boundary conditions.

2.2 Circulation and the formation of wakes

2.2.1 Circulation

From the Kutta-Joukowsky theorem for the lift force L

$$L = \rho U \Gamma, \tag{2.16}$$

with U the free stream velocity, we know that a non-zero circulation Γ exists around two-dimensional lifting bodies in free space. The circulation is defined as

$$\Gamma = \oint_C \underline{v} . d\underline{s}, \tag{2.17}$$

in which C can be any contour enclosing the body, and s is the distance along C. Substituting eq. (2.12) into eq. (2.17) gives

$$\Gamma = \oint_C d\phi, \tag{2.18}$$

which equals zero unless a discontinuity in the potential is present at a position along the curve. The obvious way to add circulation around a body in potential flow theory is by introducing a pair of infinitesimally close lines extending from some point on the body to the external boundary of the domain. A constant potential jump $\Delta\phi$, equal to Γ, is prescribed over this so-called slit line. In this way circulation is added while at the same time the velocity remains continuous across the slit line.

For two-dimensional problems, the introduction of slit lines is also necessary from a mathematical point of view. The solution of the Laplace equation is not uniquely determined for a multiply connected region. The reason is that pure circulation of an arbitrary strength is a solution to the Laplace equation. A unique solution can however be obtained using suitable slit-lines which convert the region into a singly connected domain.

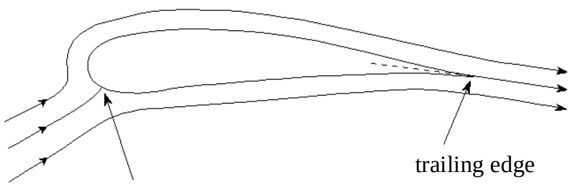

Fig. 2.1: Stagnation points and stream direction for an airfoil in steady flow.

Fig. 2.2: "Giesing-Maskell" flow condition for unsteady flow at trailing edge stagnation point depends on sign of trailing vortices: clockwise (left) or anti-clockwise.

2.2.2 Kutta condition

Airfoils

The amount of circulation around lifting airfoils is determined from the Kutta condition, which states that an unseparated flow leaves a sharp trailing edge of an airfoil smoothly. This can only be the case if the trailing edge is a stagnation point. The pressure is continuous in the vicinity of the edge and the velocity will have a finite value. However, stated like this, the Kutta condition by no means enforces a unique solution. The direction in which the flow leaves the trailing edge is still undetermined. It is common practice, though, to extend the condition with the hypothesis that the stagnation streamline bisects the wedge angle of the airfoil, in case of steady flow (fig. 2.1). This is often referred to as the "classical Kutta condition". Experimental data indicate that this is indeed the case, even for trailing edges which are somewhat rounded.

For unsteady flow, it is argued by Giesing (1969) and Maskell (1971) that the flow leaves the trailing edge along extensions of either the lower surface or the upper surface, depending on the sign of the change in bound circulation (fig. 2.2). Basu and Hancock (1978) came to the same conclusion after performing inviscid flow computations for an airfoil oscillating in pitching motion. Some experimental evidence in support of the Giesing-Maskell condition is supplied by Poling and Telionis (1986).

Impeller blades

The situation for impeller blades is quite different from the well defined trailing edge geometries mentioned above. A two-dimensional view of a typical impeller geometry is given in fig. 2.3. It is postulated by the author that the separation point (called *the* trailing edge, from now on) is located on the suction side of the blade. Following the above mentioned arguments, the direction of the stagnation streamline should either be along the bisection of the trailing edge angle for steady relative flow, or alternating between directions along the suction surface and the blunt trailing edge surface in

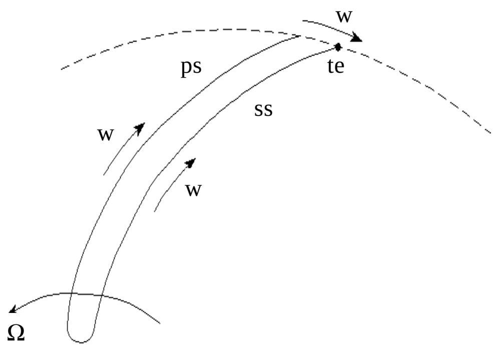

Fig. 2.3: Typical impeller blade geometry, showing position of stagnation point and relative velocity direction. (PS=pressure side, SS=suction side)

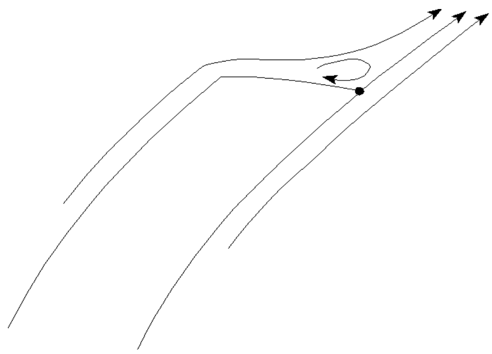

Fig. 2.4: Presumed flow direction at the trailing edge of an impeller blade.

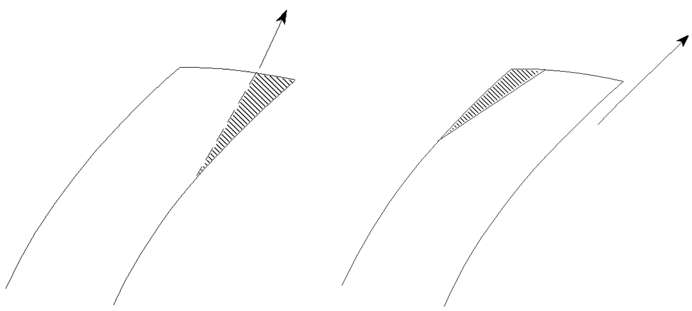

Fig. 2.5: Blade profiling at trailing edge section (hatched area shows removed material): underfiling (left) and overfiling.

case of unsteady flow. However, a severe shortcoming of this model is the fact that it does not reduce to the limiting case of an impeller blade with negligible thickness. To overcome this drawback, it is assumed that the stagnation streamline is always in the direction of the suction surface (fig. 2.4). Boundary layer separation at the blunt trailing edge surface might well be the underlying mechanism. A flow direction like this is in agreement with one-dimensional and two-dimensional methods for the estimation of delivered head, which have been used successfully for many years. Practical experience with blade profiling at the trailing edge appears to support the assumption that it is essentially the suction surface which determines the local flow direction. The method of ‘underfiling’ the blade is known to increase the head-value, while ‘overfiling’ has a negligible effect on delivered head (fig. 2.5).

An attempt to measure the flow direction near the blade trailing edge of a free mixed-flow impeller at the University of Twente is described in section 6.4. However, a conclusive answer in favor of a flow direction either along the suction surface or along the wedge angle cannot be given on the basis of these measurements.

In the current model, the Kutta condition is formulated as a vanishing normal velocity at the trailing edge separation point

$$\underline{w}.\underline{n}\big|_{te} = 0, \tag{2.19}$$

with $\underline{w}$ the velocity relative to the airfoil and $\underline{n}$ the outward unit vector normal to the suction surface. In case of rotating impeller blades the relative velocity is related to the absolute velocity $\underline{\mathrm{v}}$ by

$$\underline{w} = \underline{\mathrm{v}} - \underline{\Omega} \times \underline{r}, \tag{2.20}$$

where $\underline{\Omega}$ is the angular velocity of the rotor, and $\underline{r}$ is the position vector with respect to the axis of rotation. Thus, condition (2.19) can be written in terms of the absolute velocity as

$$\underline{\mathrm{v}}.\underline{n}\big|_{te} = (\underline{\Omega} \times \underline{r}) \cdot \underline{n}\big|_{te}, \tag{2.21}$$

It should be stressed that velocity components tangential to the blade are allowed to be discontinuous at the trailing edge, as long as the pressure remains continuous.

2.2.3 Wakes

For a three-dimensional airfoil the circulation will in general vary along its span. In unsteady flow the circulation will also vary in time. From Kelvin’s circulation theorem for inviscid flow

$$\frac{D\Gamma}{Dt} = 0, \tag{2.22}$$

it follows that a varying circulation around the airfoil must result in a continuous shedding of vortices with strength equal and opposite to the change in bound vortic-

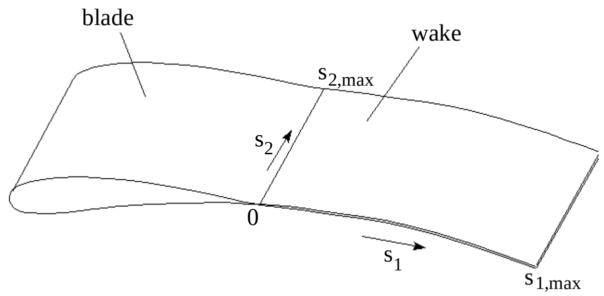

Fig. 2.6: Representation of a vortex sheet behind a blade

ity. The vortex sheet which is thus formed is called a wake.

Consider a wake in which vortices are shed by a blade. Fig. 2.6 shows a blade and its wake, which is represented by two coinciding surfaces. It is assumed that lines with constant s_2-coordinate represent streamlines of the average flow. The potential values at position (s_1,s_2) are denoted by $\phi^+(s_1,s_2)$ and $\phi^-(s_1,s_2)$, where the plus and minus sign correspond to upper and lower wake surface respectively. Upper and lower values of the potential differ by a certain amount

$$\phi^+(s_1, s_2) = \phi^-(s_1, s_2) + \Delta\phi(s_1, s_2)\,, \tag{2.23}$$

where $\Delta\phi(s_1,s_2)$ is called the potential jump at position (s_1,s_2). The blade circulation in a point at the trailing edge is defined as

$$\Gamma_{s_2} = \Delta\phi(0, s_2)\,. \tag{2.24}$$

The value of the potential jump in the wake depends on whether the flow is steady or unsteady.

Steady flow along a blade

For a blade in steady flow, the blade circulation will not vary in time, although a variation along the trailing edge is possible. According to Kelvin's circulation theorem, the circulation Γ around a blade and its wake is a conserved quantity for potential flows. This means that the potential jump will be constant along a streamline. Referring to fig. 2.6, values of the potential jump can be expressed as

$$\Delta\phi(s_1, s_2) = \Delta\phi(0, s_2) = \Gamma_{s_2}. \tag{2.25}$$

Unsteady flow along a blade

In unsteady flow the value of the blade circulation will be time-dependent, as well as dependent on the position along the trailing edge. From the unsteady Bernoulli equation (2.14) it follows that

$$\frac{\partial}{\partial t}(\phi^+ - \phi^-) = -\frac{1}{2}\left(\left|\nabla\phi^+\right|^2 - \left|\nabla\phi^-\right|^2\right), \tag{2.26}$$

since the pressure is continuous across the wake. After linearization we obtain

$$\frac{\partial}{\partial t}(\phi^+ - \phi^-) + U_{s_1}\frac{\partial}{\partial s_1}(\phi^+ - \phi^-) = 0, \tag{2.27}$$

with U_{s1} the mean fluid velocity in streamwise direction. This means that vortices which are shed at trailing edge points are convected downstream with the mean velocity along the wake.

2.3 Computational domain and boundary conditions

Consider a typical pump geometry consisting of (part of) the inlet pipe, the impeller, the discharge region (diffuser or volute) and the exit pipe. Different types of conditions apply at its boundaries:

Inlet and outlet surfaces

For boundaries representing the inlet or outlet of the turbomachine a uniform normal velocity is prescribed

$$\frac{\partial\phi}{\partial n} = \pm\frac{Q}{A}, \tag{2.28}$$

where Q is the flow rate and A is the area of the surface. It is assumed that the surfaces are located far enough up- and downstream to justify the application of undisturbed and uniform flow conditions.

Impeller blade surfaces

At the impeller blade surfaces (both pressure and suction side) the Neumann boundary condition takes the form

$$\frac{\partial\phi}{\partial n} = (\underline{\Omega} \mathrm{x} \underline{r}) \cdot \underline{n}, \tag{2.29}$$

where $\underline{\Omega}$ is the rotational velocity of the impeller and $\underline{n}$ is the outward unit normal vector.

Stator vane surfaces, hub, shroud and volute walls

At boundaries which do not move, and at the hub and the shroud, normal velocity components vanish

$$\frac{\partial\phi}{\partial n} = 0. \tag{2.30}$$

Wake surfaces

At stream surfaces emanating from trailing edges, vortices are shed downstream. These vortex sheets (or wakes) are a result of both nonuniform blade loading (variation of the circulation along the blade's span) and time-dependent variation of the blade circulations. Within the potential flow model, wakes can be modelled by the boundary conditions

$$\begin{aligned} \phi^+(s_1, s_2) &= \phi^-(s_1, s_2) + \Delta\phi\,(s_1, s_2) \\ \frac{\partial}{\partial n}\,\phi^+(s_1, s_2) &= -\frac{\partial}{\partial n}\,\phi^-(s_1, s_2) \end{aligned} \tag{2.31}$$

where plus and minus signs denote both sides of the wake, and s_1 and s_2 are coordinates along the wake. By allowing the potential jump $\Delta\phi$ to vary along the vortex sheet, a discontinuity of the tangential velocity across the wake is introduced.

2.4 Multi-block approach

The presence of a rotor and a stator part naturally suggests using a multi-block approach. The flow region of interest is divided into non-overlapping subdomains or blocks, all having a topologically cubic shape. For a free rotor computation one block will usually suffice, although a division into more blocks is possible. However, for a flow computation inside a complete pump a number of blocks are required (see fig. 2.7).

An advantage of the multi-block approach is the greater ease in creating a good mesh for the complex three-dimensional geometries that are considered here. It also constitutes an important component of the numerical method that is described in the next chapter. It will be shown that a considerable reduction of computing time can thus be achieved.

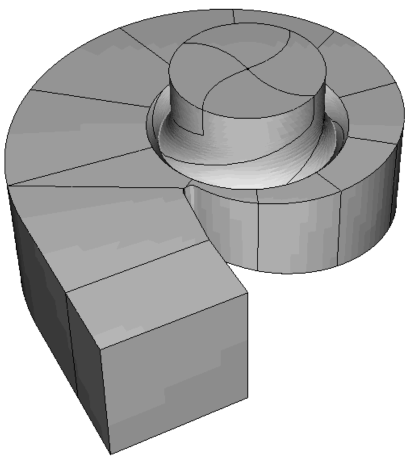

Fig. 2.7: Example of a pump geometry divided into blocks.

In the multi-block approach additional boundary conditions have to be formulated that apply at the artificial internal boundaries between blocks. The velocity field at these internal boundaries should be continuous. Therefore, the values of the potential at corresponding nodes can differ only by a fixed amount and normal velocities are opposite. This means that the boundary conditions for such internal boundaries are the same as those for wakes (eq. 2.31), with $\Delta\phi$ constant:

$$\begin{aligned} \phi^{+}(s_1, s_2) &= \phi^{-}(s_1, s_2) + \Delta\phi \\ \frac{\partial}{\partial n}\phi^{+}(s_1, s_2) &= -\frac{\partial}{\partial n}\phi^{-}(s_1, s_2) \end{aligned} \tag{2.32}$$

Periodic boundary conditions, for a free rotor computation, are also of this type.

2.5 Rotor-stator interaction

When considering configurations of complete pumps, the rotating motion of the impeller relative to the stator has to be taken into account. In order to achieve this without having to create a new mesh for each time step (as was done in Miner et al., 1992), blocks in the rotor and the stator are separated by a cylindrical or conical surface. In this way the impeller blocks are allowed to rotate freely while sliding along the stator blocks (see fig. 2.7). Both the rotor and the stator part have a coordinate system attached to it. Local time-derivatives of the potential (necessary to compute the pressure) are determined relative to these coordinate systems, by a differencing scheme central in time.

The material derivative of the scalar quantity ϕ is invariant for rotation:

$$\left.\frac{D\phi}{Dt}\right|_R = \frac{D\phi}{Dt}, \tag{2.33}$$

with subscript R denoting the time derivative of the absolute potential relative to the rotating frame of reference. From eq. (2.33) it can be shown that local time derivatives are related as

$$\frac{\partial\phi}{\partial t} = \left.\frac{\partial\phi}{\partial t}\right|_R + (\underline{w} - \underline{v}) \cdot \nabla\phi, \tag{2.34}$$

since the nabla operator is invariant as well. Substituting eqs. (2.12), (2.20) and (2.34) into eq. (2.14) gives the Bernoulli equation in the rotating frame of reference

$$\left.\frac{\partial\phi}{\partial t}\right|_R - (\underline{\Omega} \times \underline{r}) \cdot \underline{v} + \frac{1}{2}\underline{v} \cdot \underline{v} + \frac{p}{\rho} + gz = c(t), \tag{2.35}$$

which is used to compute the pressure at positions in the rotor part. This method is identical to the method employed in Badie et al. (1994) and Jonker and Van Essen (1997).

The division of the geometry into a rotor and stator part poses an important consequence on the modelling of wakes behind impeller blades. It is discussed in the next section.

2.6 Modelling wakes

The flow in a pump is unsteady. Especially at off-design conditions the blade circulations will vary in time. One way of computing this unsteady flow would be to incorporate the varying blade circulations, but to neglect the shedding of trailing vortices in the wakes. This is called the quasi-steady approach. Contrary to this method, a fully unsteady computation requires the shed vortices to be taken into account as well. The distribution of potential jumps in the wake is now time-dependent and can only be determined by performing a number of calculations at successive time steps. As a result, the latter method will require much more computing time.

If wake surfaces could be placed from the trailing edges down to the exit pipe of the pump, the distribution of potential jumps in the wakes would be properly described by the equations in section 2.2.3. However, in the current model these surfaces cannot be extended beyond the cylindrical or conical surface which divides the rotor part from the stator part of the pump. Therefore, the varying potential jump across the wake must take a constant value upon reaching this point.

In this section both methods are described. Anticipating the results of computations which will be presented in chapter 6, some results concerning the difference between both methods and the influence of a finite wake length are given as well.

2.6.1 Quasi-steady wake computation

If the shedding of trailing vortices is neglected, potential jumps in the wakes should follow eq. (2.25). However, the finite wake length in the current model requires the varying potential jumps across the wake to be averaged out gradually. Referring to fig. 2.6, the value of the potential jump in the wake can now be described by

$$\Delta\phi(s_1, s_2) = \Gamma_{s_2} + \frac{s_1}{s_{1,max}}(\Gamma_{\mathrm{av}} - \Gamma_{s_2}), \tag{2.36}$$

where Γ_{av} is the average blade circulation, given by

$$\Gamma_{\mathrm{av}} = \frac{1}{s_{2,max}} \int_0^{s_{2,max}} \Gamma_{s_2}\, ds_2. \tag{2.37}$$

The second term at the right hand side of eq. (2.36) can be considered as the deviation from Kelvin's circulation theorem (eq. 2.22). However, when averaging this deviation over s_2, the deviation reduces to zero. At the end of the wake, the averaged value of the potential jump is guided to the outer wall of the pump, along part of the conical surface and some block boundaries located in the stator region of the pump.

It should be noted that a quasi-steady wake computation like this will show local discontinuities in the static pressure at the wake surfaces and at internal block boundaries along which the potential jump is guided to the pump's outer wall. This can be seen from eqs. (2.14) and (2.35) where one should realize that the potential jump across the surfaces will generally vary in time, whereas the velocity will be continuous. To prevent this, blade circulations should be frozen temporarily during the time interval used for computing the pressure.

2.6.2 Unsteady wake computation

In computations where the convection of shed vortices (eq. 2.27) is taken into account, the potential distribution should also be averaged in case of a wake of finite length. Suppose that the non-averaged potential jump distribution at a given position (s_1,s_2) is denoted by $\Delta\Phi(s_1,s_2)$. An averaging procedure similar to eq. (2.36) can now be introduced. This yields

$$\Delta\phi\,(s_1, s_2) \;=\; \Delta\Phi\,(s_1, s_2) + \frac{s_1}{s_{1,\,max}}\,[\Delta\Phi_{\mathrm{av}}\,(s_1) - \Delta\Phi\,(s_1, s_2)]\,, \tag{2.38}$$

where the average value of the potential jumps at constant s_1-coordinate is given by

$$\Delta\Phi_{\mathrm{av}}\,(s_1) \;=\; \frac{1}{s_{2,\,max}}\int_0^{s_{2,\,max}} \Delta\Phi\,(s_1, s_2)\; ds_2. \tag{2.39}$$

Again the deviation from the exact solution reduces to zero when averaging over locations at constant s_1. As was the case in the quasi-steady approach, the potential jump is equal to the local blade circulation at points on the trailing edge. Upon reaching the end of the wake, the jump takes a value which is constant for all s_2. This averaged value is guided to the outer wall of the pump, along part of the conical surface and some block boundaries located in the stator region of the pump.

Similar to quasi-steady wake computations, the static pressure at block boundaries along which potential jumps are guided to the outer wall will show a discontinuity. In this case it is caused by the varying strength of vortices which reach the end of the wakes at the sliding interface. It can be prevented by eliminating those free vortices which are to leave the wakes within the time-interval used to compute the pressure.

2.6.3 Results

To show the difference between the quasi-steady and the fully unsteady wake model in actual computations, some of the results for a mixed-flow impeller-volute combination (presented in section 6.5) are given here. In fig. 2.8 head-capacity curves are given for both methods, where it is seen that differences increase at off-design condi-

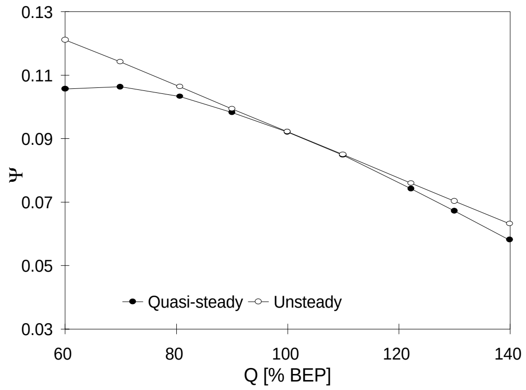

Fig. 2.8: QH-curves for a mixed-flow pump, using the quasi-steady and the fully unsteady wake model.

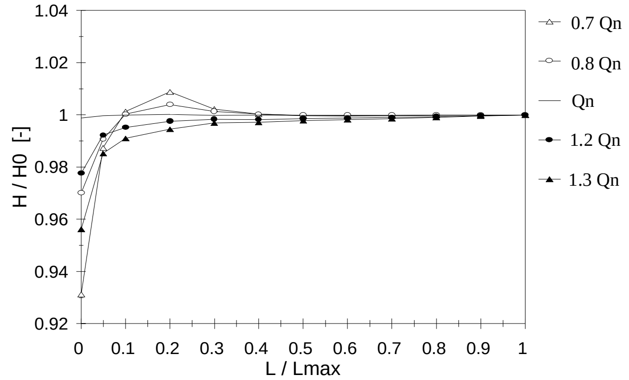

Fig. 2.9: Influence of wake length on computed head for several flow rates. Wake length L is given as a fraction of maximum wake length L_{max}. H_0 is the head at maximum wake length.

tions. It shows that quasi-steady wake computations may suffice at best efficiency point, but at off-design the fully unsteady wake model should be employed. For this case, the results of both methods coincide at optimum flow rate. The reason is that the volute matches the impeller at this flow rate. As a consequence the time-variation of the blade circulation is very small compared to conditions at higher and lower flow rate.

The distance between the impeller blade trailing edge and the tongue of the volute determines the maximum wake length L_{max}. The influence of the wake length is given in fig. 2.9. It appears that results for nominal flow rate are nearly independent of wake length. This is in agreement with the results presented in fig. 2.8. At off-design conditions the minimum required wake length increases, but it shows that, at least in this case, L_{max} is still sufficiently large.

2.7 Viscous losses

In section 2.1 it was shown that the flow through hydraulic pumps can be characterized by an inviscid main flow where the influence of viscosity is small and confined to local areas, like thin boundary layers and wakes. For this reason a computational method based on an inviscid model, like the potential flow model, will yield good results. A valid subsequent step in the approximation is to incorporate the effect of viscosity by using additional models to quantify the viscous losses. Many of these models exist, most of which are based on input variables obtained by an inviscid analysis.

In chapter 5, a number of additional models will be presented. Among these are models to quantify leakage flow, disc friction and hydraulic losses. It should be regarded as a first step in the direction of an accurate viscous analysis.

Nomenclature

A	Area
D	Impeller diameter
$\underline{F}$	External force
L	Blade length / Lift force
Ma	Mach number
Q	Flow rate
Re	Reynolds number
S	Impeller passage width at trailing edge radius
Tu	Turbulence intensity
U,u'	Characteristic velocity and fluctuation
g	Gravitational acceleration
$\underline{n}$	Unit normal vector
p	Static pressure
p_0	Stagnation pressure
$\underline{r}$	Radius vector
s_1,s_2	Local surface coordinates
t	Time
$\underline{v}$	Absolute velocity vector
$\underline{w}$	Relative velocity vector
x	Local blade coordinate
z	Height

Greek symbols

Γ	Circulation
$\underline{\Omega}$	Angular velocity vector
δ^*	Boundary layer displacement thickness
ϕ	Velocity potential
ν	Kinematic viscosity
ρ	Fluid density

Subscripts

R	Rotating frame of reference
av	Average value
te	At blade trailing edge
x	Based on local blade coordinate

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Chapter 3
Numerical algorithm

Many investigators have presented numerical methods for computing the potential flow inside a rotor channel in two or three dimensions, and in complete rotor-stator configurations in two dimensions. See for example the work of Daiguji (1983a,b), Worster and Worster (1983), Maiti et al. (1989), Chen and Sue (1993), Badie et al. (1994), and Van Essen (1995). In order to impose the Kutta condition of smooth flow at the blades' trailing edges they either superimpose a number of subpotentials (reflecting unit flow rate, unit rotation and line vortices of unit strength, shed from points at the trailing edges) or determine the (varying) circulations iteratively.

In this chapter, a fast method is presented for computing the unsteady three-dimensional potential flow inside an entire pump configuration (Van Esch et al., 1995, Kruyt et al., 1996). It takes full account of the unsteadiness induced by rotor-stator interaction. The flow field is solved by means of a finite element method using an extension of the super-element technique (Zienkiewicz and Taylor, 1989). In this technique internal degrees of freedom of all super-elements are eliminated from the discretized governing equation. The extension of the method presented here deals with an analogous elimination of the internal degrees of freedom from the discretized Kutta conditions. The method consists of two steps:

- elimination of internal degrees of freedom from the system of equations (Laplace equation and Kutta conditions), for all blocks separately. This leads to the formulation of the super-elements.
- assemblage of the super-elements. After solving the resulting global system of equations, the previously eliminated degrees of freedom are obtained.

In section 3.1, the finite element method is briefly introduced. The actual numerical method is presented in sections 3.2 and 3.3. Some aspects of wake model implementation are given in section 3.4. The numerical method is integrated in a much larger information system for pump performance analysis. This system will be described in section 3.5. Finally, a comparison in terms of computer resources between the current numerical method and a method based on subpotentials will be given in section 3.6.

3.1 Finite Element Method

Consider the Laplace equation defined in domain V with Neumann boundary conditions at surface S_N and Dirichlet conditions at surface S_D:

$$\begin{aligned} \nabla^2\phi &= 0 \qquad \text{in} \quad V \\ \frac{\partial\phi}{\partial n} &= \mathrm{v}_n \qquad \text{at} \quad S_N \\ \phi &= \phi_0 \qquad \text{at} \quad S_D \end{aligned} \tag{3.1}$$

where surfaces S_N and S_D are non-overlapping and $S = S_N \cup S_D$, with S the surface enclosing volume V. The outward normal direction is denoted by n. If it is presumed that the potential ϕ satisfies the Dirichlet boundary condition, eq. (3.1) is equivalent to stating

$$\int_V \psi\,(\nabla^2\phi)\,dV + \int_{S_N} \psi^*\left(\frac{\partial\phi}{\partial n} - \mathrm{v}_n\right)dS = 0, \tag{3.2}$$

for arbitrary weight functions ψ and ψ^*. The first term of eq. (3.2) can be written as

$$\int_V \psi\,(\nabla^2\phi)\,dV = -\int_V \nabla\psi\,.\nabla\phi\,dV + \int_S \psi\frac{\partial\phi}{\partial n}dS \tag{3.3}$$

from integration by parts. Substituting eq. (3.3) into (3.2) and choosing

$$\psi^* = -\psi, \tag{3.4}$$

which is allowed since both functions are arbitrary, eq. (3.2) can be written as

$$\int_V \nabla\psi\,.\nabla\phi\,dV - \int_{S_N} \psi\,\mathrm{v}_n\,dS - \int_{S_D} \psi\frac{\partial\phi}{\partial n}dS = 0. \tag{3.5}$$

Note that by performing a partial integration, the weight functions ψ are required to be continuous. On the other hand, the order of the governing differential equation is reduced by one. Hence, we only assume that ϕ is once differentiable, which is equivalent to the assumption that the velocity exists. Eq. (3.5) is called the weak formulation of the original boundary value problem (eq. 3.1).

The approximate solution to the potential ϕ is obtained by requiring eq. (3.5) to be satisfied for a given set of weight functions only. It means that local derivations (or residuals) to the governing equation and boundary condition may exist, as long as the weighted integral equals zero, hence the name “weighted residual method”. Noting that the potential ϕ is presumed to satisfy the Dirichlet boundary condition on surface S_D, weight functions ψ may be restricted to functions which are zero on boundary S_D.

Thus eq. (3.5) changes into

$$\int_V \nabla\psi \, . \, \nabla\phi \, dV = \int_{S_N} \psi \, \mathrm{v}_n \, dS. \tag{3.6}$$

In the finite element method the domain is divided into small subdomains or elements. For each element a number of nodal points are defined where the potential value is given by ϕ_i, with i the node number. The approximate solution ϕ can be given in terms of nodal values by

$$\phi = \sum_{i=1}^{n} N_i \, \phi_i \, , \tag{3.7}$$

where n is total number of nodes in the domain. Functions N_i are called interpolation functions or shape functions.

In the method of Galerkin, the set of weight functions ψ is taken equal to the set of interpolation functions N. Using this method and inserting eq. (3.7) into eq. (3.6) gives a system of equations

$$\lceil L \rfloor \, \{\phi\} = \{F\} \, , \tag{3.8}$$

in which $\{\phi\}$ is the vector of unknowns ϕ_i (i=1..n) and matrix coefficients L_{ij} and vector elements F_i are given by

$$L_{ij} = \int_V \nabla N_i \, . \, \nabla N_j \, dV \tag{3.9}$$

and

$$F_i = \int_{S_N} N_i \, \mathrm{v}_n dS. \tag{3.10}$$

Note that the matrix [L] is independent of the externally applied Neumann boundary conditions, the influence of which is restricted to the right-hand-side vector {F}.

If interpolation functions N_i are confined to individual elements (subdomain collocation) the above integrations are restricted to element interiors and boundaries. As a result the system matrix will be sparse.

In the current three-dimensional method, elements are tetrahedral in shape. The four nodes of each element have linear interpolation functions which take a unit value at their corresponding node, while being equal to zero at all other nodes of the element.

3.2 Super-element formulation

In the super-element (or substructuring) technique, the entire domain is divided into subdomains or blocks. For each block, the Laplace equation for the velocity potential together with the Neumann and Dirichlet boundary conditions (if any) is discretized according to the standard finite-element method (section 3.1), resulting in a system of linear equations

$$[L]\,\{\phi\} = \{F\} + \{R\}\,, \tag{3.11}$$

where $[L]$ is a positive-definite matrix reflecting the discretized Laplace operator, $\{\phi\}$ is the vector of degrees of freedom (DOFs) and $\{F\}$ is the vector corresponding to flow rates through block boundaries resulting from Neumann type boundary conditions. Vector $\{R\}$ is the equivalent of $\{F\}$ and is related to the (still unknown) flow rates at internal block boundaries.

For each block, the discretized Kutta conditions can be expressed in terms of potential values in the block using the modified gradient operator $[K]$

$$[K]\,\{\phi\} = \{\,(\underline{\Omega} x \underline{r})\cdot\underline{n}\big|_{te}\}\,. \tag{3.12}$$

Applied at the values of the potential, operator $[K]$ gives the normal velocities at the trailing edges.

The basic idea of the super-element technique is to express eqs. (3.11) and (3.12) in terms of DOFs at internal block boundaries (called the “master” DOFs), by eliminating the remaining interior “slave” DOFs. For this purpose eqs. (3.11) and (3.12) are partitioned as follows

$$\begin{bmatrix} [L^{ss}] & [L^{sm}] \\ [L^{ms}] & [L^{mm}] \end{bmatrix} \begin{Bmatrix} \{\phi^s\} \\ \{\phi^m\} \end{Bmatrix} = \begin{Bmatrix} \{F^s\} \\ \{F^m\} + \{R^*\} \end{Bmatrix}, \tag{3.13}$$

$$\begin{bmatrix} [K^s] & [K^m] \end{bmatrix} \begin{Bmatrix} \{\phi^s\} \\ \{\phi^m\} \end{Bmatrix} = \{\,(\underline{\Omega} x \underline{r})\cdot\underline{n}\big|_{te}\}\,. \tag{3.14}$$

Superscripts s and m denote “slaves” and “masters” respectively. Vector $\{R^*\}$ denotes the non-zero part of vector $\{R\}$ in eq. (3.11).

By extracting the slave DOFs from eq. (3.13) it follows that

$$\{\phi^s\} = [L^{ss}]^{-1}\{F^s\} - [L^{ss}]^{-1}[L^{sm}]\,\{\phi^m\}\,. \tag{3.15}$$

The resulting “super system” is obtained after substituting eq. (3.15) into eqs. (3.13) and (3.14)

$$\left[L^{sup}\right]\{\phi^m\} = \{F^{sup}\} + \{R^*\} \tag{3.16}$$

$$\left[K^{sup}\right]\{\phi^m\} = \{G^{sup}\} \tag{3.17}$$

where

$$\left[L^{sup}\right] = \left[L^{mm}\right] - \left[L^{ms}\right]\left[L^{ss}\right]^{-1}\left[L^{sm}\right] \tag{3.18}$$

$$\left[K^{sup}\right] = \left[K^{m}\right] - \left[K^{s}\right]\left[L^{ss}\right]^{-1}\left[L^{sm}\right] \tag{3.19}$$

and

$$\{F^{sup}\} = \{F^m\} - \left[L^{ms}\right]\left[L^{ss}\right]^{-1}\{F^s\} \tag{3.20}$$

$$\{G^{sup}\} = \{ (\underline{\Omega} x \underline{r}) \cdot \underline{n}\big|_{te}\} - \left[K^{s}\right]\left[L^{ss}\right]^{-1}\{F^s\} \tag{3.21}$$

Note that column i of -$[L^{ss}]^{-1}[L^{sm}]$ in eqs. (3.18) and (3.19) can be interpreted as the values for the slave potentials corresponding to master DOF i equal to 1 and all other master DOFs equal to zero. Similarly, the term $[L^{ss}]^{-1}\{F^s\}$ in eqs. (3.20) and (3.21) represents the effect of Neumann boundary conditions on slave DOFs, while keeping the master DOFs equal to zero. Note that these potentials can be computed by simple back-substitutions once the matrix $[L^{ss}]$ has been decomposed. For the terms in eq. (3.17) a clear physical interpretation can be given: matrix $[K^{sup}]$ contains normal velocity components at trailing edge points induced by unit master DOFs, while the last term in eq. (3.21) represents normal velocities induced by Neumann boundary conditions.

In principle this procedure must be carried out for all blocks that form the entire geometry. However, an important observation is that the super-matrices and right-hand-side vectors are invariant under rotation for the scalar equations considered. Therefore the symmetry of the rotor can be exploited, as in general all rotor channels are geometrically identical. This means that the superelement formulation step has to be performed for the blocks of a single rotor channel only. Furthermore, in a time-dependent computation, the superelement formulation step will have to be carried out only once.

The assemblage of blocks, which can be regarded as super-elements, is part of the second step. This is described in the following section.

3.3 Assemblage of super-elements

In the global solution step, the values of master DOFs of all participating blocks are determined by assembling and solving the global system of equations.

A complicating factor in the computation of the master DOFs is the fact that blade circulations and, as a consequence, the potential jumps at nodes in the wakes are still unknown. Therefore the values of blade circulations are regarded as additional variables to be determined along with the nodal DOFs (see also Baskharone and Hamed, 1981). The circulation at a blade section is given by the potential jump at the trailing edge node across the wake. The vector of unknowns in the global problem is now denoted by

$$\{u\}^T = \{\phi_1, ..., \phi_{n_\Phi}, \Gamma_1, ..., \Gamma_{n_\Gamma}\} = \{\{\Phi\}^T \{\Gamma\}^T\} . \tag{3.22}$$

where n_Φ is the number of nodes in all block connections (coinciding nodes counted as one), n_Γ is the number of unknown blade circulations (i.e. the total number of nodes at trailing edges), $\{\Phi\}$ is the vector of unknown master DOFs for the potential and $\{\Gamma\}$ is the vector of unknown blade circulations.

The master DOFs can now be expressed in terms of global DOFs. Note that master DOFs may also involve potential jumps in order to add circulation to the flow (section 2.2). Depending on the type of computation (quasi-steady or unsteady) and the location within the wake, potential jumps are composed of unknown blade circulation values and/or free vortices of which the strength is known from previous time steps. All master DOFs of block b are now written formally as

$$\{\phi_b^m\} = \left[T_b^\Phi\right]\{\Phi\} + \{\Delta\phi_b\} , \tag{3.23}$$

where $\{\Delta\phi_b\}$ is the potential jump vector, given by

$$\{\Delta\phi_b\} = [W_b]\{\Gamma\} + \{\gamma_b\} , \tag{3.24}$$

and

$[T_b{}^\Phi]$ = matrix which gives the transformation of global equation numbers of nodal DOFs to the local numbering of master DOFs in block b. Each row contains exactly one nonzero coefficient, with value 1.

$[W_b]$ = matrix which gives the equation numbers of global blade circulation DOFs for the masters of block b. It also accounts for the “averaging” of potential jumps in the wakes.

$\{\gamma_b\}$ = known values of potential jumps at boundaries of block b. These jumps are present in computations including unsteady wakes.

Similar to the way in which element matrices and right-hand-side vectors are assembled to form the large system of equations, the super-element matrices and right-hand-side vectors of eqs. (3.16) and (3.17) are assembled into a global system of equations for the Laplace equation and for the Kutta conditions

$$\sum_{b=1}^{n_b} \left[T_b^{\Phi}\right]^T \left[L_b^{sup}\right] \{\phi_b^m\} = \sum_{b=1}^{n_b} \left[T_b^{\Phi}\right]^T \{F_b^{sup}\} \tag{3.25}$$

$$\sum_{b=1}^{n_b} \left[T_b^{\Gamma}\right]^T \left[K_b^{sup}\right] \{\phi_b^m\} = \sum_{b=1}^{n_b} \left[T_b^{\Gamma}\right]^T \{G_b^{sup}\} \tag{3.26}$$

where matrices $[T_b^{\Gamma}]$ take care of the transformation of the global blade circulation DOFs to the equation numbers of the local numbering of trailing edge nodes and n_b is the total number of participating blocks. Note that the contribution of the vectors $\{R_b^*\}$ in eq. (3.16) cancel at internal block boundaries.

Substituting $\{\phi_b^m\}$ from eq. (3.23) into eqs. (3.25) and (3.26) gives the global system of equations

$$\begin{bmatrix} [M^{\Phi\Phi}] & [M^{\Phi\Gamma}] \\ [M^{\Gamma\Phi}] & [M^{\Gamma\Gamma}] \end{bmatrix} \begin{Bmatrix} \{\Phi\} \\ \{\Gamma\} \end{Bmatrix} = \begin{Bmatrix} \{H^{\Phi}\} \\ \{H^{\Gamma}\} \end{Bmatrix}, \tag{3.27}$$

where

$$\left[M^{\Phi\Phi}\right] = \sum_{b=1}^{n_b} \left[T_b^{\Phi}\right]^T \left[L_b^{sup}\right] \left[T_b^{\Phi}\right] \tag{3.28}$$

$$\left[M^{\Phi\Gamma}\right] = \sum_{b=1}^{n_b} \left[T_b^{\Phi}\right]^T \left[L_b^{sup}\right] \left[W_b\right] \tag{3.29}$$

$$\left[M^{\Gamma\Phi}\right] = \sum_{b=1}^{n_b} \left[T_b^{\Gamma}\right]^T \left[K_b^{sup}\right] \left[T_b^{\Phi}\right] \tag{3.30}$$

$$\left[M^{\Gamma\Gamma}\right] = \sum_{b=1}^{n_b} \left[T_b^{\Gamma}\right]^T \left[K_b^{sup}\right] \left[W_b\right] \tag{3.31}$$

and

$$\{H^{\Phi}\} = \sum_{b=1}^{n_b} \left[T_b^{\Phi}\right]^T (\{F_b^{sup}\} - \left[L_b^{sup}\right] \{\gamma\}) \tag{3.32}$$

$$\{H^{\Gamma}\} = \sum_{b=1}^{n_b} \left[T_b^{\Gamma}\right]^T (\{G_b^{sup}\} - \left[K_b^{sup}\right] \{\gamma\}) \tag{3.33}$$

Once the global system is solved, the solution for the potential for a block is obtained by first computing the values for the master DOFs (using eq. 3.23) and subsequently performing a back-substitution to determine the slave DOFs from eq. (3.15). This procedure is carried out using the decomposed matrix $[L^{ss}]$ which was stored on disc during the elimination step.

3.4 Wake models

3.4.1 Quasi-steady wake computation

In the quasi-steady wake model the shedding of trailing vortices is neglected. This means that at any instant the wake is assumed to be steady. Thus, the flow field can be determined for an isolated time step, without any information from previous time steps. The potential jump distribution $\Delta\phi$ over a wake surface was given in section 2.6.1. In fig. 3.1, a schematic representation is given. Eq. (2.36) is applied at each individual node in the wake. In this way the potential jump can be written as a linear combination of the circulation DOFs $\{\Gamma\}$, and substituted into eq. (3.23).

The algorithm to perform a quasi-steady wake computation is as follows:

- Determine the super-element system (eqs. 3.16 and 3.17) for all blocks.
- For all time steps in a blade revolution period:
  - Construct the potential jump distribution in the wakes, according to eq. (2.36) and insert into eq. (3.24).
  - Solve the global system of equations (3.27).
  - Determine the master DOFs for all blocks from eq. (3.23).
  - Obtain the slave DOFs for all blocks from eq. (3.15).

To prevent pressure discontinuity across internal block boundaries (refer to section 2.6.1), potential jump distributions should be kept constant during time-intervals used for computing the pressure.

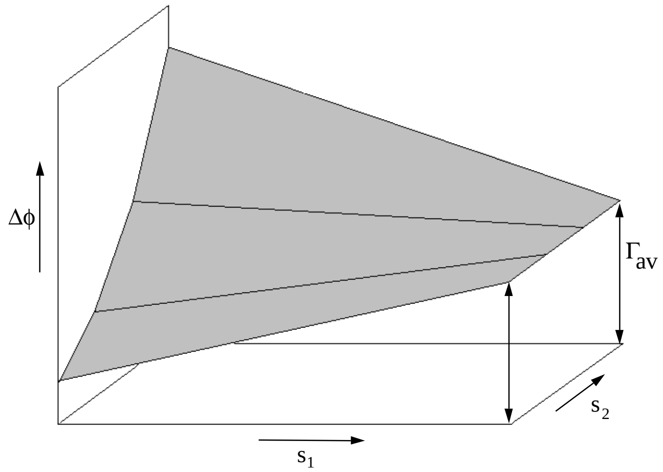

Fig. 3.1: Schematic representation of the potential jump distribution over a quasi-steady wake surface with local coordinate system (s_1,s_2). Average blade circulation is indicated by Γ_{av}.

3.4.2 Unsteady wake computation

For truly unsteady wake computations the transportation of free vortices along the wake (as described in section 2.2.3) should be taken into account. Fig. 3.2 shows how a potential jump distribution $\gamma(s_1,s_2,t_1)$ along a streamline at time step t_1 is transported downstream the wake to its new position at time step t_2. The strength of the free vortices is known from previous time steps. During this time interval the change in blade circulation results in the formation of a new trailing vortex with strength dΓ. Its length l_v is equal to the time interval Δt=t_2-t_1 times the local relative velocity. Thus, for all DOFs in the wake, the potential jump $\Delta\phi$ can be given in terms of circulation DOFs $\{\Gamma\}$ and known potential jumps γ (from convecting free vortices), and inserted into eq. (3.24).

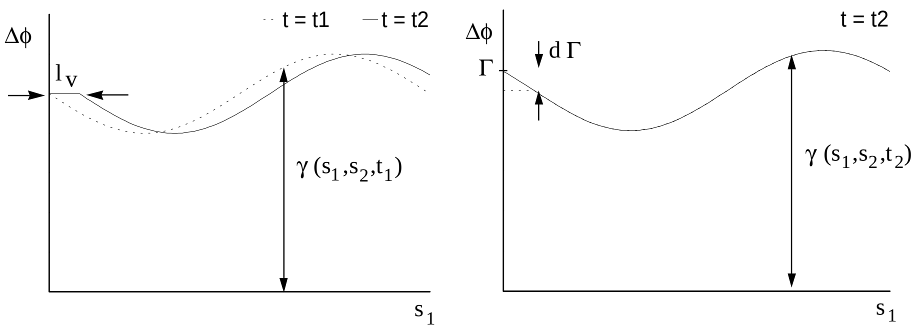

Fig. 3.2: Transportation of free vortices along a wake streamline (left) and formation of trailing vortex with strength dΓ.

An obvious way to perform a computation with fully unsteady wakes would be:

- Determine the super-element system (eqs. 3.16 and 3.17) for all blocks.
- Compute the mean velocity distribution relative to the wakes for all time steps in a blade revolution period, using the quasi-steady wake model, and store velocities on disc.
- For each time step (starting with ‘empty’ wakes):
    - Read the wake velocities for the current and the next time step from disc and determine the average of these two values.
    - Compute the length l_v of all trailing vortices.
    - Compute the potential jump distribution in the wakes, according to eq. (2.38) and insert into eq. (3.24).
    - Compute the strengths $\{\Gamma\}$ of all trailing vortices by solving the global system of equations (3.27).
    - Use the velocities along the wakes to convect the trailing vortices downstream, during the next time interval (eq. 2.27).

  This iteration procedure should be continued until a solution is obtained which is periodic within a certain desired accuracy.
- For all time steps in a blade revolution period:
    - Determine the master DOFs for all blocks from eq. (3.23).
    - Obtain the slave DOFs for all blocks from eq. (3.15).

However, as in general a large number of iterations are required (typically a number of shaft revolutions), computing time for this method is large.

A much faster algorithm has been implemented in the current method. It is based on the fact that the contribution of known potential jumps γ to the global system of equations is restricted to its right-hand side, as seen from eqs. (3.27) to (3.33). Once the system matrix has been decomposed, solutions for different wake vortex distributions can be obtained by (relatively fast) back-substitutions. This process might be performed for the case of ‘empty’ wakes, except for one single node where the potential jump has the value one. In fact, similar back-substitutions may be performed for all nodes located in wakes. From the solutions which are thus obtained, the strength of trailing vortices resulting from an arbitrary wake vortex distribution can simply be derived by superposition. In other words: once all the solutions for unit potential jumps are computed for a given time step, solutions for arbitrary vortex distributions are obtained without solving the global system of equations. The complete algorithm is as follows:

- Determine the super-element system (eqs. 3.16 and 3.17) for all blocks.
- Compute the mean velocity distribution relative to the wakes for all time steps in a blade revolution period, using the quasi-steady wake model, and store velocities on disc.
- Compute solutions for all unit potential jumps in the wakes, for all time steps in a blade revolution period, and store corresponding trailing vortex strengths on disc.
- For each time step (starting with ‘empty’ wakes):
    - Read the wake velocities for the current and the next time step from disc and

determine the average of these two values.
- Compute the length l_v of all trailing vortices.
- Compute the potential jump distribution in the wakes, according to eq. (2.38) and insert into eq. (3.24).
- Read the vortex strengths due to unit potential jumps in the wakes from disc, for the current time step.
- Compute the strengths $\{\Gamma\}$ of all trailing edge vortices by superposition of 'unit-potential-jump-strengths'.
- Use the velocities along the wakes to convect the trailing vortices downstream, during the next time interval (eq. 2.27).

The iteration should be continued until a periodic solution is obtained.

• For all time steps in a blade revolution period:
    - Insert the periodic potential jump distributions (eq. 3.24) into eq. (3.23) and solve the global system of equations (3.27)
    - Determine the master DOFs for all blocks from eq. (3.23).
    - Obtain the slave DOFs for all blocks from eq. (3.15).

To prevent pressure discontinuity across internal block boundaries (refer to section 2.6.2), potential jump distributions should be adapted to prevent free vortices to leave the wakes within a time-interval used for computing the pressure.

3.5 Implementation aspects

Second-order accuracy for potential, velocities and pressures is obtained by employing linear elements in combination with the SPR-technique (Zienkiewicz and Zhu, 1992) for the determination of the gradient of the potential, i.e. the velocity. In the global solution step the system of equations is solved using a direct method. An algorithm based on the method of characteristics has been implemented to solve the transport eq. (2.27) for the unsteady wake model. The correct implementation of both methods will be verified in chapter 4. The profile width of the sparse global matrix is reduced using a spectral renumbering technique (Hendrickson and Leland, 1995).

The numerical method has been implemented as a flow solver in the information system COMPASS (Van Esch and Kruyt, 1996). COMPASS is an acronym for Centrifugal Or Mixed-flow Pump Analysis Software System. It consists of a number of programs, of which the most important are:

• An interactive program for the (parametric) description of arbitrary impeller geometries and the subsequent block division and mesh generation (Van Os, 1996).
• A similar program for volute geometries.
• A multi-block potential flow solver as described in this chapter.
• A program for the analysis of computed results (post-processing), offering the possibility to compute user defined functions of the potential, velocity components and geometric quantities. Integration of these functions over lines, surfaces and volumes can also be performed.
• An interactive program for visualizing geometries, meshes, and user defined

quantities like velocity vectors, pressure values, etc.

Software has been developed using the Borland C++ programming environment. Implementation is done in the ANSI-C programming language. It allows the use of various platforms, ranging from personal computers to supercomputers.

3.6 Computer resources

In this section the use of computer resources (both computing time and internal memory requirements) is considered for the method proposed in this thesis. Results are compared with an analogous method based on subpotentials, in which no use is made of a multi-block approach. The latter method is in fact a three-dimensional extension of the two-dimensional method of Badie et al. (1994) and Van Essen (1995).

3.6.1 Method of subpotentials

Quasi-steady wake computation

Let N be the total number of nodes in the three-dimensional mesh of an impeller-volute configuration. If this geometry is modelled as a cylinder of radius r and height $r/2$ containing an equidistant mesh, then the half bandwidth B of the system matrix can be estimated as

$$B = \frac{1}{2} N^{2/3} . \qquad (3.34)$$

A direct solution method for a symmetric matrix like this would take time

$$T_{LU} = NB^2 / V \qquad (3.35)$$

for the decomposition and an additional

$$T_B = 2NB / V \qquad (3.36)$$

for the subsequent backsubstitution to solve the total matrix-vector equation. In these expressions V is the computational speed.

If this solution method were to be used to perform a quasi-steady impeller-volute computation, the total time required would be

$$T = t\,[T_{LU} + (kn_b + 2)\,T_B] , \qquad (3.37)$$

where t is the number of time steps, n_b the number of impeller blades and k the number of nodes at each trailing edge. The number of back-substitutions is equal to the number of subpotentials (kn_b line vortices in the wakes and one for the external Neumann boundary conditions) plus one for the total solution. Note that for one time step the system matrix remains unchanged for all subpotentials.

An iterative method can be used instead to solve the system of equations. The time needed can be expressed as

$$T_i = N n_c n_i / V, \tag{3.38}$$

where n_c is the number of non-zero coefficients on a matrix row, and n_i is the number of iterations. Based on potential flow calculations with SEPRAN (Segal, 1992), n_i is estimated as

$$n_i = 14 N^{1/3} \tag{3.39}$$

using a conjugate gradient method (accuracy: 10^{-10}). It should be stressed that estimates of the number of iterations are extremely difficult to make, as it highly depends on the type of problem, the initial solution and the required accuracy. A quasi-steady computation would then take time

$$T = t(kn_b + 1) T_i \tag{3.40}$$

for a converged solution.

Unsteady wake computation

An unsteady wake computation proceeds as follows. First a quasi-steady computation is performed to determine the velocity distributions relative to the wakes for all time steps in one blade revolution period. During this process the subpotentials for unit vortices shed from trailing edge nodes are determined as well. The last step is an iteration over a number of blade revolution periods in which trailing edge vortices are convected downstream the wakes. Only two back-substitutions are required for each time step; one for the subpotential (the combined effect of external Neumann boundary conditions and free vortices in the wakes) and one for the eventual solution. For a direct solution method the total computing time will be

$$T = t[T_{LU} + (2kn_b + 1) T_B] + tI(T_{LU} + 2T_B), \tag{3.41}$$

where I is the number of blade revolutions needed for a converged solution. The time for the quasi-steady solution (eq. 3.37) to determine the wake velocity is recognized in the first term of eq. (3.41). The additional kn_b back-substitutions reflect the unit vortices shed from trailing edge nodes. It is assumed that decomposed system matrices for different time steps are too large to be kept in internal memory (or stored on disc) for later use.

An iterative method will take time

$$T = t(2kn_b + 1) T_i + tIT_i . \tag{3.42}$$

Internal memory required is

$$M = n_r NB \tag{3.43}$$

for a direct solution method, and

$$M = n_r N \frac{1}{2} n_c \tag{3.44}$$

for an iterative solution method (factor ½ accounts for symmetry). In these expressions n_r is the length of a real number (in bytes) and M is the internal memory required (in bytes).

3.6.2 Multi-block method

Consider an impeller with vaneless diffuser which is divided into n_s blocks of equal dimensions $(n_1 \times n_2 \times n_3)$, where n_1 is in radial direction (e.g. along the impeller blade), n_2 is in circumferential direction and n_3 is in axial direction. Assume dimension n_1 to be largest. The number of blocks in the rotor and the stator region are both equal to $n_s/2$. It is further assumed that each impeller channel contains one block. The number of master DOFs n_m for one block can be estimated as

$$n_m = n_1 n_3 + n_2 n_3, \tag{3.45}$$

where it is assumed that impeller blades extend over $n_1/2$ nodes in radial direction.

Super-element formulation

In the super-element procedure, the system matrix for one block is first decomposed, followed by n_m backsubstitutions to obtain the super-element system. The (half) bandwidth B will be

$$B = n_2 n_3. \tag{3.46}$$

The total time for one block can then be expressed as

$$T_S = T_{S,LU} + n_m T_{S,B} , \tag{3.47}$$

where $T_{S,LU}$ and $T_{S,B}$ are the times required for decomposition and back-substitution respectively, and are given by

$$T_{S,LU} = NB^2/V \tag{3.48}$$

and

$$T_{S,B} = 2NB/V, \tag{3.49}$$

with matrix dimension

$$N = n_1 n_2 n_3. \tag{3.50}$$

Internal memory requirement is given by eq. (3.43).

Global procedure

The system matrix for the global solution procedure has dimension

$$N = \frac{1}{2} n_s n_m . \tag{3.51}$$

This is easily understood by realizing that the system matrix is constructed by assembling n_s super-systems with n_m nodes. Roughly half of the DOFs are eliminated as coinciding nodes are counted as one. Note that the number of circulation DOFs is not considered, as this number is small compared to N. After some calculus the half bandwidth B can be estimated as

$$B = \sqrt{25 - \left(\frac{100}{3n_s}\right) + 3n_s} \cdot \frac{n_m}{3} . \tag{3.52}$$

The times $T_{G,LU}$ and $T_{G,B}$ for decomposition and back-substitution are given by expressions analogous to eqs. (3.48) and (3.49). Thus the total computing time needed (1) to decompose the global system matrix, (2) to perform a subsequent back-substitution and (3) to derive the slave DOFs for all blocks by back-substitutions is now

$$T_G = T_{G,LU} + T_{G,B} + n_s T_{S,B} . \tag{3.53}$$

Internal memory requirement is again given by eq. (3.43).

Quasi-steady wake computation

For a quasi-steady wake computation the total computing time is simply given by

$$T = \left(1 + \frac{n_s}{2}\right) T_S + t\, T_G , \tag{3.54}$$

where the super-element procedure is applied at only one impeller block and all diffuser blocks.

Unsteady wake computation

An unsteady wake computation consists of a number of consecutive steps (see section 3.4.2). First of all the velocities relative to the wakes are computed by a quasi-steady computation. The second step concerns the computation of the strength of trailing edge vortices as a result of unit potential jumps at master DOFs in wakes. The results of these two steps enable us to perform an iteration over blade revolutions without having to solve the system of equations even once. As this iteration procedure is very fast, the total computing time is virtually independent of the number of iterations. The final step concerns the computation of master and slave DOFs based on the potential jump distribution in the wakes. The total computing time can thus be expressed as

$$T = \left(1 + \frac{n_s}{2}\right) T_S + t\,T_G + t\left(T_{G,\,LU} + \frac{1}{2} n_s n_w T_{G,\,B}\right) + t\,T_G, \tag{3.55}$$

where the time for a quasi-steady computation (eq. 3.54) is recognized in the first two terms at the right hand side of eq. (3.55). The third term represents the time to compute the strength of trailing edge vortices; one system matrix decomposition and $n_s n_w/2$ back-substitutions, where $n_s/2$ is the number of wakes, and n_w the number of master DOFs at one wake, estimated as

$$n_w = \frac{1}{4} n_1 n_3. \tag{3.56}$$

3.6.3 Comparison

Computer resources required for the two methods described above will be compared for the case of an impeller containing five blades with a vaneless diffuser. For both methods the total number of DOFs is 64,000. The case is defined by

k = 10
n_b = 4
t = 15
I = 10
V = 60 Mflops
n_r = 8 bytes

and additionally for the method of subpotentials

n_c = 13

and for the multi-block method

n_{b} = 8
n_1 = 40
n_2 = 20
n_3 = 10

The estimated computer resources are given in table 3.1. Computing time is given in hours, internal memory in Mbytes. It shows that the proposed multi-block method is by far superior to the method of subpotentials in terms of computing time, even if an iterative solution method is used. Internal memory usage is larger for the multi-block method when compared to an iterative solution for the subpotential method, but still within the limits of workstations and today's personal computers.

	Subpotentials (direct)	Subpotentials (iterative)	Multi-block
T quasi-steady	3.1 h	1.4 h	0.4 h
T unsteady	29.2 h	3.3 h	1.1 h
Internal memory	410 Mb	3.6 Mb	super: 13 Mb global: 26 Mb

Table 3.1: Comparison of computer resources

Nomenclature

$[L]$ Laplace operator matrix
$\{F\}$ Laplace operator right hand side vector
$\{R\}$ Laplace operator right hand side vector
$[K]$ Kutta condition matrix
$\{G\}$ Kutta condition right hand side vector
$[M]$ Global system matrix
$\{H\}$ Global system right hand side vector
$[T]$ Transformation matrix of nodal DOFs
$[W]$ Transformation matrix of circulation DOFs
B Half bandwidth of matrix
I Number of blade revolutions
N Dimension of matrix
N_i Interpolation function
M Internal memory (in bytes)
S Surface
T Computing time (in sec)
V Volume / Computational speed (in flops)
k Number of nodes at a trailing edge
$\underline{n}$ Unit normal vector
n_b Number of impeller blades
n_c Number of non-zero coefficients on matrix row
n_i Number of iterations
n_m Number of master DOFs
n_r Length of real (in bytes)
n_s Number of blocks in multi-block method

n_w Number of nodes in a wake
n_1 Number of nodes in radial direction
n_2 Number of nodes in circumferential direction
n_3 Number of nodes in axial direction
$\underline{r}$ Radius vector
t Number of time steps per blade revolution period
u Global DOF
v Absolute velocity

Greek symbols
Φ Global DOF
Γ Circulation DOF
$\underline{\Omega}$ Angular velocity vector
ϕ Velocity potential
γ Potential jump between block boundaries
ψ, ψ^* Weight functions

Subscripts
LU LU-decomposition
B Backsubstitution
S Super-element procedure
G Global procedure
av Average value
b Block
n Normal direction
i Iterative
te At trailing edge

Superscripts
T Transpose
m Master
s Slave
sup Super system
Φ Nodal DOFs
Γ Circulation DOFs

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Chapter 4
Verification of the numerical implementation

The subject of this chapter is verification of the implementation of the numerical method. Although numerous tests have been performed to check the correctness of the code, verification of only two of the implemented submodels will be discussed: the method to derive the velocity from a given potential field (section 4.1), and the wake model (sections 4.2 and 4.3). It should be stressed that the validation of the underlying flow model is by no means demonstrated. This will be the subject of chapter 6.

The velocity is computed using a special smoothing technique, called Super Convergent Patch Recovery (SPR, Zienkiewicz and Zhu, 1992). It results in a higher order accuracy for the derivatives of the potential solution. Verification is done by comparing the results with analytic solutions for two-dimensional impellers. Furthermore, the accuracy of the SPR method is compared with the accuracy obtained by an ordinary nodal averaging procedure.

The correct implementation of the wake model is verified using two test-cases for which analytic potential flow solutions exist. First the three-dimensional wake behind a steady wing is considered. Then the two-dimensional unsteady wake behind a flat plate in oscillatory motion is compared with analytic solutions. The combination of both cases is assumed to be sufficient for three-dimensional unsteady wake verification.

4.1 The SPR method

In this section attention is directed towards the accuracy of the solution and its derivatives obtained by a finite element procedure. A smoothing method, called Super Convergent Patch Recovery (SPR), for increasing the accuracy of derivatives of the solution is compared with an ordinary nodal averaging procedure, using elements with linear and quadratic interpolation functions.

4.1.1 Convergence rates using polynomial interpolation functions

If interpolation functions, which are used to approximate the exact solution in each subdomain, consist of a complete polynomial of degree p, then the maximum error E

in the approximation for an element of size h can be written as

$$E = O(h^{p+1}) . \tag{4.1}$$

The error in the approximation of the d-th derivative of the solution will be

$$E = O(h^{p+1-d}) . \tag{4.2}$$

See for instance Zienkiewicz and Morgan (1983).

4.1.2 Smoothing procedures

Calculation of derivatives from the finite element approximation is done by differentiating the interpolation functions directly. It results in lower order accuracy and discontinuities at element boundaries. One way of resolving the discontinuity at element boundaries is to employ smoothing by simple nodal averaging. A disadvantage of this technique is that the accuracy is low for nodes at the boundary of the domain and for internal nodes when the mesh is not equidistant.

A smoothing method which improves the accuracy of derivatives by one order of magnitude, is the Superconvergent Patch Recovery (SPR) method (Zienkiewicz and Zhu, 1992). In this technique the derivative is expressed as a polynomial expansion, restricted to an element patch surrounding each node. The coefficients of the expansion are determined from a least square minimization over superconvergent points. As this is a local process, the additional computational cost is small.

4.1.3 Quadratic elements vs. linear elements using SPR

We now focus on the solution of the potential flow in an isolated impeller with straight blades of zero thickness. The problem is solved in three dimensions. Three different cases are distinguished:

i) Linear interpolation functions; tetrahedral elements with 4 nodes; the velocity is obtained by simple nodal averaging.
ii) As in case (i), but the SPR smoothing technique is used to recover the velocity.
iii) Quadratic interpolation functions; isoparametric tetrahedral elements with 10 nodes; and simple nodal averaging to obtain the velocity.

We concentrate on the accuracy of the computed value of the blade circulation (see Visser et al., 1994, for analytic solutions). While the interpolation functions are fixed for each element (constant p), the size h of the elements near the trailing edges of the blades is progressively reduced by increasing the total number of elements or by refining the elements using non-uniform grids (Fried and Yang, 1972). This process is called *h-convergence* (Szabo and Mehta, 1978), as opposed to *p-convergence* where the mesh is fixed and the order of the polynomial interpolation functions is increased.

Bearing in mind that the computed circulation value depends on the determination of the velocity at the trailing edge, and applying eqs. (4.1) and (4.2), we can predict the rate of convergence for the potential and circulation values (table 4.1).

case	order of polynomial	accuracy of potential	accuracy of circulation
i	1	$O(h^2)$	$O(h)$
ii	1	$O(h^2)$	$O(h^2)$
iii	2	$O(h^3)$	$O(h^2)$

Table 4.1: Expected accuracy

Results of computations can be found in fig. 4.1. Although the results for case (ii) (linear interpolation functions with SPR) and case (iii) (quadratic interpolation functions and nodal averaging) are nearly identical, as was expected, the two methods appear to be only first order accurate, where second order accuracy was expected.

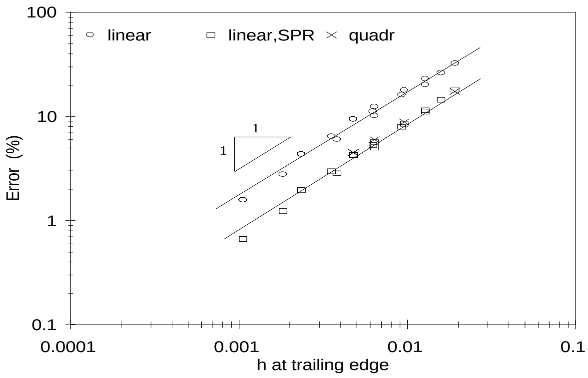

Fig. 4.1: Error in computed blade circulation value for an isolated rotating impeller with straight thin blades as a function of element size h near the trailing edge.

4.1.4 Influence of singularity on accuracy

The reason for the reduced convergence rate is the existence of a singularity in the velocity field. Similar behaviour is observed when dealing with crack analysis, a well-known problem in fracture mechanics (Tong and Pian, 1973, Gallaghar, 1978, Szabo and Metha, 1978). It is shown in Zienkiewicz and Taylor (1989), that the singularity is reduced when the crack tip in this kind of problems is rounded. Indeed the rate of convergence approaches its theoretical value when doing so. It appears that in problems with singularities the rate of convergence for the d-th derivative is

$$O\left(h^{\min(p+1-d,\,\lambda)}\right), \tag{4.3}$$

where λ is a number depending on the intensity of the singularity.

Apparently, in case of the rotating impeller with straight thin blades, the value for λ is close to one. Removing the singularity altogether can be achieved by setting the rota-

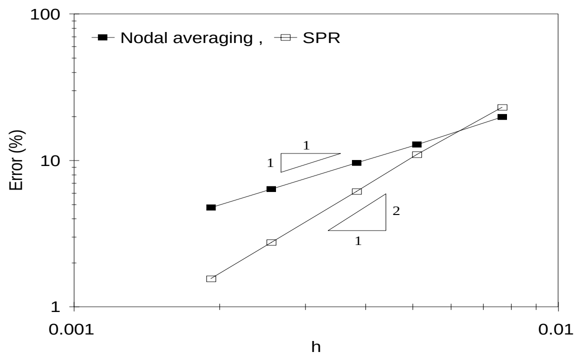

Fig. 4.2: Error in computed radial velocity for an isolated impeller with straight thin blades, as a function of element size *h* near the trailing edge. Elements have linear interpolation functions. Rotational speed is zero.

tional velocity to zero. In fig. 4.2, the error in the computed radial velocity is plotted for a point at the impeller blade. The SPR method is compared with nodal averaging (linear interpolation functions for both cases). It is clear that in this case the theoretically predicted rate of convergence is obtained.

4.1.5 Removing singularities by introducing blade thickness

Another way to remove, or at least reduce, the singularity at the blade's trailing edge is to consider blades with a certain thickness. Analogously to Zienkiewicz and Taylor (1989), the singularity is now reduced by rounding the trailing edge. The results are plotted in fig. 4.3. The differences in the computed circulation on consecutive grids are plotted for various element sizes, as the analytical solution is not available. The rate of convergence approaches its theoretical value when the trailing edge is rounded

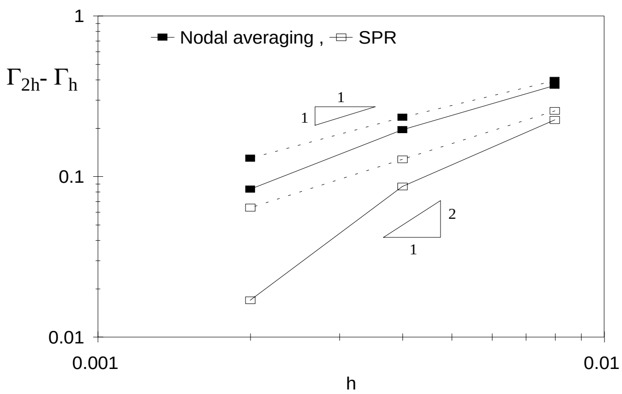

Fig. 4.3: Error in circulation value Γ for an isolated, rotating, radial impeller with blade thickness, as a function of element size *h* near the trailing edge. Elements have linear interpolation functions. Dashed lines refer to the case without blade thickness.

(straight lines in fig. 4.3). Note that convergence is not uniform as the geometry of the problem changes slightly when the mesh is refined.

4.1.6 Conclusions

The SPR method to compute the velocity has been implemented correctly. Results show that second order accuracy for the first derivative of the solution is obtained with linear elements, provided there is no singularity near the location of interest. The SPR method offers an increased accuracy for the velocity at a negligible extra computing time. Thus, the need for huge numbers of gridpoints or elements with higher order interpolation functions is avoided. Recently, Boroomand and Zienkiewicz (1997) proposed a new superconvergent recovery technique based on weighted equilibrium in patches. Its performance along boundaries is improved, although computing time is somewhat larger.

4.2 The steady three-dimensional wake

4.2.1 Theory

Analytic solutions for the lift of a finite wing in steady flow can only be given for a restricted number of simple cases. One of these is the untwisted wing with elliptic planform (fig. 4.4). For this geometry a small-disturbance solution can be derived using a simple model, first suggested by Prandtl (see Glauert, 1959). According to this method the circulation at any spanwise station is considered to be concentrated in a single vortex placed along the wing's quarter-chord line. It can be shown that a positioning like this automatically satisfies Kutta's condition of smooth flow at the wing's trailing edge. The strength of the vortices in the wake is related to the change of circulation of the vortex bound to the wing by Helmholtz's law. If we consider an untwisted finite span wing of large aspect ratio at small angle of attack α, then the

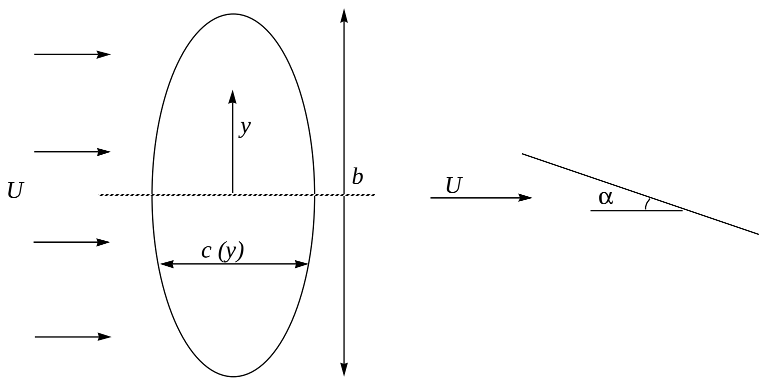

Fig. 4.4: Zero-thickness untwisted wing with elliptic planform, at angle of attack α. Left figure: top view. Right figure: side view.

condition of zero normal flow at the wing's surface can be written as

$$\frac{-\Gamma(y)}{\pi c(y) U} - \frac{1}{4\pi U}\int_{-b/2}^{b/2} \frac{\frac{d\Gamma(y_0)}{dy} dy_0}{y - y_0} + \alpha = 0 \quad , \tag{4.4}$$

where U is the free stream velocity, Γ(y) the local circulation and $c(y)$ the local chord length at station y along the wing with span b. Note that the integral must be evaluated according to Cauchy's principal value, since it is singular at y=y_0. The trailing vortices that together form the wake are regarded as semi-infinite lines located along the streamlines of mean flow. The starting vortex is assumed to be far enough downstream for the flow to be steady. In the first term the analytic expression for the two-dimensional flat plate circulation is recognized

$$\Gamma = \pi c U \alpha \ . \tag{4.5}$$

Equation (4.4) can be interpreted as

$$-\alpha_{eff} - \alpha_{wake} + \alpha = 0 \ , \tag{4.6}$$

where the effective angle of attack α_{eff} is smaller than the actual angle of attack α by α_{wake}, which is the result of downwash induced by the wake. Both α_{eff} and α_{wake} depend on y. If the chord distribution $c(y)$ has the elliptic form

$$c(y) = c_0\left[1 - \left(\frac{y}{b/2}\right)^2\right]^{1/2}, \tag{4.7}$$

with c_0 the chord at the centre of the wing, it can be shown that the circulation has an elliptic distribution as well

$$\Gamma(y) = \Gamma_0\left[1 - \left(\frac{y}{b/2}\right)^2\right]^{1/2}, \tag{4.8}$$

with maximum circulation value, taken at midspan,

$$\Gamma_0 = \frac{2bU\alpha}{1 + \frac{2b}{\pi c_0}} \ . \tag{4.9}$$

See e.g. Katz and Plotkin (1991) for a derivation.

4.2.2 Computations

Domain

A structured grid consisting of four blocks is placed around a flat wing with elliptic planform, tilted by an angle of five degrees. The spanwidth b is 2 m, its maximum

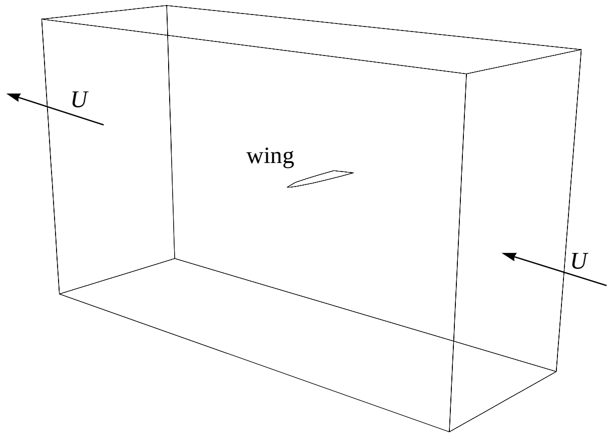

Fig. 4.5: Position of elliptic wing within the computational domain.

chordlength c_0 is 0.3 m. The wake length is ten times the maximum chordlength. Due to symmetry, only half of the wing is considered. Moderate grid refinement is used near the wing's surface, especially towards the leading edge, the trailing edge and the wing tip. The total number of gridpoints is 15,000. Fig. 4.5 shows the boundaries of the computational domain and the wing.

Boundary conditions

The wing is placed at an angle of five degrees to the horizontal plane. A uniform flow field corresponding to $U = 1.0\,\mathrm{m/s}$ is imposed at the inflow and outflow boundaries of the domain (fig. 4.5). Zero normal velocity is prescribed at the wing's upper and lower surfaces and at the remaining external surfaces of the domain.

4.2.3 Results

In fig. 4.6 the computational result for the spanwise circulation distribution is compared to the analytic solution for small disturbances, using eqs. (4.8) and (4.9). It

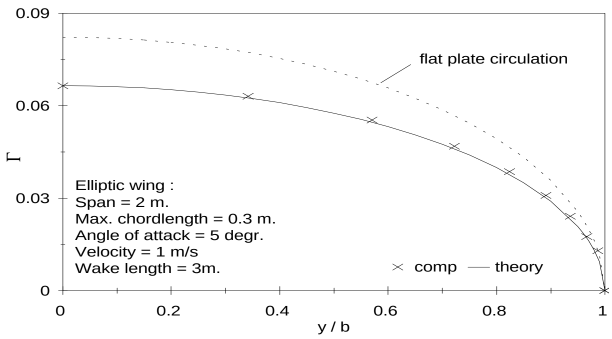

Fig. 4.6: Spanwise distribution of circulation Γ. Comparison of analytic small disturbance solution and computations. Dashed line gives flat plate circulation of eq. (4.5).

shows a very good agreement. The analytic circulation for the two-dimensional case of a flat plate (eq. 4.5) is given as well. The difference between the two lines shows the influence of the wake.

4.3 The unsteady two-dimensional wake

The next step in the verification process of a wake model would be to investigate its ability to represent truly unsteady wakes. Such wakes are present, for instance, behind airfoils in non-uniform motion. However, a general analytic solution for arbitrary airfoils in unsteady motion does not exist, not even in the small disturbance approximation. Only for very simple geometries undergoing specific unsteadiness have investigators reached some approximate solutions.

Perhaps the most famous of all is the flat plate in oscillatory motion, investigated by Glauert (1929) and, later, by Theodorsen (1935), Küssner (1936) and Von Kármán and Sears (1938). Much is owed to the work of Von Kármán and Sears who set up their theory using “only the basic conceptions of the vortex theory familiar to the modern engineer” (McCune and Tavares, 1993).

The next sections will concentrate on the case of a flat plate in oscillatory heaving motion. A short introduction to the corresponding theory will be given, mainly following the work of Von Kármán and Sears, although using a different notation. Subsequently, results of numerical computations are presented. It will be shown that the wake model based on the method of characteristics for solving the transport equation (section 3.5) is sufficiently accurate for a wide range of oscillation frequencies. Two methods to compute the aerodynamic forces are compared: direct pressure integration over the plate as opposed to using a control-volume method.

4.3.1 Theory

Consider a flat plate of length c moving at constant speed U in a stationary fluid with density ρ. The plate exhibits a periodic heaving motion (with frequency ω) of which the vertical displacement h can be denoted by

$$h = h_0 \sin \omega t, \tag{4.10}$$

where the amplitude h_0 is sufficiently small to justify the assumption of a wake distributed along a straight line. Wake roll-up is neglected. Following the work of Von Kármán and Sears we can write this as a complex quantity

$$h = -ih_0 e^{i\omega t}, \tag{4.11}$$

and the plate’s vertical velocity as

$$h' = h_0 \omega e^{i\omega t}. \tag{4.12}$$

The lift force acting on the plate is composed of three contributions,

$$L = L_0 + L_1 + L_2, \tag{4.13}$$

where

$$L_0 = -\pi\rho U\omega c h_0 e^{i\omega t}, \tag{4.14}$$

$$L_1 = -i\frac{\pi\rho\omega^2 c^2}{4} h_0 e^{i\omega t}, \tag{4.15}$$

and

$$L_2 = \pi\rho U\omega c h_0 e^{i\omega t}\frac{K_0(ik)}{K_0(ik) + K_1(ik)}. \tag{4.16}$$

In the above equations K_0 and K_1 are two modified Bessel functions of the second kind, and k is the so-called reduced frequency,

$$k = \frac{\omega c}{2U}. \tag{4.17}$$

Thus the total lift can be written as

$$L = -\pi\rho U\omega c h_0 e^{i\omega t}\left(C(k) + i\frac{k}{2}\right), \tag{4.18}$$

where $C(k)$ is called the (complex) Theodorsen lift deficiency function

$$C(k) = 1 - \frac{K_0(ik)}{K_0(ik) + K_1(ik)}. \tag{4.19}$$

The first contribution L_0 is often referred to as the quasi-steady lift. It is the lift a plate would encounter if the instantaneous velocity were permanently maintained. As seen from the body coordinate system the incoming fluid would then have an apparent angle of attack α where

$$\alpha = -\frac{h_0\omega}{U}e^{i\omega t}, \tag{4.20}$$

and the lift of a flat plate at an angle of attack α is

$$L = \pi\rho\alpha U^2 c. \tag{4.21}$$

Substitution of eq. (4.20) into eq. (4.21) gives eq. (4.14).

The second contribution L_1 represents the reaction of the accelerated fluid. It is the force which would act on the plate if no circulation, and hence no wake, were produced. It is called the contribution of the apparent mass and can be written as minus the product of the acceleration of the plate

$$h'' = ih_0\omega^2 e^{i\omega t}, \tag{4.22}$$

and the apparent mass of the system

$$m_a = \frac{1}{4}\pi\rho c^2, \tag{4.23}$$

which gives eq. (4.15).

The third contribution L_2 results from the influence of the vortices which are shed from the trailing edge of the plate into the wake. It can be regarded as a lift deficiency due to the non-uniform motion of the plate.

The moment which is applied to the plate by the fluid can easily be derived from the expressions for the lift. The quasi-steady lift contribution L_0, as well as the wake influence L_2, act at the so-called quarter-chord point, while the apparent mass contribution acts at the centre of the plate. The moment M_{le} about the plate's leading edge will then be

$$M_{le} = \frac{\pi\rho U\omega c^2}{4} h_0 e^{i\omega t}(C(k) + ik), \tag{4.24}$$

while for the moment M_c about the plate's centre the latter contribution vanishes

$$M_c = -\frac{\pi\rho U\omega c^2}{4} h_0 e^{i\omega t} C(k). \tag{4.25}$$

By expressing the lift and moment as complex quantities it is possible to construct vector diagrams. These are particularly useful for understanding the phase and magnitude of the individual contributions. Fig. 4.7 shows how the amplitude of the lift may be composed of its three contributions, leading the velocity of the plate by a phase angle γ.

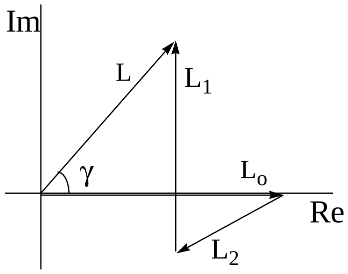

Fig. 4.7: Vector diagram for the lift L of an oscillating flat plate

According to Küssner (1936), the pressure difference below and above a plate in periodic heaving motion is given by

$$\Delta C_p(\theta) = 8\frac{h_0}{c}\left[ikC(k)\tan\left(\frac{\theta}{2}\right) - k^2\sin\theta\right], \tag{4.26}$$

where the pressure coefficient C_p is defined as

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U^2}, \tag{4.27}$$

with p the static pressure and p_∞ the static pressure at infinity. The spatial coordinate x along the plate's surface is related to θ by

$$x = \frac{1}{2}c\cos\theta. \tag{4.28}$$

Thus the plate's leading edge corresponds to $\theta = \pi$, and its trailing edge to $\theta = 0$. See also Försching (1974) for a review.

4.3.2 Computations

Computational domain

A flat horizontal plate is placed in the middle of an orthogonal grid (fig. 4.8). Although the geometry is two-dimensional, a third dimension is necessary in order to make it suitable for a three-dimensional method. This is achieved by taking only one element in spanwise direction. Grid refinement is applied near the plate, towards its leading and trailing edges and in the wake. The length of the wake is approximately five chordlengths. The computational domain is divided into two blocks; one above and one below the plate. The three-dimensional grid contains 4,400 nodes.

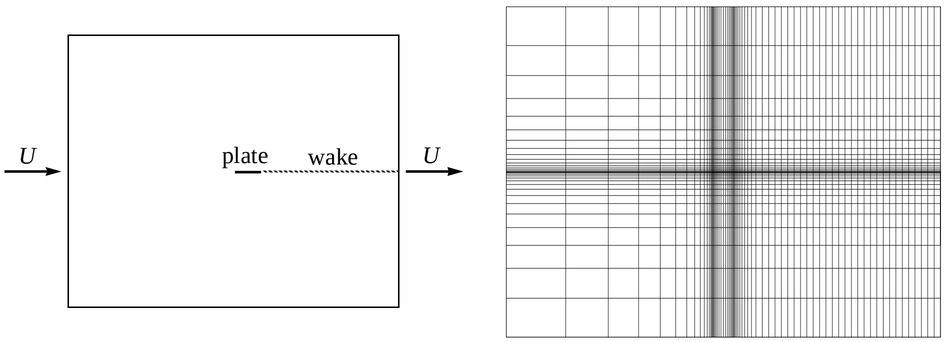

Fig. 4.8: Schematic view of the computational domain (left) showing the plate and the wake. Right figure shows a two-dimensional view of the grid.

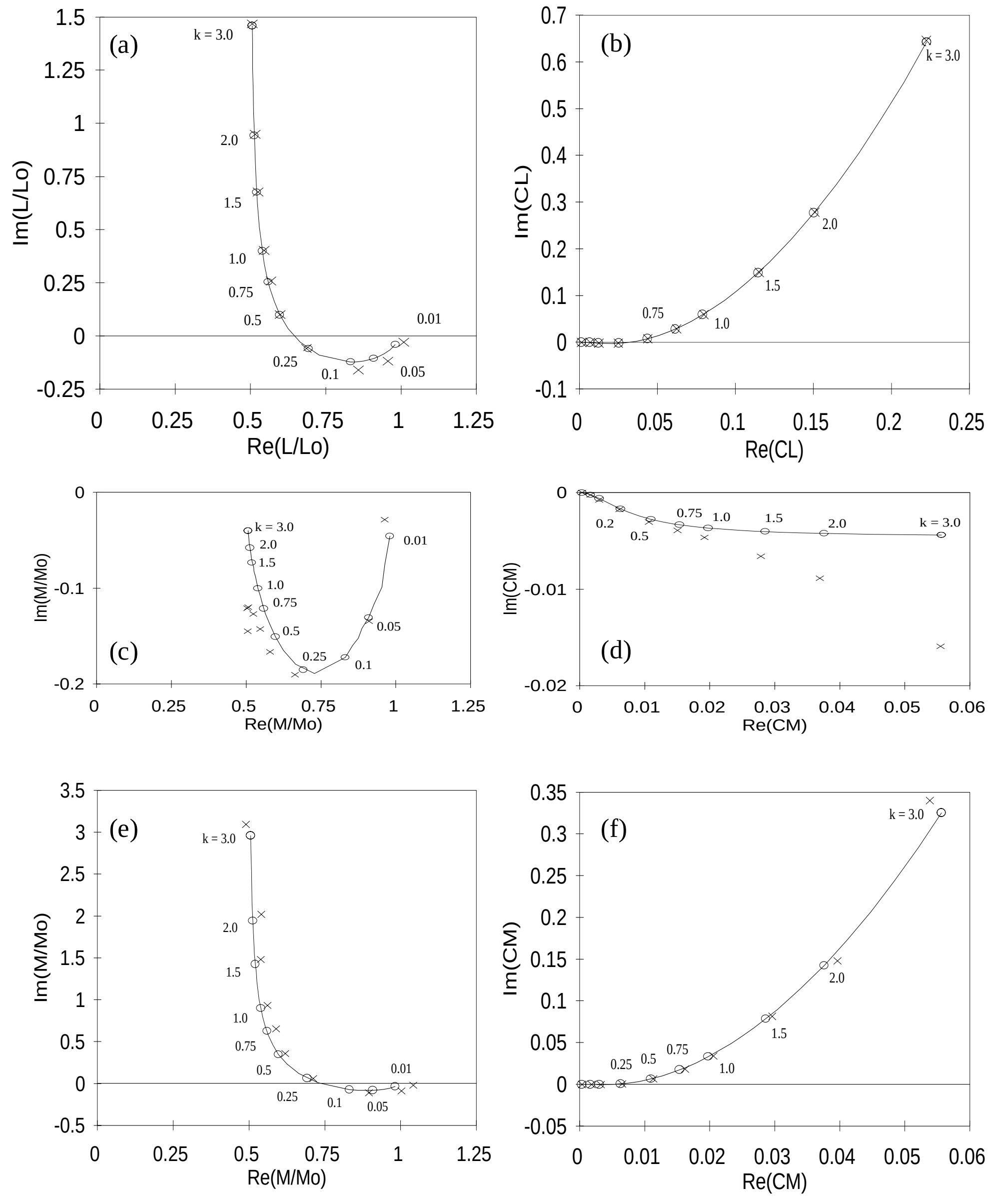

Fig. 4.9: Vector diagrams for the lift (a;b), the moment with respect to the centre of the plate (c;d), and the moment with respect to the plate's leading edge (e;f), for the control volume method. The figures on the left show quantities normalized with their quasi-steady part (L_0 and M_0). Figures on the right show lift and moment coefficients C_L and C_M. Reduced frequencies k (eq. 4.17) are plotted near the curves. (o : analytical, x : numerical).

Boundary conditions and time-stepping

The heaving motion of the plate is simulated by prescribing the appropriate normal velocity at the plate's upper and lower surfaces, rather than physically changing its position. This is allowed since we seek small disturbance solutions to the problem. For this reason the apparent displacement, reflected in the amplitude of the plate's velocity, is never more than one percent of its chordlength. A uniform flow field is imposed at the inflow and outflow boundary of the domain. Zero normal velocity is prescribed at the upper and lower surfaces. A number of reduced frequencies k are considered[1)], ranging from 0.01 to 3.0. Forty time steps are taken in each oscillation period.

4.3.3 Results

Control volume method

Fig. 4.9 shows the results of computations for the lift and the moment as well as the analytic solutions, for a wide range of reduced frequencies. The forces are calculated using a control volume method.

It can be seen from fig. 4.9 that the lift forces and the moment with respect to the plate's leading edge agree very well with the analytic solutions, especially for reduced frequencies higher than 0.25. Numerical values for the moment with respect to the centre of the plate, however, show less agreement with analytic solutions. It appears to be the phase of the moment which is miscalculated, not so much the magnitude. This is illustrated by fig. 4.10, where the amplitude and the phase are plotted separately as a function of reduced frequency. The reason might be the strong singularity at the plate's leading edge, as will be explained in the remainder of this section.

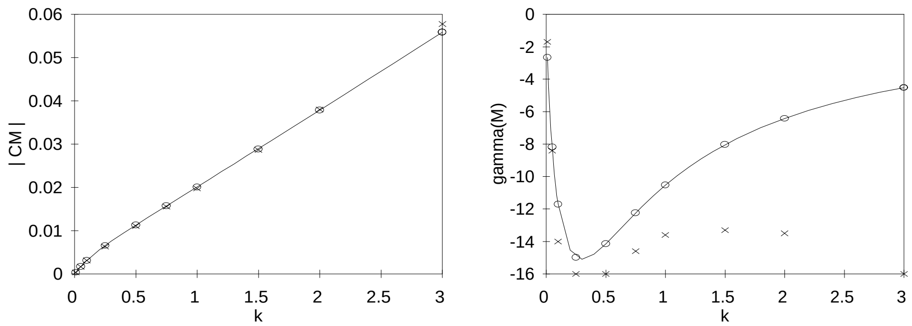

Fig. 4.10: Magnitude (left) and phase γ of the moment coefficient C_M, with respect to the centre of the plate, as a function of reduced frequency k. The control volume method was used to compute the moment. (θ : analytical, x : numerical).

1. For wakes behind impeller blades the reduced frequency is of order 1. This value is based on shaft revolution frequency Ω, blade trailing edge radius r and an estimated velocity along the wake of $\frac{1}{2}\Omega r$.

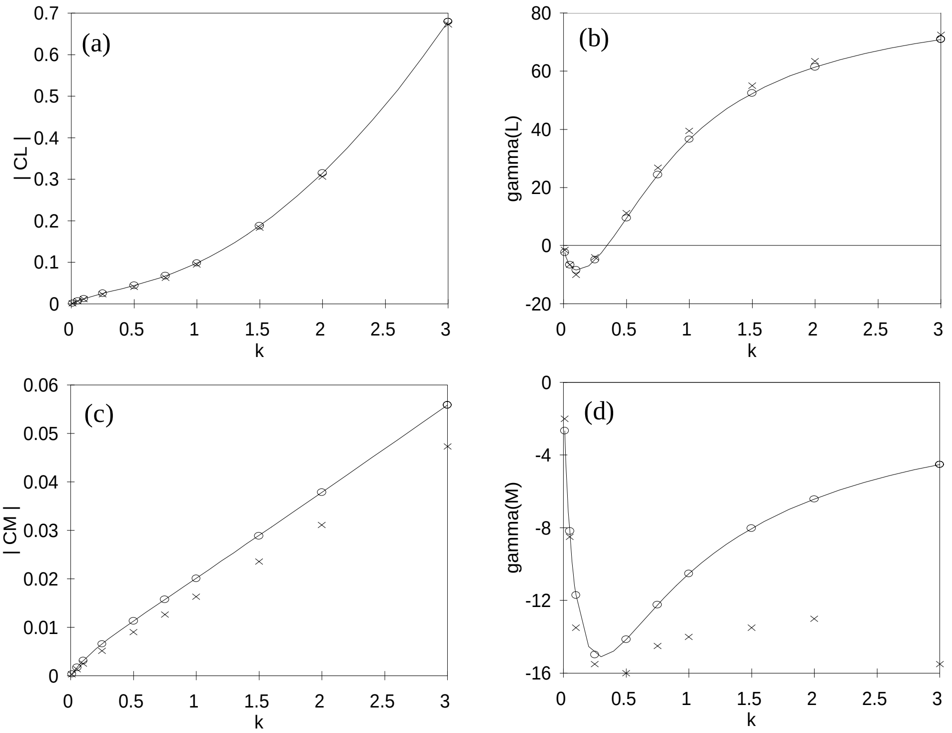

Fig. 4.11: Amplitude (left) and phase γ (right) for the lift (a;b) and moment M_C with respect to the centre of the plate (c;d), as a function of reduced frequency k. The method of direct pressure integration was used. (ө : analytical, x : numerical).

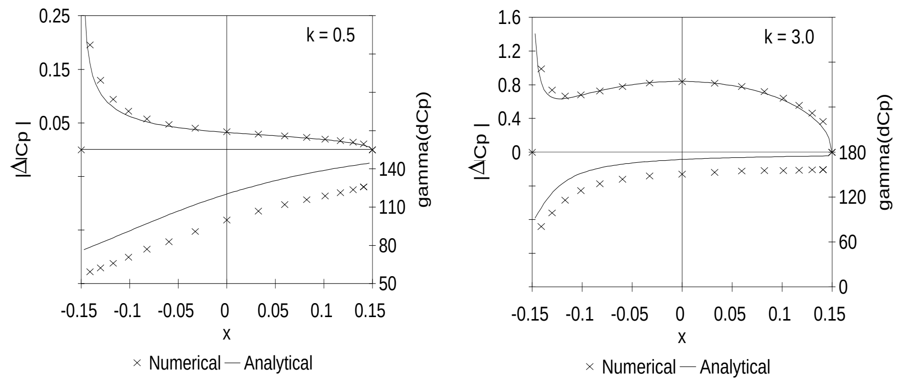

Fig. 4.12: Computed amplitude and phase γ of the pressure difference distribution over the plate compared to analytic values, for two reduced frequencies k.

Pressure integration

Lift and moment forces have also been computed by direct pressure integration over the plate's surface. Results show that, on the average, pressure integration is less accurate than using a control volume method. This is illustrated by fig. 4.11, where the amplitude and the phase of lift and moment coefficient are plotted as a function of reduced frequency. Both the amplitude and the phase deviate from analytic values. This is especially true for the moment coefficient. The reason is that pressure integration becomes inaccurate near the plate's leading edge, where it encounters a strong singularity in velocity and thus in pressure. The moment with respect to the plate's centre is much more sensitive to this inaccuracy, than the lift. This can be illustrated by comparing the computed pressure difference distribution over the plate to the analytic expression (eq. 4.26). Fig. 4.12 shows the results for two reduced frequencies. It can be concluded that deviations for the magnitude of the pressure difference are largest near the leading edge. The phase of the pressure difference deviates by a constant value over the entire plate.

4.3.4 Conclusions

The generation of unsteady vortices behind a flat plate in unsteady motion and the subsequent transportation along the wake is simulated satisfactorily by the implemented model. Control volume computations of the lift and moment with respect to the leading edge show a good agreement with analytically obtained values. The amplitude of the moment with respect to the centre of the plate is in excellent agreement as well. However, its phase shows a large deviation. The method of pressure integration along the plate surface is inferior to the control volume method, as a result of the leading edge singularity.

Nomenclature

C_L	Lift coefficient: $L/(\frac{1}{2}\rho U^2 c)$
C_M	Moment coefficient: $M/(\frac{1}{2}\rho U^2 c^2)$
C_p	Pressure coefficient
$C(k)$	Theodorsen lift deficiency function
E	Error
L	Lift force per unit span
M_c	Moment per unit span about plate's centre
M_{le}	Moment per unit span about plate's leading edge
U	Free stream velocity
b	Blade span
c	Chord length
h	Element size / Vertical displacement
k	Reduced frequency: $\omega c/(2U)$
p	Static pressure / degree of polynomial

t	Time
x	Chordwise coordinate
y	Spanwise coordinate

Greek symbols

Γ	Circulation
α	Angle of attack
γ	Phase shift
ω	Angular frequency
ρ	Density

Subscripts

0	Maximum value

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Chapter 5
Viscous losses

Inviscid methods will always overpredict a pump's head, while being unable to predict its efficiency. The obvious reason is that viscous losses are not taken into account. A very efficient way to incorporate the effects of viscosity is to use an inviscid method with additional models to estimate the viscous losses. A method like this is allowed since the flow through hydraulic pumps can be characterized by an inviscid main flow where the influence of viscosity is small and confined to local areas, like thin boundary layers and wakes (see chapter 2). Many models to quantify viscous losses exist, for which most of the input can be supplied by an inviscid analysis.

The power P_{sh} which is applied at the shaft of the pump is only partly converted into net power P_{net}, used to increase the total pressure (or head) of the fluid between inlet and outlet of the pump. A sometimes substantial part of the shaft power is used to overcome losses in the pump

$$P_{sh} = P_{net} + \Delta P, \tag{5.1}$$

where ΔP denotes the overall loss, for example by mechanical losses in bearings, by dissipation in boundary layers and mixing areas, by disc friction at the impeller external surfaces, and the loss which can be attributed to the leakage flow through seals. In this chapter, the influence of mechanical losses is neglected. The shaft power is related to the shaft torque M_{sh} by

$$P_{sh} = \Omega M_{sh}, \tag{5.2}$$

while the net power is defined as the total pressure difference between the inlet and outlet of the pump, times the flowrate Q. In terms of the pump's head H it is given by

$$P_{net} = \rho g Q H. \tag{5.3}$$

The pump's efficiency is simply defined as the ratio of net power and shaft power

$$\eta = \frac{P_{net}}{P_{sh}}. \tag{5.4}$$

In this chapter the additional models to quantify the viscous losses are presented. These include models for boundary layer dissipation, mixing processes, expansion and contraction, disc friction and leakage flow. A method to compute boundary layer quantities like displacement thickness, momentum thickness and skin friction will be discussed as well. But first the relation between shaft power, impeller torque and delivered head and efficiency is explained, as well as the way in which inviscid analyses are to be used in this context.

5.1 Shaft power

Consider a control volume enclosing the impeller as depicted in fig. 5.1. The control volume V is bounded by surface A, which consists of open boundaries A_1 and A_2, and solid boundaries A_3 and A_4. Conservation of angular momentum in integral form states that (e.g. Fox and McDonald, 1985)

$$M_{sh} + \int_A rF_{s,\theta}\mathrm{dA} = \frac{\partial}{\partial t}\int_V \rho r \mathrm{v}_\theta \mathrm{dV} + \int_A \rho r \mathrm{v}_\theta (\underline{\mathrm{v}}.\underline{n})\,\mathrm{dA}\,, \tag{5.5}$$

where $\underline{n}$ is the outward unit normal vector, $\underline{\mathrm{v}}$ the absolute velocity vector, r the radius, and F_s the surface force, which in this case of axisymmetric surfaces is restricted to the wall shear force τ_w in circumferential direction θ only. The time derivative in the above equation can be ignored since we are interested in the time-average of a periodic quantity. Secondly, the contribution of shear stress at boundaries A_1 and A_2 is negligible. If we furthermore neglect the influence of boundary layers on the velocity profiles at boundaries A_1 and A_2, eq. (5.5) can be written as

$$M_{sh} = M_{E,\,Q+Qleak} + M_{df}\,, \tag{5.6}$$

with

$$M_{E,\,Q+Qleak} = \rho \int_{A_1+A_2} r\,\mathrm{v}_\theta (\underline{\mathrm{v}}.\underline{n})\,\mathrm{dA} \tag{5.7}$$

$$M_{df} = -\int_{A_3+A_4} r\tau_w \mathrm{dA}\ . \tag{5.8}$$

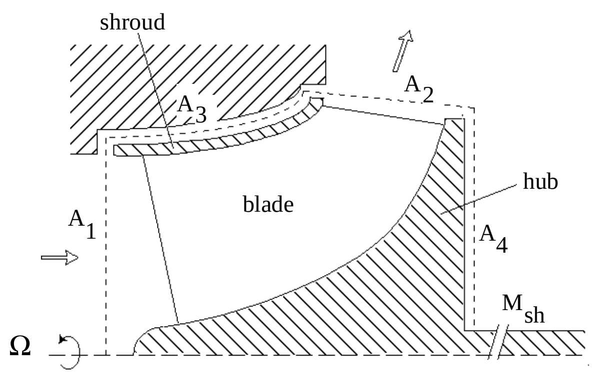

Fig. 5.1: Position of control volume enclosing the impeller.

The inviscid Euler moment $M_{E,Q+Qleak}$ can be considered as the moment which is exerted by the impeller on the internal fluid by means of pressure forces at the blade surfaces. Note that account has been given to the larger flowrate $Q+Q_{leak}$ through the impeller as a result of leakage flow. Quantifying $M_{E,Q+Qleak}$ is complicated as in general leakage flow, which leaves and re-enters the impeller, will carry angular momentum. The moment M_{df} is related to the loss in shaft power resulting from shear stress exerted by the fluid at the external surfaces of the impeller.

5.2 Delivered head and efficiency

A head value $H_{E,Q+Qleak}$ may be associated with the impeller torque $M_{E,Q+Qleak}$

$$H_{E,\,Q+Qleak} = \frac{\Omega M_{E,\,Q+Qleak}}{\rho g\,(Q+Q_{leak})} \,. \tag{5.9}$$

It is the head which would result if the impeller torque $M_{E,Q+Qleak}$ could be converted into dynamic pressure without loss. In practice, a lower head value will be found due to viscous losses in the internal flow. Thus

$$H = H_{E,\,Q+Qleak} - \Delta H_{hydr} \,, \tag{5.10}$$

where irreversible pressure losses due to boundary layer dissipation, mixing processes, expansion and contraction are all gathered in ΔH_{hydr}. Suitable models to quantify the various sources of hydraulic loss will be presented in the sequel of this chapter. It will prove to be of use in the remaining of this section to divide the head loss in a part generated in the impeller, $\Delta H_{hydr,\,\mathrm{i}}$, and one generated in the volute, $\Delta H_{hydr,\,\mathrm{v}}$:

$$\Delta H_{hydr} = \Delta H_{hydr,\,\mathrm{i}} + \Delta H_{hydr,\,\mathrm{v}} \,. \tag{5.11}$$

By combining eqs. (5.2)-(5.4), (5.6), and (5.9)-(5.11), the efficiency of the pump can be written as

$$\eta = \frac{\Omega M_{E,\,Q+Qleak} \cdot \dfrac{Q}{(Q+Q_{leak})} - \rho g Q\,(\Delta H_{hydr,\,\mathrm{i}} + \Delta H_{hydr,\,\mathrm{v}})}{\Omega\,(M_{E,\,Q+Qleak} + M_{df})} \,. \tag{5.12}$$

Alternatively, referring to eqs. (5.1), (5.2) and (5.4), this expression can be written as

$$\eta = \frac{\Omega M_{sh} - \Delta P}{\Omega M_{sh}} \,, \tag{5.13}$$

where it is seen from eqs. (5.6) and (5.12) that ΔP can be given by

$$\Delta P = \Delta P_{leak} + \Delta P_{hydr} + \Omega M_{df} \,, \tag{5.14}$$

with

$$\Delta P_{leak} = \rho g \, (H_{E,\,Q+Qleak} - \Delta H_{hydr,\,\mathrm{i}}) \, Q_{leak} \ , \tag{5.15}$$

and

$$\Delta P_{hydr} = \rho g \, (Q + Q_{leak}) \ \Delta H_{hydr,\,\mathrm{i}} + \rho g Q \ \Delta H_{hydr,\,\mathrm{v}} \ , \tag{5.16}$$

where the first term in eq. (5.16) denotes the hydraulic loss in the impeller, and the second term the loss in the volute of the pump.

5.3 Relation to inviscid theory

Determining $M_{E,Q+Qleak}$ from eq. (5.7) using the velocity field of an inviscid computation is not entirely correct since the computed velocity field deviates slightly from the actual one, due to the presence of boundary layers. The integration gives the inviscid moment of momentum (or Euler momentum), resulting from pressure forces, and neglects the contribution from shear forces at the impeller blades and hub and shroud surfaces. One way to obtain the shear stress contribution M_τ to the moment of momentum separately, would be to compute it as

$$M_\tau = -\int_{A_{int}} (\underline{r} \times \underline{\tau}_w)_a \, \mathrm{dA} \ , \tag{5.17}$$

where the axial direction is denoted by a and the wall shear stress is given by vector $\underline{\tau}_w$. Integration is performed over all internal surfaces A_{int} (impeller blades, hub and shroud). For the shear stress moment M_τ to be determined, the skin friction coefficient along the impeller blades and the hub and shroud surfaces should be known. A one-dimensional integral boundary layer analysis along several streamlines (as presented in section 5.7) would suit this purpose. However, it should be noted that the pressure distribution along the blades will have changed as well, due to the presence of boundary layers. Furthermore, boundary layers will influence the direction of the flow near blade trailing edges. A better alternative would therefore be to study the effect of boundary layers on the velocity distribution at the control volume boundary, and to use eq. (5.7) to compute $M_{E,Q+Qleak}$. In the current investigation the effect of shear layers on fluid moment is considered small enough to be disregarded as a first approximation.

Even if the flow in the suction pipe is free of angular momentum, the integral over surface A_1 in eq. (5.7) is in general non-zero. The reason is that the returning leakage flow will carry angular momentum. From conservation of momentum it follows that the contribution of surface A_1 to the integral of eq. (5.7) is equal to the moment of momentum M_{leak} of the leakage flow upon re-entrance, which is

$$M_{leak} = \rho R_s \mathrm{v}_\theta \, Q_{leak}, \tag{5.18}$$

with R_s the radius of the seal exit and v_θ the average circumferential velocity of the leakage flow. As in practice the slip for pre-rotation is very small (Visser et al., 1994) the integral over surface A_2 in eq. (5.7) may be determined without leakage flow pre-rotation (but with the larger flow rate Q+Q_{leak}).

5.4 Hydraulic losses

Irreversible pressure losses in the internal flow itself are called hydraulic losses. The most important sources of hydraulic loss are energy dissipation in boundary layers along walls, losses through mixing processes, and losses due to sudden expansions and contractions in through-flow area:

$$\Delta P_{hydr} = \Delta P_{diss} + \Delta P_{mix} + \Delta P_{exp} + \Delta P_{contr}, \tag{5.19}$$

where ΔP denotes power loss. Models to quantify these losses are presented in this section.

5.4.1 Boundary layer dissipation

Due to friction in attached boundary layers the flow is subject to an irreversible pressure loss and, as a consequence, to a power loss ΔP_{diss} . In two dimensions the change in power of the steady flow along a solid surface of a stationary passage (see fig. 5.2) can be written as

$$\frac{dP}{dx} = \frac{d}{dx}\int_0^\delta \left(p + \frac{1}{2}\rho \mathrm{v}^2\right)\mathrm{v}\, dy, \tag{5.20}$$

where v is the velocity tangential and with respect to the wall, and δ is the boundary layer thickness. Note that the integral is restricted to the boundary layer as any contribution vanishes at larger distance. Stating that just outside of the boundary layer the static pressure p_e is related to the velocity v_e by conservation of stagnation enthalpy

$$\frac{dp_e}{dx} = -\frac{d}{dx}\left(\frac{1}{2}\rho \mathrm{v}_e^2\right), \tag{5.21}$$

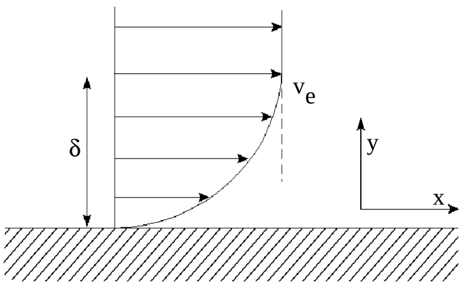

Fig. 5.2: Velocity profile in two-dimensional boundary layer

and assuming that the pressure is independent of y-coordinate within the boundary layer, eq. (5.20) transforms to

$$\frac{dP}{dx} = -\frac{d}{dx}\int_0^\delta \frac{1}{2}\rho\,(v_e^2 - v^2)\,v\,dy. \tag{5.22}$$

This expression can be written as

$$\frac{dP}{dx} = -\frac{1}{2}\rho\frac{d}{dx}\,(v_e^3\delta_3)\,, \tag{5.23}$$

for incompressible fluids, with δ_3 the energy thickness of the boundary layer defined by

$$\delta_3 = -\frac{1}{v_e^3}\int_0^\delta (v_e^2 - v^2)\,v\,dy. \tag{5.24}$$

Analogous to Schlichting (1979, p. 676) we can write

$$\frac{d}{dx}\,(v_e^3\delta_3) = v_e^3 c_D\,, \tag{5.25}$$

with c_D the energy dissipation coefficient. Combining eqs. (5.23) and (5.25) gives:

$$\frac{dP}{dx} = -\frac{1}{2}\rho v_e^3 c_D\,. \tag{5.26}$$

See also Cumpsty (1989) for a derivation. This expression may be extended to three dimensions and integrated over the wall surface S to obtain the total power loss due to boundary layer dissipation

$$\Delta P_{diss} = \frac{1}{2}\rho\int_S c_D v_e^3 dS. \tag{5.27}$$

An analogous expression can be derived for steady flow with respect to a rotating passage. From the unsteady Bernoulli equation it can be shown that the rothalpy

$$I = \frac{p}{\rho} + \frac{1}{2}\,(w^2 - \Omega^2 r^2) \tag{5.28}$$

is a conserved quantity for inviscid, adiabatic flow. See the work of Lyman (1993) for a discussion. In this expression w is the velocity relative to the rotating surface. The loss which occurs in the rotating passage is not related to the change in enthalpy but, instead, to the change in rothalpy

$$\frac{dP}{dx} = \frac{d}{dx}\int_0^\delta \left[p + \frac{1}{2}\rho\,(w^2 - \Omega^2 r^2) \right] w\,dy. \tag{5.29}$$

Outside of the boundary layer the static pressure is related to the velocity by conservation of rothalpy

$$\frac{dp_e}{dx} = \frac{d}{dx}\left(-\frac{1}{2}\rho w_e^2 + \frac{1}{2}\rho\Omega^2 r^2 \right). \tag{5.30}$$

Assuming the pressure to be independent of y-coordinate, eq. (5.30) can be substituted into eq. (5.29) to obtain

$$\frac{dP}{dx} = -\frac{d}{dx}\int_0^\delta \frac{1}{2}\rho\,(w_e^2 - w^2)\,w\,dy. \tag{5.31}$$

A derivation similar to the one for stationary passages leads to the following expression for the total power loss over a surface S of a rotating passage

$$\Delta P_{diss} = \frac{1}{2}\rho\int_S c_D\, w_e^3\, dS. \tag{5.32}$$

According to Denton (1993) the dissipation coefficient for turbulent boundary layers is relatively insensitive to the detailed state of the boundary layer (accelerating or decelerating). Denton suggests an average value of 0.0038 for turbulent boundary layers with Re_θ of order 1000, but it should be stressed that actual values vary between 0.002 for accelerating flows and 0.005 for flows encountering an adverse pressure gradient. The loss in a laminar boundary layer is also not very sensitive to its state but it does very much depend on its thickness (Truckenbrodt, 1952). As a first guess boundary layers, both in the impeller and in the volute, are assumed to be turbulent. Dissipation coefficients are taken constant for different flow rates.

In computing the loss in the impeller, the higher flow rate due to leakage should be taken into account.

5.4.2 Mixing losses

Two types of mixing losses are important in pumps:

- Mixing of the returning leakage flow with the undisturbed entrance flow.
- Mixing of shed vortices from the trailing edges of impeller blades with the ambient flow.

For both types of mixing, simple models can be derived.

Mixing of leakage flow

From conservation of axial momentum it follows that

$$(p_2 + \rho v_{a,2}^2)\, A_2 = (p_3 + \rho v_{a,3}^2)\, A_3, \tag{5.33}$$

with p the static pressure, v_a the axial velocity component and A the through-flow area. Subscripts 2 and 3 denote the stations before and after the mixing area (fig. 5.3). Assuming that the main flow is free of rotation, the eventual circumferential velocity component of the fluid v_θ at station 3 can be derived from

$$v_{\theta,\,\mathrm{leak}} Q_{\mathrm{leak}} = v_{\theta,\,3}\,(Q + Q_{\mathrm{leak}})\,. \tag{5.34}$$

With equations (5.33) and (5.34), and assuming that A_2 equals A_3, the loss in total pressure can be computed as

$$\Delta p_0 = \frac{1}{2}\rho v_{a,3}^2 \frac{(2\varepsilon + \varepsilon^2)}{(1+\varepsilon)^2} - \frac{1}{2}\rho v_{\theta,\,leak}^2 \left(\frac{\varepsilon}{1+\varepsilon}\right)^2 \tag{5.35}$$

with

$$\varepsilon = \frac{Q_{leak}}{Q}\,. \tag{5.36}$$

Fig 5.4 shows that the first term of eq. (5.35) is in agreement with values obtained by experiments, for pipe flow T-junctions with negligible branch area A_1 (Miller, 1978).

The power loss through mixing, which is defined by

$$\Delta P_{mix} = (Q + Q_{\mathrm{leak}})\,\Delta p_0, \tag{5.37}$$

can thus be expressed as

$$\Delta P_{mix} = \frac{1}{2}\rho v_{a,3}^2 Q_{leak} \frac{2+\varepsilon}{1+\varepsilon} - \frac{1}{2}\rho v_{\theta,\,leak}^2 Q_{leak} \frac{\varepsilon}{1+\varepsilon}\,. \tag{5.38}$$

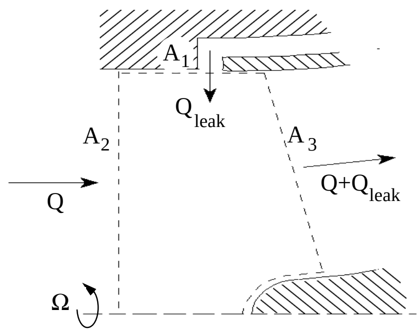

Fig. 5.3: Position of control volume at impeller suction side.

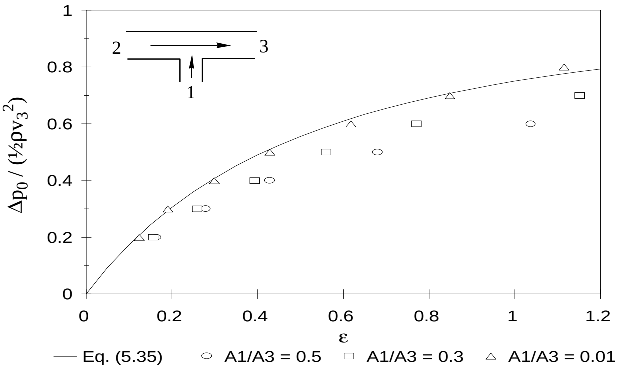

Fig. 5.4: Loss in total pressure due to mixing (based on dynamic pressure of combined flow) as a function of leakage flow rate ε, for different ratios of cross-sectional areas A_1 and A_3 (Miller, 1978).

Mixing of trailing vortices

An important type of mixing is the mixing of trailing vortices in the wake behind a blade. According to Denton (1993) the mixing loss may be given by

$$\Delta P_{mix} = (Q + Q_{leak}) \frac{1}{2}\rho v_{\text{ref}}^2 \left[-\frac{C_{pb} b}{W} + \frac{2\theta}{W} + \left(\frac{\delta^* + b}{W}\right)^2 \right], \tag{5.39}$$

with b the blade thickness and W the blade pitch, b/W representing the blockage (fig. 5.5). The sum of the momentum thickness at pressure and suction side, at a station near the trailing edge, is given by θ. The same applies to δ^*, the sum of the displacement thicknesses. The base pressure coefficient C_{pb} is defined by

$$C_{pb} = \frac{p_b - p_{ref}}{\frac{1}{2}\rho v_{\text{ref}}^2}, \tag{5.40}$$

with p_b the static pressure just after the blunt trailing edge, and p_{ref} and v_{ref} the pressure and velocity at a reference position on the blade, just before the trailing

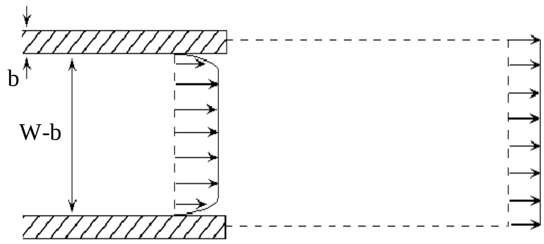

Fig. 5.5: Position of control volume at impeller discharge

edge. In general, the value of C_{pb} will be negative, as the pressure directly behind the trailing edge is usually smaller than the ambient pressure. Typical values are close to -0.15. The first term between brackets in eq. (5.39) defines the loss resulting from the low pressure acting on the trailing edge, the second term represents the mixing loss out of boundary layers and the third term represents the loss due to the combined blockage effect of the blade and the boundary layers. In three dimensions the fractions θ/W and δ^*/W may be replaced by the actual fractions in through-flow area, thus accounting for the boundary layers at the hub and shroud surfaces.

5.4.3 Expansion and contraction losses

Total pressure loss in flows may result from sources other than wall friction and mixing. One of these mechanisms is flow through abrupt changes in cross section, like sudden expansions or contractions. In the process kinetic energy is dissipated (e.g. Streeter, 1961).

Expansion

For an abrupt expansion a momentum balance analysis leads to an expression for the dissipated kinetic energy

$$\Delta P_{exp} = \frac{1}{2}\rho Q\,(v_1 - v_2)^2, \tag{5.41}$$

with v_1 and v_2 the through-flow velocities before and after the expansion respectively. This is known as the “Borda-Carnot loss”. Velocities v_1 and v_2 can be computed from the continuity of mass and are related by

$$v_2 = v_1\gamma_e, \tag{5.42}$$

with γ_e the ratio of cross sectional areas A_1/A_2. Eq. (5.41) is in agreement with experiments performed by Gibson (1952) for abrupt expansions. For losses at gradual expansions see e.g. Gibson (1952) for conical diffusers, and Reneau et al. (1967) for two-dimensional, straight-wall diffusers.

Contraction

To quantify the energy loss at a contraction is much more difficult. A sudden contraction leads to the formation of so-called “vena contracta” just behind the contraction. Most of the kinetic energy loss occurs at the expansion region just after these “vena contracta”. The power loss can be written as

$$\Delta P_{contr} = K_c(\gamma_c)\,\frac{1}{2}\rho Q v_2^2, \tag{5.43}$$

where v_2 is the velocity in the contracted region and K_c is the dissipation coefficient for sudden contractions, which is a function of γ_c, the ratio of cross sectional areas A_2/A_1 (fig. 5.6). K_c varies between the value 0.5 for $\gamma_c = 0$ (contraction from infinite to some finite cross section) and zero for $\gamma_c = 1$ (no contraction).

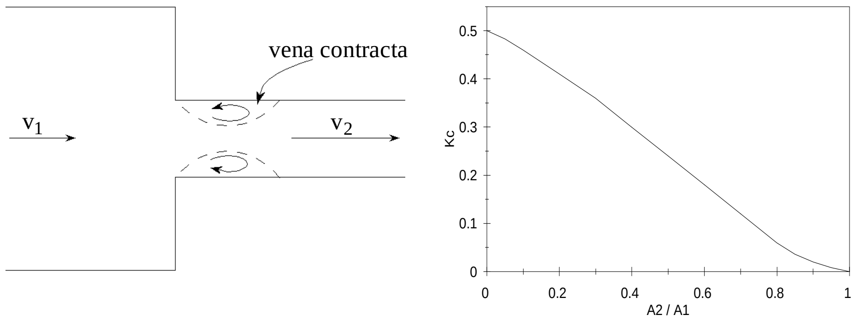

Fig. 5.6: Schematic drawing of sudden contraction showing vena contracta (left) and loss coefficient K_c for abrupt contraction as a function of ratio of cross sectional areas A_2/A_1 (Streeter, 1961)

5.5 Leakage flow

In most hydraulic machines seals are used to minimize leakage flow. Without seals between the shaft and the pump housing for instance, the fluid would squirt out of the pump because of its high pressure compared with the ambient pressure. In pumps with closed impellers another type of leakage flow occurs. The fluid which is brought under high pressure at the impeller outlet tends to flow back to the inlet pipe through the gap between the shroud of the impeller and the housing of the pump (fig. 5.8). In unshrouded impellers a different type of flow exists between the tip of the impeller blades and the housing of the pump (tip-clearance flow) Although clearly a volumetric loss, this is not normally regarded as a leakage flow.

Much work has been done to investigate the flow through seals and to predict the rate of leakage, both experimentally (Yamada 1962, Trutnovsky 1964, Wagner 1972) and numerically (Hirs 1970/1973, Childs 1989/1993, Baskharone 1989). A complicating aspect in the analysis is formed by the region between the impeller outlet and the actual seal. Especially for radial impellers, the height of this cavity can be quite large, resulting in a complicated flow pattern (secondary flow). In order to be able to predict the leakage flow rate, it is nevertheless important that the often considerable pressure drop which occurs in cavities is known.

In this section we focus on leakage flows in pumps with shrouded impellers. The height of the leakage area is assumed to be sufficiently small for the flow to be modelled as a thin lubricant film. A bulk-flow model is presented for a leakage flow area with a conical shape and a constant height. Many of the leakage areas can be modelled as a series of such regions. A method to compute the bulk-flow field through arbitrary cavity-seal combinations has been implemented which accounts for the loss (or gain!) in static pressure due to abrupt expansions and contractions. Referring to a study on tip leakage loss at shrouded turbine blades, Denton (1993) concludes that the swirl component of the leakage flow may be taken constant on passing through abrupt expansions and contractions.

It should be noted that what is presented in this section is more than a method to compute the volumetric loss only. As a by-product one obtains the leakage flow circumferential velocity component upon re-entrance as well as the values of the shear stress at the impeller external surface. The former is needed to compute the rate of pre-rotation of the incoming flow (section 5.3) and the mixing loss (subsection 5.4.2), while the latter will be used to compute part of the disc friction losses (section 5.6).

As a bulk-flow model like this is very approximate, its results should be used with care in cases were the leakage flow rate is large compared to the main flow rate through the impeller. A more sophisticated model should be employed for these cases.

A last remark concerns another, still unresolved, influence of the leakage flow on the main flow upon re-entrance. The leakage flow will almost certainly alter the development of the boundary layer along the shroud of the impeller; either stabilizing it or, possibly, leading to separation.

5.5.1 Bulk-flow model for conical leakage area

To compute the bulk-flow through a narrow conical gap of constant height h, the momentum equations for a conical coordinate system can be derived as:

$$\rho\left(\mathrm{v}_s\frac{\partial \mathrm{v}_s}{\partial s}+\sin\alpha\frac{\mathrm{v}_\varphi^2}{r}\right) = -\frac{\partial p}{\partial s}-\frac{1}{r}\frac{\partial(r\tau_{st})}{\partial t} \tag{5.44}$$

$$\rho\left(\mathrm{v}_s\frac{\partial \mathrm{v}_\varphi}{\partial s}-\sin\alpha\frac{\mathrm{v}_s\mathrm{v}_\varphi}{r}\right) = -\frac{1}{r^2}\frac{\partial(r^2\tau_{\varphi t})}{\partial t} \tag{5.45}$$

for through flow direction s and circumferential direction ϕ respectively. Refer to fig. 5.7 for directions s, t and ϕ. The flow is assumed to be steady. Derivatives in circumferential direction ϕ as well as velocity components in perpendicular direction t are neglected.

These expressions deviate somewhat from the expressions given by Childs (1989). His expressions do not converge to the circumferential and axial momentum equa-

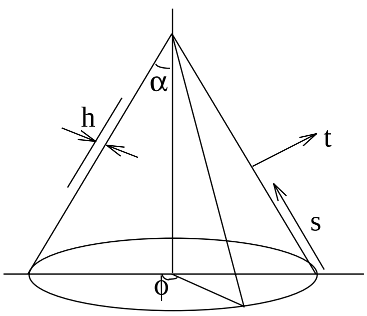

Fig. 5.7: Conical coordinate system.

tions in case α reduces to zero (see e.g. Bird et al., 1965). For a thin film the spatial derivatives of the shear stress components can be expressed as

$$\frac{\partial \tau_{st}}{\partial t} = \frac{\tau_{st}^S + \tau_{st}^R}{h} \tag{5.46}$$

and

$$\frac{\partial \tau_{\varphi t}}{\partial t} = \frac{\tau_{\varphi t}^S + \tau_{\varphi t}^R}{h}, \tag{5.47}$$

where superscripts S and R denote the steady and rotating wall respectively.

The flow in the conical gap is of a combination of two basic types; “pressure flow”, driven by a pressure gradient, and “drag flow”, due to the sliding motion of one of the surfaces. Hirs (1970, 1973) shows that the correlation between wall shear stress and bulk-flow velocity weakly depends on the nature of the flow in a given fluid film. He gives expressions for the shear stress for both types of flow, as well as a method to obtain the shear stress values for combinations of pressure flow and drag flow, both in parallel and in mutually perpendicular directions.

If the difference in shear stress between pressure and drag flow is neglected, the shear stress components can be written as

$$\begin{aligned}
\tau_{\varphi t}^S &= C_f^S \frac{1}{2}\rho \mathrm{v}_\varphi \mathrm{v}^S \\
\tau_{\varphi t}^R &= C_f^R \frac{1}{2}\rho\,(\mathrm{v}_\varphi - \Omega r)\,\mathrm{v}^R \\
\tau_{st}^S &= C_f^S \frac{1}{2}\rho \mathrm{v}_s \mathrm{v}^S \\
\tau_{st}^R &= C_f^R \frac{1}{2}\rho \mathrm{v}_s \mathrm{v}^R
\end{aligned} \tag{5.48}$$

with v^{S} and v^{R} the bulk-flow velocities relative to the stator and rotor wall respectively:

$$\begin{aligned}
\mathrm{v}^S &= \sqrt{\mathrm{v}_s^2 + \mathrm{v}_\varphi^2} \\
\mathrm{v}^R &= \sqrt{\mathrm{v}_s^2 + (\mathrm{v}_\varphi - \Omega r)^2}
\end{aligned} \tag{5.49}$$

The factors C_f^{S} and C_f^{R} denote the stator and rotor skin friction coefficients.

Both Childs (1989) and Nelson and Nguyen (1987) consider the case of a combination of pressure flow and drag flow in mutually perpendicular directions, but they use

Hirs' expression for the shear stress of a pressure gradient driven flow. The model of Moody is based on pipe-friction measurements, where account has been given to the influence of surface roughness. The following empirical expression for the skin friction coefficient was derived (see Nelson and Nguyen, 1987)

$$C_f = 0.001375\left[1 + \left(10^4 \frac{e}{h} + \frac{10^6}{2\,Re}\right)^{1/3}\right], \tag{5.50}$$

where the pipe diameter has been substituted by $2h$ and e is the absolute roughness of the stator or rotor surface. The Reynolds number Re is defined by

$$Re = \frac{Uh}{\nu}, \tag{5.51}$$

where U represents the velocity of the bulk-flow relative to the wall. Thus velocities v^S and v^R should be substituted to obtain factors C_f^S and C_f^R respectively.

Eqns. (5.44) and (5.45) can be solved provided the leakage flow rate and the initial value of the circumferential velocity are known. The circumferential velocity component of the flow entering the cavity can be computed by an inviscid analysis. Starting from an initial estimate for the flow rate Q, the momentum equations should be solved iteratively until convergence is obtained.

5.5.2 Examples

The flow through a simplified cavity-seal region of the CalTech centrifugal pump (Adkins and Brennen, 1988, see also section 6.6) is computed using Moody's skin friction model as described in the previous subsection. The leakage flow area is given in fig. 5.8. Two seal clearances are considered; h = 0.14 mm and h = 0.78 mm. Unless stated otherwise the circumferential velocity of the fluid at cavity entrance is equal to half the circumferential velocity of the rotor tip, the static pressure difference between cavity inlet and seal outlet is 30 kPa, the rotor rotational speed is 100 rad/s,

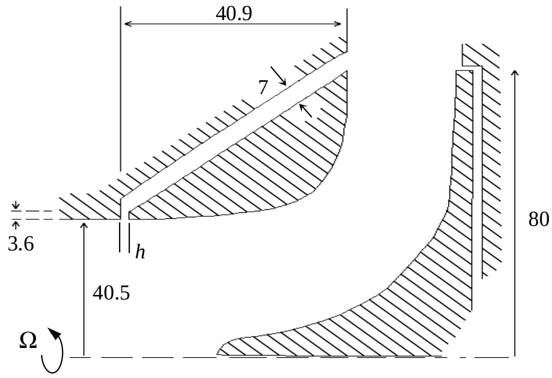

Fig. 5.8: Simplified cavity-seal region of the CalTech centrifugal pump (length in mm).

and the rotor and stator walls are smooth

In fig. 5.9 the distribution of the circumferential velocity from cavity inlet to seal outlet is given for both seal clearances and for different rotor and stator surface roughness. For convenience curves indicating (half) the rotor wall velocity are depicted in the figures as well. The results clearly show that the circumferential velocity of the flow is subject to two different driving mechanisms. The first is the inviscid effect of larger circumferential velocity at smaller radius, like in a free vortex. The second effect, the wall shear stress, tends to force the circumferential velocity towards half the velocity of the rotating wall. Thus, the longer the flow remains within the leakage flow area (e.g for small seal clearance and thus small leakage flow rates), the larger the effect of shear forces. See for example the work of Schubert (1988), who found this phenomenon from experiments. The importance of the wall shear stress also increases with surface roughness. Note that a circumferential velocity of the bulk-flow exceeding the value Ωr means that the fluid is actually driving the impeller by shear forces. It has been observed in practice that disc friction may reduce to zero, or even become negative given large enough flow rate and initial circumferential velocity. A second consequence of a larger fluid circumferential velocity is a lower leakage flow rate due to a larger centrifugal pressure build-up. However, the leakage flow will have a higher angular momentum upon re-entering the impeller leading to a reduced head.

In fig. 5.10 the distribution of the static pressure in the seal-cavity is given for three pressure differences between cavity inlet and seal outlet. It shows that for larger pressure differences the relative pressure drop at the contraction between cavity and seal increases. The reason is that a larger leakage flow rate leads to a higher static pressure drop at contractions. Fig. 5.11 shows the influence of inlet circumferential velocity on the static pressure and circumferential velocity distribution of the leakage flow. An initial circumferential velocity of 6 m/s reduces the leakage flow rate by more than 30% when compared to an initial value of 4 m/s. This can also clearly be seen from the much lower static pressure drop at the contraction. Disc friction loss is even reduced by a factor 6.

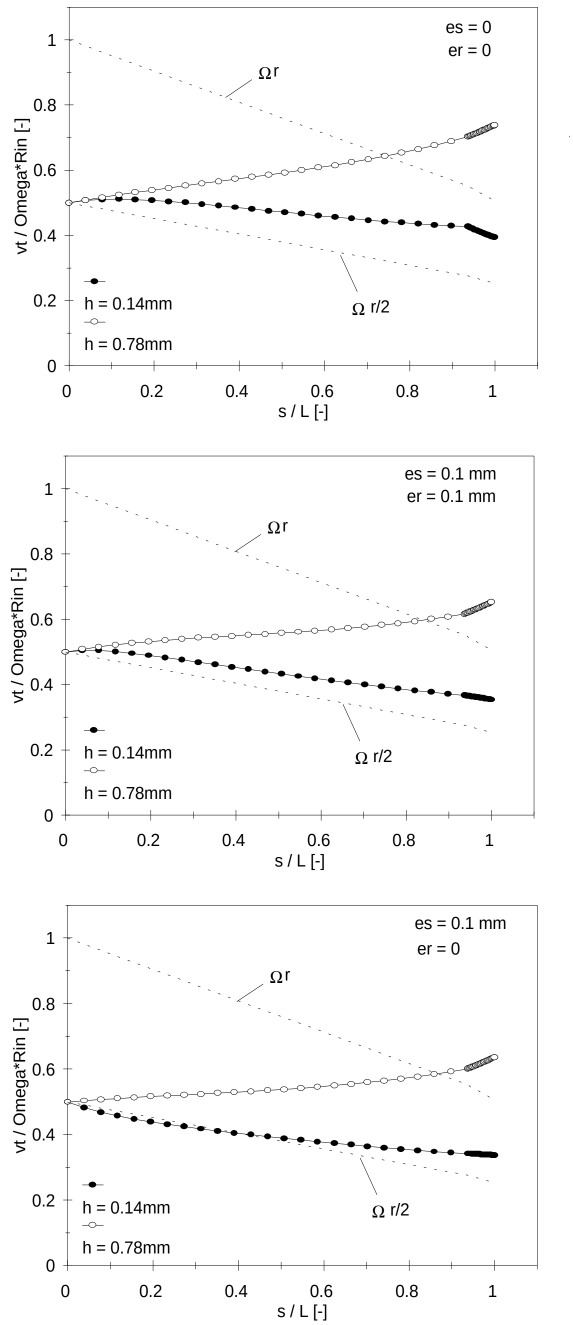

Fig. 5.9: Circumferential velocity distributions in the leakage flow area of the CalTech centrifugal pump, for two seal clearances *h* and different rotor and stator surface roughness (*er* and *es* respectively). *Rin* is the radius of the cavity entrance, and *s* the coordinate in through-flow direction. Static pressure difference is 30 kPa.

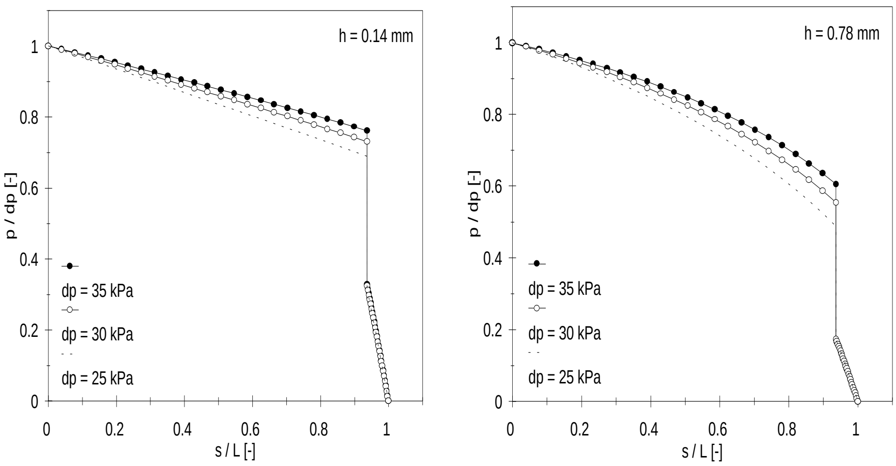

Fig. 5.10: Static pressure distributions in the leakage flow area of the CalTech centrifugal pump, for two seal clearances *h*. The static pressure difference between inlet and outlet is denoted by *dp*.

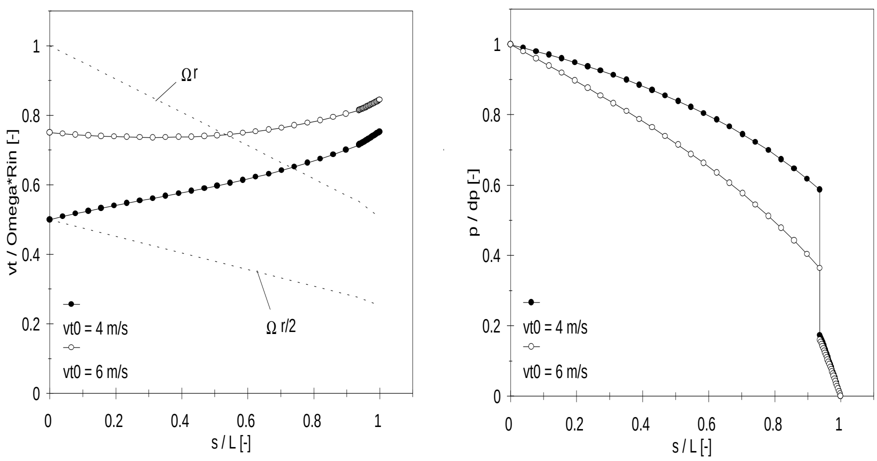

Fig. 5.11: Circumferential velocity and static pressure distributions in the leakage flow area of the CalTech centrifugal pump, for two initial circumferential velocities. Seal clearance *h*=0.78 mm. Static pressure difference is 30 kPa.

5.6 Disc friction losses

The torque which is experienced by a rotating radial disc by means of disc friction is given by

$$M_{df} = \int_A \tau\, r dA \ , \tag{5.52}$$

with τ the wall shear stress in circumferential direction and A the wetted surface. If the shear stress values are not given by a method like the one presented in section 5.5 and through-flow is negligible (e.g. at the external hub surface), empirical values obtained by Daily and Nece (1960) may be used instead.

The wall shear stress can be expressed as

$$\tau = \frac{1}{2}\rho \mathrm{v}^2 C_f \,, \tag{5.53}$$

with v the relative fluid velocity outside of the boundary layer and C_f a skin friction coefficient. Inserting eq. (5.53) into eq. (5.52) and integrating over a disc of radius R, assuming the skin friction coefficient to be constant, leads to

$$M_{df} = \frac{1}{5}\pi\rho C_f \Omega^2 R^5 , \tag{5.54}$$

for a disc wetted at one side. The local fluid velocity v is taken equal to Ωr. Daily and Nece based their measurements on a similar expression with disc friction coefficient C_m

$$M_{df} = \frac{1}{2}\rho C_m \Omega^2 R^5 , \tag{5.55}$$

where

$$C_m = \frac{2\pi}{5} C_f \,. \tag{5.56}$$

An extensive study on friction of enclosed rotating (smooth) discs resulted in expressions for C_m which actually depend on disc radius R, rotational speed Ω and clearance gap h. This contradicts the assumption of a constant skin friction coefficient. They distinguished between four possible flow regimes (see also Owen and Rogers, 1989):

- Laminar flow through small clearances (regime 1);
- Laminar flow through large clearances (regime 2);
- Turbulent flow through small clearances (regime 3);
- Turbulent flow through large clearances (regime 4).

For small clearance gaps the boundary layers at opposite solid surfaces are merged, while for large clearances two distinct boundary layers exist. The experimental results for the disc friction coefficient C_m can be summarized as follows:

- Regime 1: $C_m = \dfrac{\pi}{G\,Re}$ $\quad \begin{cases} G < 1.62\,Re^{-5/11} \\ G < 188\,Re^{-9/10} \end{cases}$
- Regime 2: $C_m = \dfrac{1.85\,G^{1/10}}{Re^{1/2}}$ $\quad \begin{cases} G > 1.62\,Re^{-5/11} \\ G > 0.57 \cdot 10^{-6}\,Re^{15/16} \\ Re < 1.58 \cdot 10^{5} \end{cases}$ (5.57)
- Regime 3: $C_m = \dfrac{0.04}{G^{1/6}\,Re^{1/4}}$ $\quad \begin{cases} G < 0.57 \cdot 10^{-6}\,Re^{15/16} \\ G < 0.402\,Re^{-3/16} \\ G > 188\,Re^{-9/10} \end{cases}$
- Regime 4: $C_m = \dfrac{0.051\,G^{1/10}}{Re^{1/5}}$ $\quad \begin{cases} G > 0.402\,Re^{-3/16} \\ Re > 1.58 \cdot 10^{5} \end{cases}$

where G is called the gap ratio, defined by

$$G = \frac{h}{R}, \tag{5.58}$$

and Re is the Reynolds number

$$Re = \frac{\Omega R^2}{\nu}. \tag{5.59}$$

In general, the shape of pump external surfaces will deviate from the well defined discs for which the above expressions were derived. In case deviations are small, one may assume that the disc friction coefficient does not change considerably, in which case the frictional torque can be obtained by

$$M_{df} = \frac{5}{4\pi}\rho C_m \int_A \Omega^2 r^3\, dA, \tag{5.60}$$

where the integration is performed over the actual wetted surface, and C_m is computed from eq. (5.57).

The power lost by disc friction is given by

$$P_{df} = \Omega M_{df} . \tag{5.61}$$

5.7 Boundary layer analysis

From the previous sections it will be clear that in order to quantify viscous losses some knowledge about the boundary layers is required. Boundary layer quantities like displacement thickness, momentum thickness, skin friction coefficient, or merely the laminar or turbulent state of a boundary layer are important factors. For this purpose a one-dimensional integral boundary layer method was developed, which is applied at individual streamlines. It is based on the method of Thwaites (1949) for the laminar part of the boundary layer and on Green's "lag entrainment method" (Green et al., 1972) for the turbulent part. Transition of the boundary layer from a laminar to a turbulent state can be initiated by three distinct mechanisms: natural transition, bypass transition or transition after laminar separation. See Mayle (1991) for an excellent review on this subject. See also Hogendoorn (1997) for an experimental analysis.

Natural transition is perhaps the one best studied. It is characterized by weak instabilities (also called Tollmien-Schlichting instabilities) which are gradually amplified and eventually develop into fully turbulent flow. This type of transition predominantly occurs in flows with low free-stream turbulence (around 1 percent) and moderate pressure gradients, as in the flow over aircraft wings. Michel's criterion (1952) is used to determine the location of transition:

$$Re_{\theta,\,t} = 1.174\left(1 + \frac{22400}{Re_s}\right)Re_s^{0.46}, \tag{5.62}$$

where $Re_{\theta,t}$ is the momentum thickness Reynolds number at the onset of transition and Re_s the Reynolds number based on the streamwise distance. This criterion is more or less equivalent to $Re_{\theta,t}$ = 400 (or Re_s =350,000) for a turbulence intensity of one percent. For very low turbulence, $Re_{\theta,t}$ may even increase to a value close to 1000.

The second type of transition is induced by external disturbances, like free stream turbulence, and completely bypasses the Tollmien-Schlichting type of instabilities, hence the name bypass transition. A correlation of experiments indicates the following criterion for bypass transition

$$Re_{\theta,\,t} = 400\,Tu^{-5/8}, \tag{5.63}$$

where the turbulence intensity is denoted by *Tu* (percents). This simple criterion appears to be valid for zero-pressure gradient flows (0.2 < Tu < 10) as well as for flows encountering adverse and favorable pressure gradients where *Tu* is larger than 3 percent.

The third type of transition occurs after laminar separation when the flow reattaches as a turbulent boundary layer to form a separation bubble of a certain length. Laminar separation is detected in the method by the criterion of Thwaites, which reads:

$$\lambda = \frac{\theta^2}{\nu}\frac{\partial U_e}{\partial s} \leq -0.082, \tag{5.64}$$

where U_e is the velocity just outside of the boundary layer and s is in streamwise direction. This type of transition may produce large losses, especially for long bubbles. Currently, no model is available to predict the extent of the bubbles. Therefore, in the numerical method the boundary layer is assumed to change to a turbulent state immediately after laminar separation. Although results for this case are not expected to be very accurate, it should merely serve as an indication that large losses may be expected, clearly a situation which should be avoided in practice.

Turbulent separation is assumed to occur if the shape factor (ratio of displacement thickness and momentum thickness) exceeds the value of 2.8.

For convex walls conditions for transition are comparable to that for a flat surface, possibly occurring somewhat later. For concave walls Liepmann (1943) found by experiments that instabilities leading to transition are caused by Görtler vortices, occurring somewhat earlier than the Tollmien-Schlichting instabilities for the flat plate. He deduced the following condition for transition:

$$G\ddot{o} = Re_{\theta,t}\sqrt{\theta/\mathrm{r}} \geq 7, \tag{5.65}$$

where Gö is called the Görtler number, and r the curvature radius. However, these measurements were conducted at very low turbulence intensities. At higher intensities transition is not influenced by curvature, except for extreme curvature $(\mathrm{r}/\theta \leq 900)$ where transition is actually delayed.

For internal flows the influence of the boundary layers will slightly change the characteristics of the bulk-flow. Therefore, an iterative procedure of (1) computing the inviscid bulk-flow, (2) a boundary layer analysis and (3) the adjustment of the wall surface according to the displacement thickness should be performed until convergence is obtained.

5.8 Conclusion

To conclude this chapter on loss models, the fundamental equations for shaft power P_{sh}, delivered head H, and efficiency η will be repeated here.

Referring to eqs. (5.2) and (5.6):

$$P_{sh} = \Omega M_{E,Q+Qleak} + \Omega M_{df}, \tag{5.66}$$

by combining eqs. (5.9) and (5.10):

$$H = \frac{\Omega M_{E,\,Q+Qleak}}{\rho g\,(Q+Q_{leak})} - \Delta H_{hydr}, \tag{5.67}$$

and by inserting eq. (5.3) into eq. (5.4):

$$\eta = \frac{\rho g Q H}{P_{sh}}. \tag{5.68}$$

In these expressions $M_{E,Q+Qleak}$ is the inviscid Euler moment (eq. 5.7), M_{df} the moment exerted by shear stress at the impeller external surfaces (eq. 5.8), and ΔH_{hydr} the head loss due to viscous losses in the internal fluid (section 5.5).

An important observation is that the shaft power can be computed without information on hydraulic losses. Thus, by comparing computed and measured values of the shaft power, the validity of the disc friction model, the leakage flow model and the method for computing the Euler moment can be checked. Information concerning the precision of the various models for hydraulic losses can subsequently be derived from a comparison of computed and measured head values. The efficiency, though an important quantity for pump designers, is not very important in the process of analyzing the computational results. As can be seen from eq. (5.68), it can be very sensitive to small deviations in either the computed shaft power, or the computed head value.

Nomenclature

A	Area
C_f	Skin friction coefficient
C_m	Disc friction coefficient
F_s	Surface force
H	Head
I	Rothalpy
M	Moment of momentum
P	Power
Q	Flow rate
Q_{leak}	Leakage flow rate
R	Disc radius
Re	Reynolds number
S	Surface
c_D	Dissipation coefficient
g	Gravitational constant
h	Clearance gap thickness

$\underline{n}$ Unit normal vector
p Static pressure
$\underline{r}$ Radius vector
t Time
$\underline{v}$ Absolute velocity vector
$\underline{w}$ Relative velocity vector

Greek symbols
$\underline{\Omega}$ Angular velocity vector
δ Boundary layer thickness
δ^* Boundary layer displacement thickness
δ_3 Boundary layer energy thickness
γ Ratio of cross-sectional area
η Efficiency
ν Kinematic viscosity
ρ Density
τ Shear stress
θ Boundary layer momentum thickness

Subscripts
e External to boundary layer
w At wall
θ Circumferential direction

Superscripts
R Rotor
S Stator

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Chapter 6
Experimental validation

In this chapter the validity of the flow model which underlies the numerical method is examined by comparing computational results with values obtained by experiments. Six test cases are considered. Four of these cases concern simplified pump models, especially designed for laboratory experiments. They are made out of perspex to allow for flow visualization, LDA measurements and accurate probe positioning. Measured values of the static pressure and velocity at many locations in the pumps will be used for validation purposes. The remaining two cases are industrial pumps mounted in a test circuit. For these configurations global characteristics, like head and efficiency, are available. As such they serve as ideal test cases for the combined application of inviscid flow computations and viscous loss models, as advocated in this thesis.

In section 6.1, the first in a series of radial impellers is considered. Static pressure and velocity measurements have been performed in the impeller, in which the flow is regarded as steady. The second case is that of a radial impeller with a vaned diffuser, presented in section 6.2. Measurements concentrated on the induced unsteadiness of the flow at impeller discharge, as a result of the downstream presence of the stator vanes. In section 6.3, a radial impeller enclosed in an unvaned spiral volute is considered. Experiments include static pressure and velocity measurements at various locations in the volute. Finally, in section 6.4, velocity measurements at a scale model of an industrial mixed-flow impeller are presented. The relative velocity field near the blade trailing edges is investigated.

Unsteady pressure measurements are available for an industrial mixed-flow pump. Comparison with computations will be presented in section 6.5. For many operating conditions, the delivered head and efficiency are measured as well. Viscous losses are estimated using the models as presented in chapter 5. The second industrial pump considered is a radial impeller with spiral volute. Although measurements primarily focus on rotordynamic forces on the impeller shaft, the delivered head has also been measured for different flow rates and for two seal clearances. Results of validation for this case can be found in section 6.6.

In section 6.7 the results are summarized and the overall performance prediction capability is discussed.

In a separate study excellent results are obtained for the prediction of cavitation inception, using the numerical method described in this thesis. For results the reader is referred to Van Os et al. (1997).

6.1 Free radial impeller

At the Société Hydrotechnique de France (SHF) a laboratory radial impeller has been designed to study recirculation phenomena. Two geometrically similar impellers have been manufactured and tested at several laboratories. At INSA (Lyon, France) Laser Doppler Velocimetry (LDV) measurements are performed on a water model with an outlet diameter of 354.4 mm (Combès and Rieutord, 1992). The impeller is placed in a vaned diffuser containing six blades. Pressure measurements are conducted at ENSAM (Lille, France) on an air model (diameter 516.8 mm) in an unvaned diffuser (Morel, 1993). Although test conditions and experimental techniques employed are different, all experiments show flow separation in the diffuser at the hub, and at the shroud for low flow rates. However, the critical flow rate depends on test conditions and experimental technique, and varies from 0.63 up to 0.84 of the flow rate at best efficiency point (Q_{BEP}). All experiments show a recirculation zone in the inlet region of the impeller at flow rates below 0.64-0.69 Q_{BEP}.

Inviscid computations for the static pressure and the velocity are compared with measurements for two operating conditions: at design flow rate Q_{BEP} and at partial flow rate $Q/Q_{\mathrm{BEP}} = 0.6$. Results of three-dimensional turbulent flow calculations with k-ε turbulence model (Combès and Rieutord, 1992) are also presented for comparison. A detailed description is found in Van Esch et al. (1995).

6.1.1 Geometry and operating conditions

The SHF impeller is a low specific speed ($n_\omega = 0.57$) shrouded impeller with 7 blades. It has an inlet diameter of 220 mm, an outlet diameter of 400 mm and an outlet blade angle of 67.5$^{\mathrm{o}}$ relative to the radius. At 1200 rpm, the nominal flow rate Q_{BEP} is 0.1118 m^3/s. Reynolds numbers (eq. 1.3c) are between $2 \cdot 10^6$ and $2 \cdot 10^7$ for the different test models. In the radial part of the impeller, the blades have a two-dimensional shape, orthogonal to the hub and the shroud. For simplicity, the influence of the diffuser and the volute on the flow through the impeller is disregarded (free impeller). Especially at off-design flow rates this assumption may not be correct.

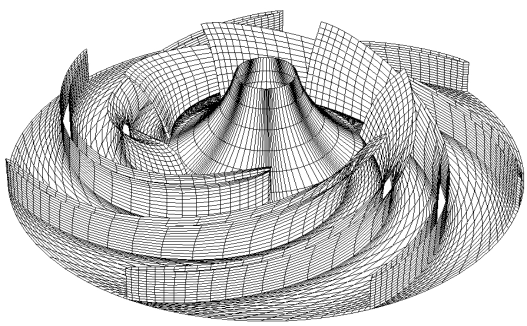

Fig. 6.1: Geometry of the SHF-radial impeller (shroud surface removed), showing surface mesh.

6.1.2 Computational aspects

The computational domain is restricted to one impeller channel and consists of 18,020 nodes, forming 79,040 linear elements. The mesh is refined near the blade, especially towards the leading and trailing edges. Fig. 6.1 shows the impeller geometry with shroud surface removed. Computations have been performed on a CONVEX-C240 vector computer, using one processor. CPU time is 30 minutes for the super-element formulation procedure, and one minute for the global process. The internal memory used is 70 Mb.

6.1.3 Results

Pressure distribution

Static pressure values are measured at a large number of positions distributed along both sides of the blade, the hub and the shroud in the impeller region. In fig. 6.2 the computed pressure distribution at blade mid-span is compared with measurements. It appears that computed pressures correspond quite well to measurements, although the agreement deteriorates towards the leading and trailing edges. At the hub and shroud surfaces a similar result is obtained. At partial flow rate the agreement is slightly less.

Velocity profiles

Fig. 6.3 shows radial and circumferential velocity profiles, for nominal and partial flow rate, near the impeller exit (r/r2 = 0.978). Points are located at three axial positions: near the hub (b/bmax = 0.05 to 0.1), at mid-channel (b/bmax = 0.5 to 0.52) and close to the shroud (b/bmax = 0.95 to 0.97). Results for the inviscid computations are compared with measurements and results obtained with the viscous flow model. Measurements at optimum flow rate reveal regions of secondary flow at the shroud and near the suction side of the blade. At partial flow rate, these secondary flow

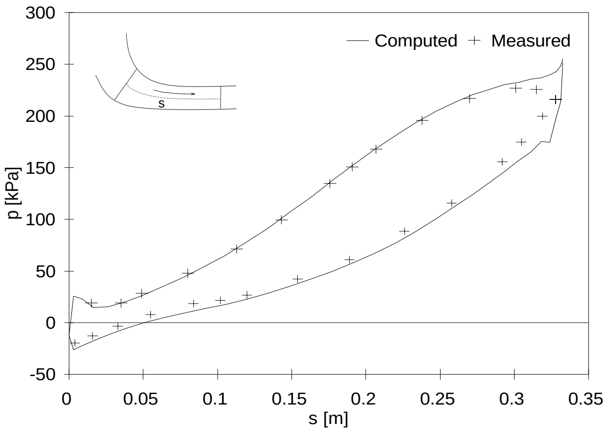

Fig. 6.2: Computed and measured values for the static pressure p along mid-span of the blade (local coordinate s), at design flow conditions.

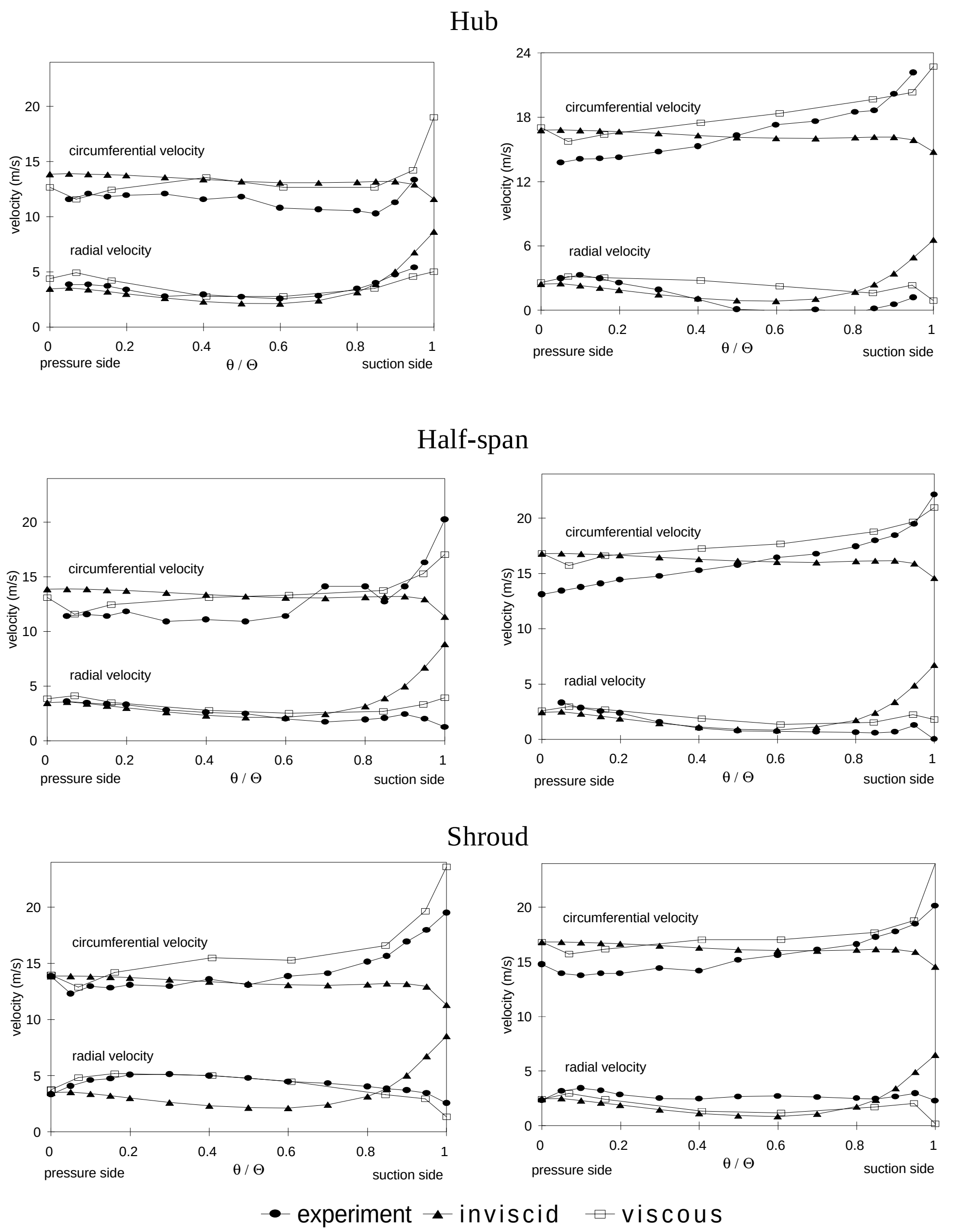

Fig. 6.3: Radial and circumferential velocity components as a function of angular coordinate θ in the impeller passage at $r/r_2 = 0.979$, for nominal flow rate Q_{BEP} (left figures) and partial flow rate $Q/Q_{BEP} = 0.6$. Shown are measured and computed velocity profiles near the hub, at blade mid-span and near the shroud. (Θ denotes angular width of impeller passage)

regions are still observed, although the velocity profiles are slightly more two-dimensional. The inviscid computations show nearly two-dimensional flow fields for both flow rates, at this radius. Outside the secondary flow regions, the computed radial velocity component agrees reasonably well with measurements, both for nominal and partial flow rate. Large deviations still occur at the shroud and near the suction side of the blade. Computed circumferential velocity profiles deviate considerably from those obtained by measurements. Viscous computations show somewhat better agreement. The shroud recirculation zone is predicted at nominal flow rate, at partial flow rate it is not. Circumferential velocities show a reasonable qualitative agreement with measurements, although deviations are still large.

6.2 Radial impeller with vaned diffuser

An interesting case is formed by the centrifugal pump with a vaned diffuser. Ubaldi et al. (1996) presented a test case for this configuration. Both velocity and static pressure measurements are performed in the small radial gap between the impeller outlet and the diffuser vanes' leading edges, at 30 different circumferential locations distributed over one diffuser vane pitch. The periodic signals are separated from the unresolved unsteadiness by the ensemble averaging technique (Lakshminarayana, 1981). Time correlation of the results for the different probe positions provide the time averaged velocity profile in the relative frame of reference, as well as the stator generated unsteadiness. In this section results of an inviscid calculation of the velocity distributions will be compared with measurements.

6.2.1 Geometry and operating conditions

The model consists of a 420 mm diameter unshrouded centrifugal impeller with seven backswept blades. The blades are untwisted and have constant thickness. Blade span in the radial part of the impeller is 40 mm, with a tip clearance of 0.4 mm. The diffuser has 12 vanes with internal and external diameters 444 mm and 672 mm respectively. The model operates in an open circuit, with air being fed to the impeller through a long straight pipe and discharged into the atmosphere directly. The experiment has been conducted at the constant rotational speed of 2000 rpm and a flow rate of 0.289 m^3/s. Reynolds number (eq. 1.3c) is $2.5 \cdot 10^6$. Velocity measurements are performed using a hot wire anemometer at positions 4 mm from the rotor blade trailing edge and 8 mm from the vane leading edge. For a detailed description of the experimental facility, instrumentation and data reduction the reader is referred to Ubaldi et al. (1996).

6.2.2 Computational aspects

A fully unsteady computation has been carried out according to the method described in section 2.5. Blade circulation values as well as the resulting unsteady wakes are considered both for impeller and stator vanes. The computational mesh is given in fig. 6.4, where the suction pipe and casing have been removed. The tip clearance between rotor blades and casing is not modelled. The domain is divided into 26 blocks, two for each rotor channel and one for each diffuser channel. Maximum use

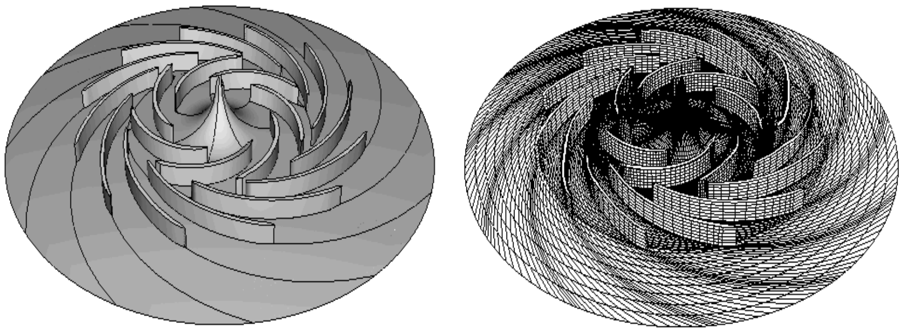

Fig. 6.4: Geometry of rotor and vaned diffuser with casing and suction pipe removed (left), and surface discretization of computational mesh.

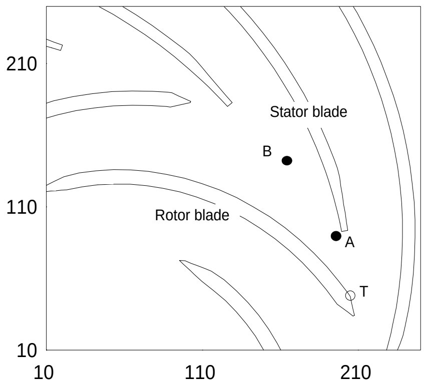

Fig. 6.5: Impeller and vaned diffuser geometry (detail), showing stationary positions *A* and *B*, and rotor tip position *T*.

can be made of symmetry as mesh generation and super-element formulation are restricted to one channel, both for the impeller and for the diffuser region. A two-dimensional view of the rotor and diffuser vanes is given in fig. 6.5. The total number of nodes is 111,000 (73,000 in the rotor), forming 525,000 linear tetrahedral elements. The number of time steps is 24 per blade passing period, which equals the number of elements in a pitch in circumferential direction.

6.2.3 Results and comparison

Instantaneous relative velocity

Experimental results of ensemble averaging for two stationary probe positions are shown in fig. 6.6. The relative velocity at mid-span is given as a function of time. The revolution of the impeller with respect to the stationary probe is clearly identified. The probe is first passed by the pressure side, showing a velocity peak, then by the wake, resulting in a velocity deficit, and finally by the impeller passage from the suction to the pressure side. When computed velocities are compared with measurements it emerges that, on the average, the computed velocities are too small. It can be

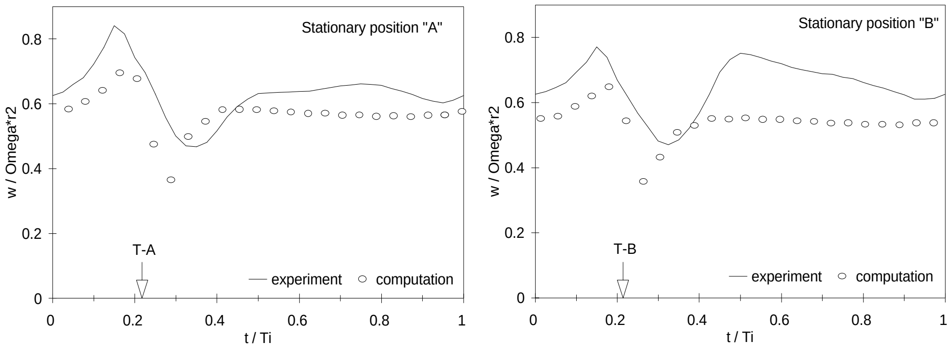

Fig. 6.6: Comparison of computed and measured relative velocity *w* as a function of time *t* for two stationary locations A and B (see fig. 6.5). The time at which rotor tip T coincides with stationary positions A and B is indicated by “T-A” and “T-B” respectively (Ubaldi, 1997). The impeller blade passing period is denoted by T_i. Computations are performed for 24 time steps per pitch.

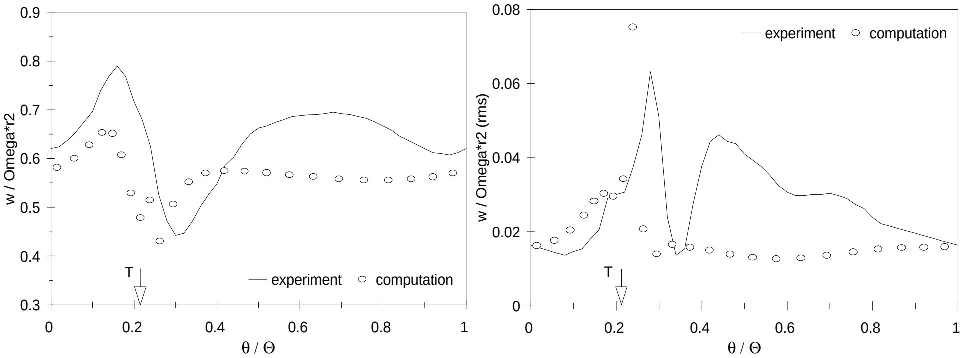

Fig. 6.7: Time averaged relative velocity *w* at the impeller outlet (left) and rms-value, as a function of angular position θ within the impeller pitch (angular width of impeller passage denoted by Θ). The location of the impeller tip *T* (see fig. 6.5) is indicated in the figures (Ubaldi, 1997). Computations are performed for 24 time steps per pitch.

explained by the head reduction due to the tip clearance between impeller vanes and casing, which is not modelled, and by the displacement effect of boundary layers at hub and shroud surfaces. For an impeller of this width, the blockage effect by boundary layers can be quite substantial. It can be concluded that the positions of the velocity peak and deficit as well as their velocity difference are in good agreement with measurements, although measured peaks appear to be somewhat broader. However, measured and computed velocity in the impeller channel area between the suction and the pressure side shows a considerable disagreement.

Relative velocity in impeller frame of reference

By combining the ensemble averaged signals of many probe locations in the diffuser vane pitch, the time averaged velocity profile in the relative frame of reference could be obtained. Furthermore, the stator-induced unsteadiness could be separated from the fluctuating part of the signal by considering only its periodic part. The averaged relative velocity profile is given in fig. 6.7 (left) together with the result of computations. Again the pressure side peak and wake deficit are clearly identified. As was the case for the stationary probe velocities, the impeller passage profile is not at all predicted correctly. Fig. 6.7 also gives the root mean square of the upstream stator-induced unsteadiness. The strong increase of unsteadiness near the impeller tip is predicted reasonably well by the inviscid method. Again the signals in the impeller passage are in disagreement

6.3 Laboratory impeller with spiral volute

At the department of Thermal Engineering of the University of Twente a test rig has been built to study the two-dimensional flow in various types of centrifugal impellers (fig. 6.8). For a detailed description of the experimental set-up the reader is referred to Visser and Jonker (1995) and Visser (1996). The main feature of the set-up is a Laser Doppler Velocimetry (LDV) measuring system which can be mounted co-rotating with the impeller. Recently the design has been adapted to include a spiral volute in order to study the interaction between the impeller and the volute of a pump (Kelder, 1996). Measurements include time-averaged values of velocity and static pressure at a large number of locations in the volute. Part of the result has been published in Van Esch et al. (1997).

6.3.1 Geometry and operating conditions

Current measurements are performed at a centrifugal pump with a specific speed n_ω of 0.4. The impeller, depicted in fig. 6.9, has seven thin blades with a constant blade angle of 70° to the radius. Its inner diameter is 320 mm, its outer diameter 640 mm, and the axial width is 25 mm. The Reynolds number (eq. 1.3c) equals $1.7 \cdot 10^6$. The volute has a trapezoidal cross-section and is designed to approximately match the impeller at design condition (Q = 0.008 m^3/s, Ω = 4.2 rad/s), according to the method of constant angular momentum (Pfleiderer, 1961). It is shown in fig. 6.10. During construction special attention was paid to the minimization of leakage flows. Fig. 6.11 shows the locations in the volute where velocity and static pressure measurements are obtained. Velocity measurements are performed using LDV. Information on the axial velocity component can not be obtained. The head-capacity curve is derived from the static pressure difference between inlet and outlet of the pump and the assumption of uniform velocity in these regions. It is not possible to measure the efficiency of the pump.

6.3.2 Computational aspects

In fig. 6.12 part of the computational mesh is shown. Each of the seven impeller channels is divided into two blocks, with a total number of 14,500 nodes. The volute is divided into six blocks, with 66,400 nodes. The total number of tetrahedral elements is 700,000.

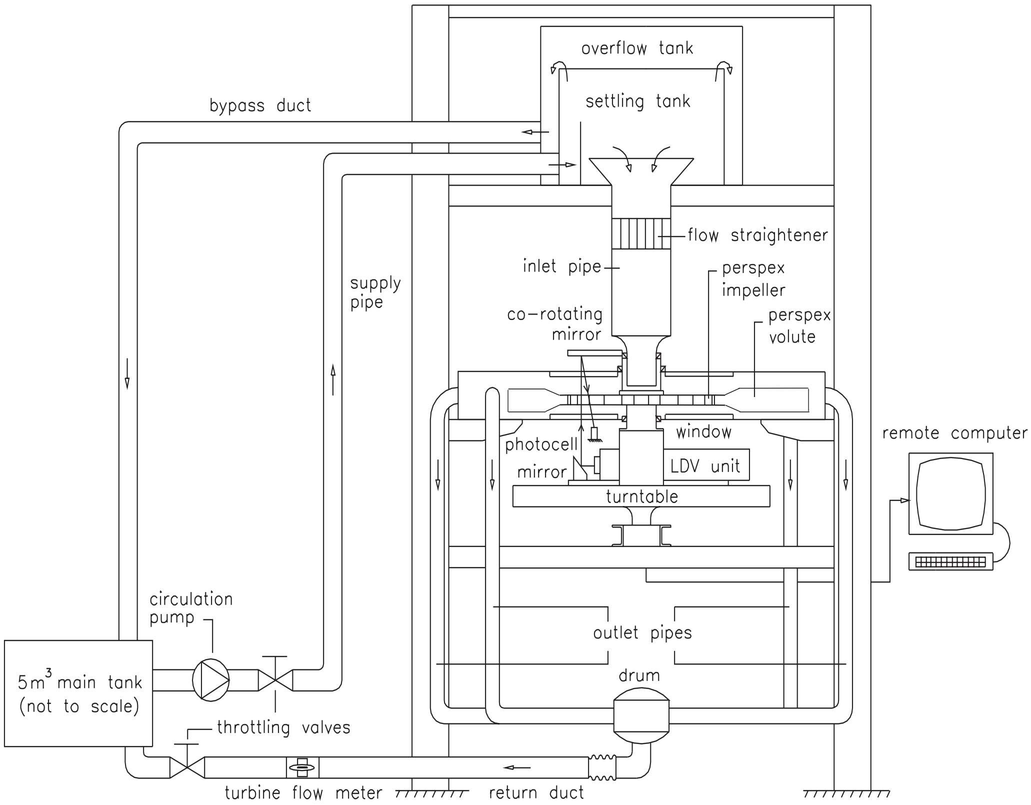

Fig. 6.8: Test rig for experimental analysis of flows through hydraulic pumps, at the University of Twente. The LDV measuring system is shown mounted a turntable which is rigidly connected to the impeller shaft. The test geometry consists of a two-dimensional impeller with spiral volute.

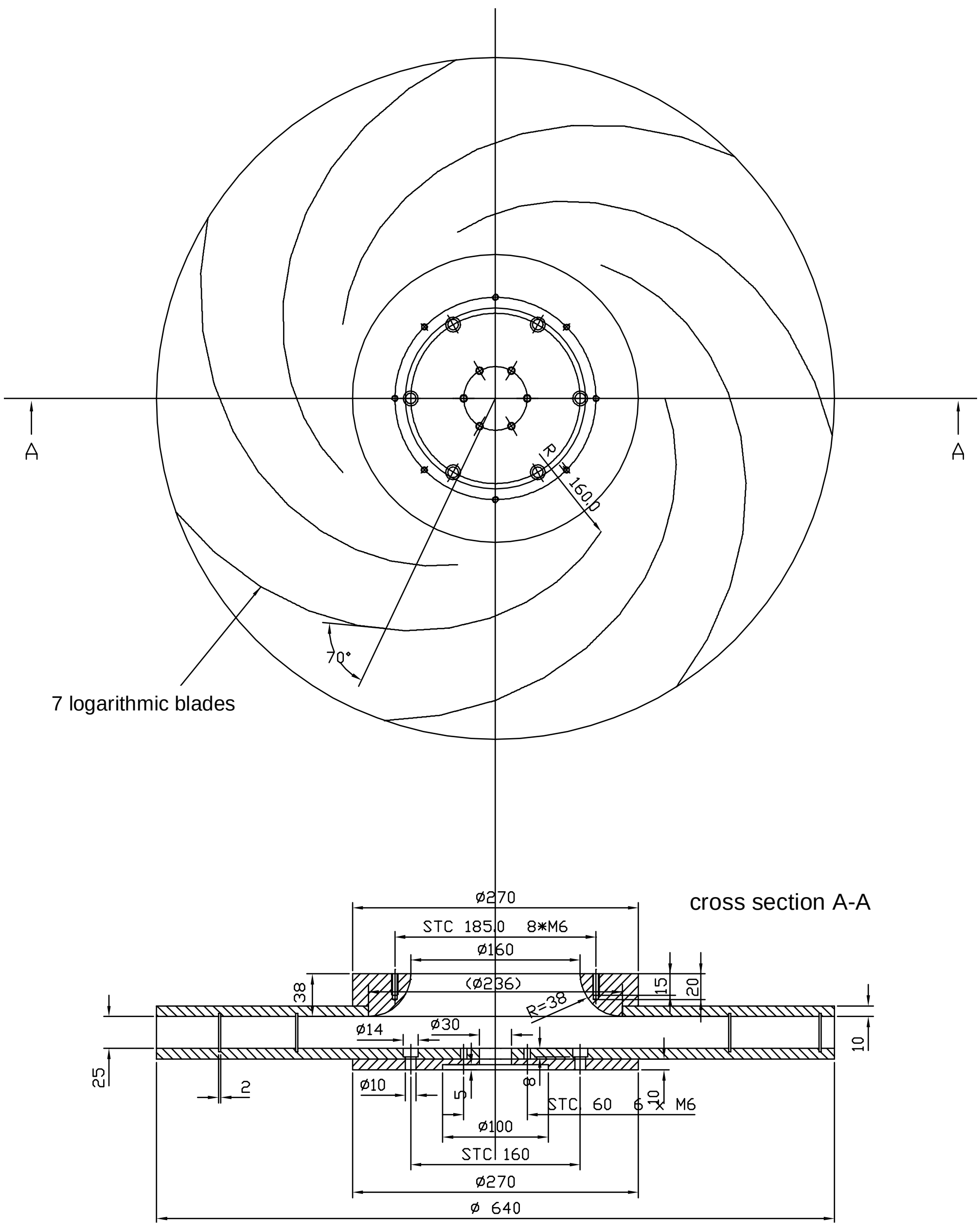

Fig. 6.9: Two-dimensional test impeller at laboratory test rig (length in mm).

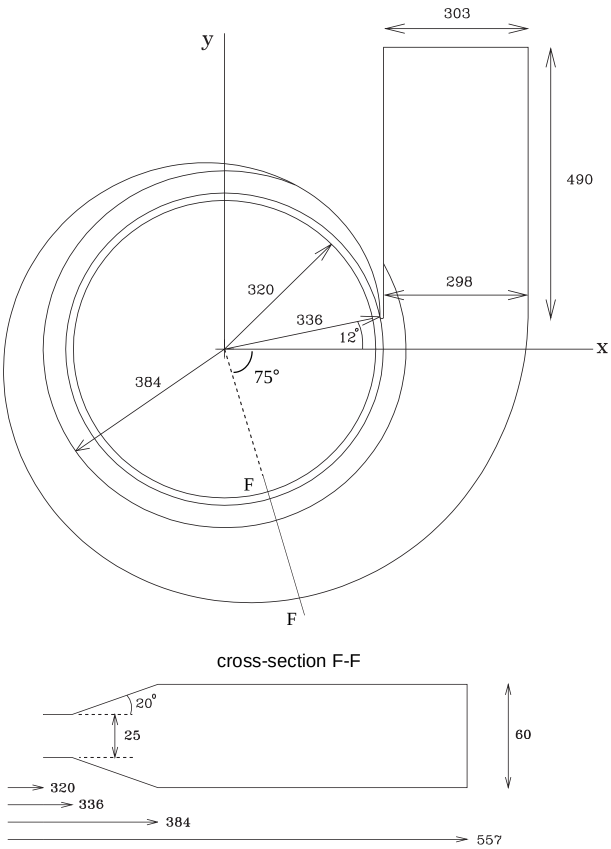

Fig. 6.10: Spiral volute at laboratory test rig (length in mm). Figures at different scales.

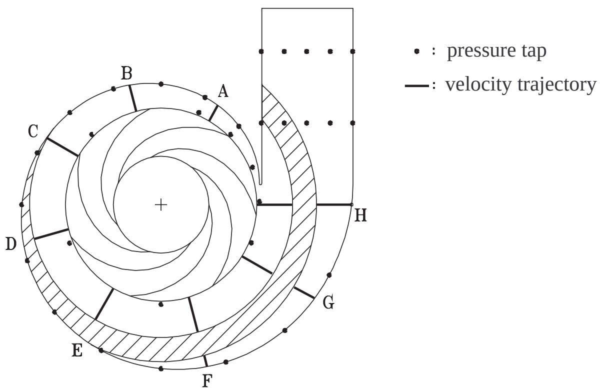

Fig. 6.11: Measurement locations in the laboratory centrifugal pump. LDV measurements are performed along trajectories *A* to *H*. Static pressure measurement locations are indicated with solid markers. Hatched area shows region which is not visually accessible.

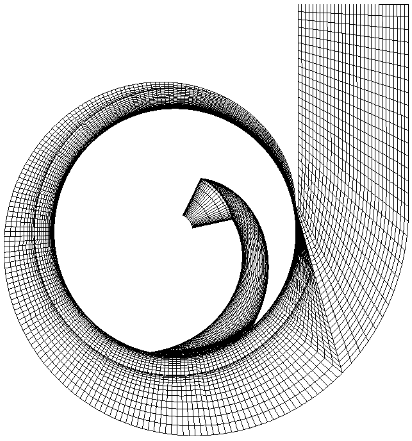

Fig. 6.12: Computational mesh for the radial impeller with spiral volute. One of the seven impeller channels is shown.

6.3.3 Results and comparison

Head curve

Fig. 6.13 shows the head-capacity curve. As leakage flow is negligible, viscous losses are restricted to boundary layer dissipation and wake mixing loss. The methods as described in section 5.4 are used to quantify the effects. The fraction which wake mixing contributes to the total head loss ranges from 10% at high mass flow to 25% at low mass flow. The fairly good agreement between computations and experiments at low and optimum flow rate implies that other sources of viscous losses are not very important in this laboratory pump. At high flow rate a larger deviation is observed between computations and experiments. A possible cause is boundary layer separation at the exit pipe inner wall, downstream of the tongue. This phenomenon is experimentally observed for high flow rates.

Velocity in the volute

Radial and circumferential velocity components are measured at half-height along a number of trajectories, named *A* to *H* (see fig. 6.11). Measurements along trajectory *F* are shown in fig. 6.14. It is seen that circumferential velocities agree well with mea-

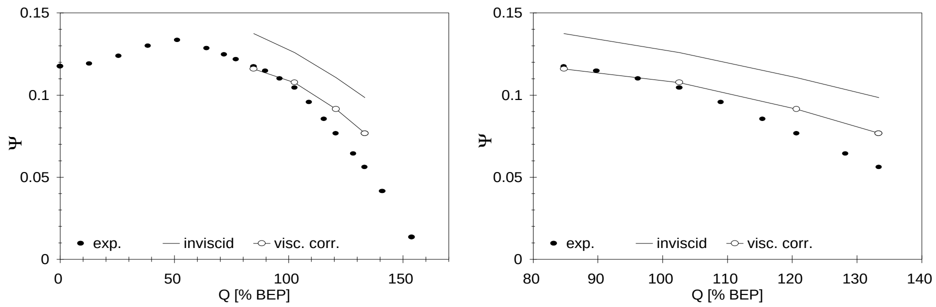

Fig. 6.13: Head-capacity curve for the laboratory centrifugal pump, showing measured and computed values, both inviscid and corrected for viscous losses. Right figure shows detail.

surements, except for high flow rate. Radial velocities are over-estimated near the impeller discharge, while under-estimated at larger radii. The results for the circumferential velocities at optimum flow conditions are typical for the agreement in a large region of the volute (60°-285° from the tongue). At low flow rate (0.825 Q_{BEP}) this agreement is restricted to a smaller region (150°-285° from the tongue). At high flow rate (1.175 Q_{BEP}) the agreement is not very good except for a small region 60°-150° from the tongue. The agreement between computed and measured radial velocities is very poor. An investigation of the axial distribution of radial velocities at a number of radial positions in the impeller and the volute reveals that a region of severe secondary flow is located in the volute. Typical radial velocity profiles are sketched in fig. 6.15. The observed convex radial velocity profile in the volute region, with negative velocities near the upper and lower surfaces, can easily be

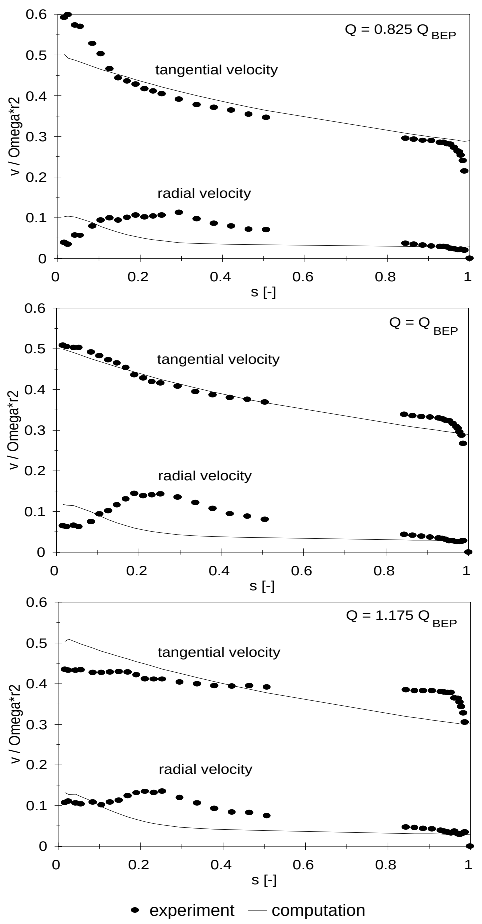

Fig. 6.14: Radial and circumferential velocity in the volute of the laboratory centrifugal pump along the radius located 285° from the tongue (trajectory F in fig. 6.11), for three flow rates. Comparison between measurements and computations. The scaled local coordinate along the trajectory is denoted by s, ranging from 0 at the impeller outer radius, to 1 at the volute outer wall. Velocities are scaled with blade tip speed.

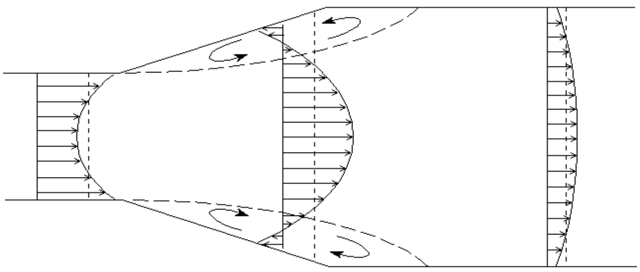

Fig. 6.15: Cross section of impeller and volute, showing measured radial velocity profiles. Average velocities are indicated with dotted lines.

explained by an analysis of pressure forces and centrifugal forces (due to curvature) in the boundary layers and the main flow. It is equivalent to the secondary flow encountered in flows through bended pipes. A similar analysis for the flow through the impeller can be made, where account should be given of the additional centrifugal force due to rotation and the Coriolis force. The equilibrium between pressure forces and (mainly) Coriolis forces in the main flow is lost in the boundary layers at hub and shroud surfaces, leading to a secondary flow in the boundary layers directed from pressure to suction side. In the main flow a reverse secondary flow direction is observed. This leads to the observed concave radial velocity profile for impellers with backward curved blades.

Static pressure

The static pressure difference p_d between the inlet of the pump and locations in the volute is measured using U-tube manometers (fig. 6.16). It is made dimensionless with the blade tip speed according to

$$p_d = \frac{p_{vol} - p_{inlet}}{\rho (\Omega r_2)^2}. \tag{6.1}$$

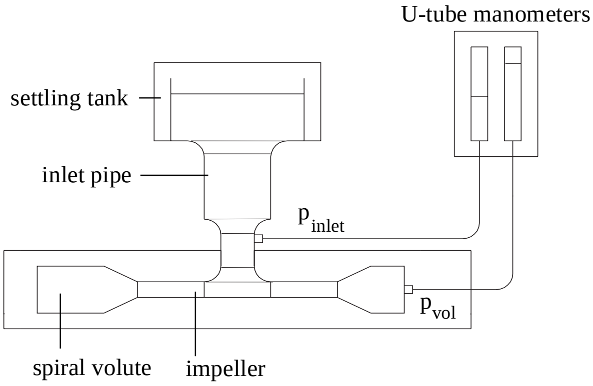

Fig. 6.16: Schematic representation of static pressure difference measurements using U-tube manometers.

In fig. 6.17 results of measurements and computations are shown for locations just outside the impeller and along the volute outer wall, at three different flow rates. The (inviscid) computations lead to pressure values which, on the average, are too high. However, in a large region, not too close to the tongue, the qualitative agreement is quite good. A constant static pressure around the impeller can be observed at design flow rate. The computed static pressure values can be corrected for viscous losses in the impeller

$$p_{d,\,corr} = p_{d,\,invisc} - \rho g\,\Delta H_{hydr,\,\mathrm{i}}\,, \tag{6.2}$$

where the hydraulic head loss in the impeller $\Delta H_{hydr,i}$ is given by eq. (5.11). By doing so the agreement is improved, although considerable deviations still occur at off-design conditions. It can be seen from the difference between measurements and computations that viscous losses build up along the volute wall.

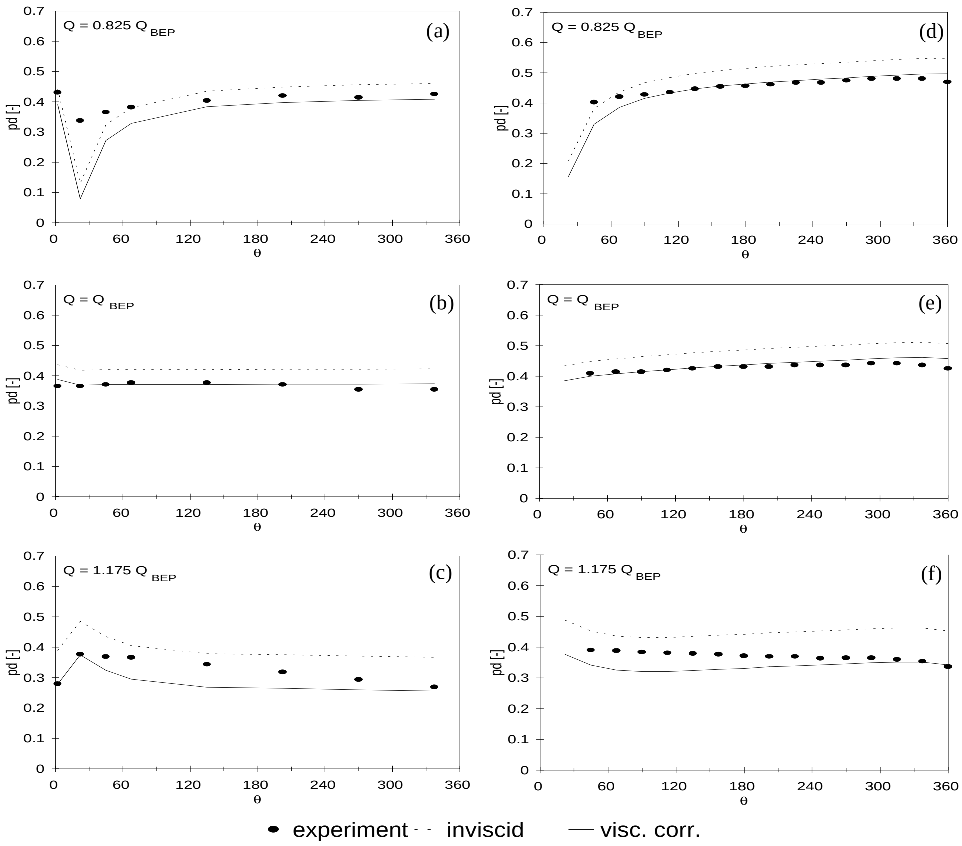

Fig. 6.17: Static pressure difference in the volute of the laboratory centrifugal pump as a function of angular distance θ (degr) from the volute tongue, for three flow rates. Figures (*a*) to (*c*) show locations just outside the impeller, figures (*d*) to (*f*) locations along the volute outer wall (see fig. 6.11).

6.4 Free mixed-flow impeller

The test facility of the University of Twente (fig. 6.8) has been modified to study the velocity field in a mixed-flow impeller using a Laser Doppler Velocimetry (LDV) system which is mounted co-rotating with the impeller (Van Os 1997, Duineveld 1997). The main objective of this investigation is to study the main flow velocity field and the boundary layer behaviour at different levels of the turbulence intensity. Some measurements have been performed to study the flow direction near the trailing edge of the blades. Computations of the velocity in this region are compared to measurements in order to validate the flow direction which was assumed in the current model (see Kutta condition, section 2.2.2).

6.4.1 Geometry and operating conditions

Measurements are performed at a model of an industrial mixed-flow impeller, operating as a free rotor. The geometry is identical to the impeller which will be described in section 6.5, but entirely made out of perspex. Operating conditions differ considerably, as rotational speed is restricted to a few revolutions per second. The Reynolds number (eq. 1.3c) is of order $8 \cdot 10^5$. Three different flow rates are considered: 75%, 100% and 150% of flow rate Q_{BEP} at best efficiency point. Fig. 6.18 (left) shows the LDV measuring plane in the region near the blade trailing edge. It is tilted by an angle of 37° with respect to the plane at constant *z*-coordinate. The velocity components in this plane are measured at ten positions, all located in a plane at constant *z*-coordinate (fig. 6.18, right).

6.4.2 Computational aspects

Computations are performed at one impeller channel. Velocity components at the measuring locations are derived by interpolation and projection onto the plane in which the LDV measurements are performed. Two different flow directions at the trailing edge stagnation points are applied: along the suction surface and along the wedge angle of the blade (fig. 6.19).

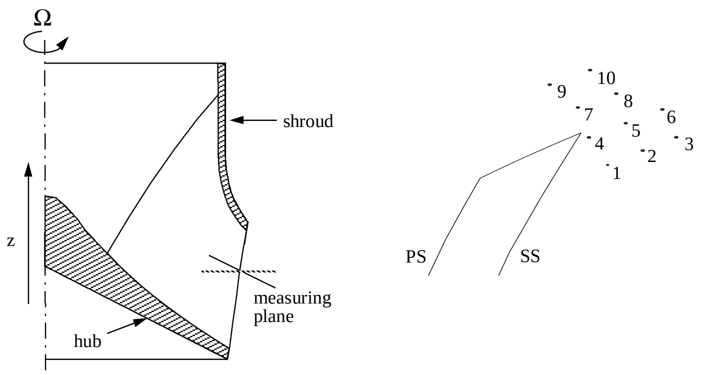

Fig. 6.18: LDV measuring locations near the trailing edge of the mixed-flow impeller. Left figure shows measuring plane in meridional view. Right figure shows the ten locations and part of the impeller blade in a plane at constant *z*-coordinate (PS = pressure side ; SS = suction side).

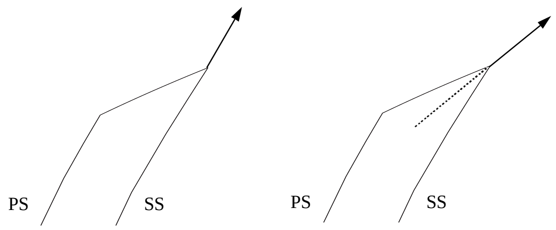

Fig. 6.19: Direction of stagnation streamline at the trailing edge: along the suction surface of the blade (left), and along the blade wedge angle (right). PS = pressure surface ; SS = suction surface.

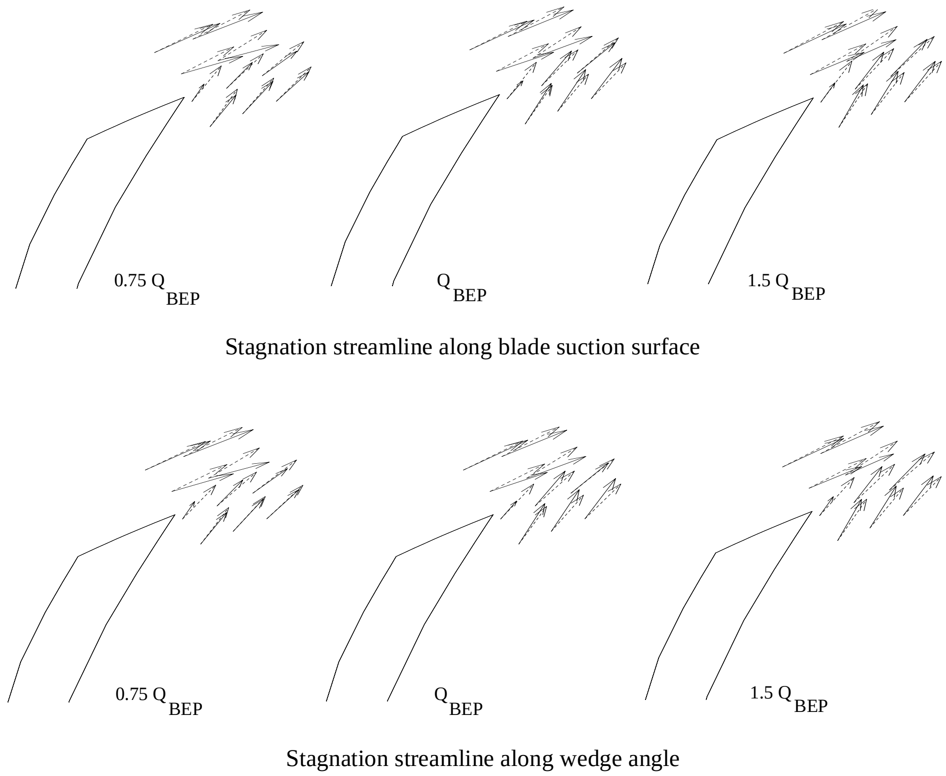

Fig. 6.20: Relative velocity near the blade trailing edge of a mixed-flow impeller for three flow rates. Solid line arrow denotes measured velocity, dashed arrow denotes computed velocity. Computations are performed for two trailing edge flow directions: along the blade suction surface (upper figures) and along the blade wedge angle.

6.4.3 Results and comparison

Results of both the computations and the measurements are given in fig. 6.20, for three flow rates. The measured velocities clearly show the existence of a wake behind the impeller blade. It appears that a stagnation streamline in the direction of the suction surface yields a slightly better correspondence to measurements than a direction along the blade wedge angle. However, a conclusive answer to the question which of the two flow directions is correct cannot be given on the basis of these measurements.

6.5 Mixed-flow impeller with volute casing

In this section results of the analysis of a mixed-flow pump will be presented. It concerns an industrial pump, made by BW/IP Stork Engineered Pumps in Hengelo (The Netherlands), used for cooling-water transportation in power stations. Computations are compared with experimentally obtained values on a small scale model. Experiments include overall performance characteristics and unsteady static pressure measurements at a number of locations on the volute wall. In section 6.5.1 the experimental setup and the measuring technique are described. Some computational aspects are given in section 6.5.2. Finally, in section 6.5.3, results of computations are compared with measured values. Part of this analysis has been published in Van Esch et al. (1997).

6.5.1 Experiments

Experiments are performed at the test facility of BW/IP Stork Engineered Pumps in Hengelo (The Netherlands). The model contains a shrouded mixed-flow impeller with four blades and a specific speed n_ω of 1.6. The volute is unvaned with a trapezoidal cross-section (figs. 6.21 and 6.24) and is designed according to the method of constant angular momentum (Pfleiderer, 1961). The model pump, driven by an electric d.c. motor, is mounted in a closed circuit (fig. 6.22).

Delivered head is derived from averaged static pressure measurements at the inlet and outlet of the pump and the assumption of uniform velocity at inflow and outflow stations. The measured shaft power is corrected for losses in bearings. Head and efficiency values are measured for a large number of flow rates, ranging from 50% to 150% of flow rate Q_{BEP} at best efficiency point.

For the static pressure at the volute wall to be measured, taps are made at 24 positions on the volute wall and at the impeller inlet, suitable for mounting pressure transducers (fig. 6.23). Two semiconductor pressure transducers are used to measure the unsteady pressure at the inlet and the volute wall taps simultaneously. In this way the instantaneous pressure difference between inlet and volute taps can be obtained. The semiconductor pressure transducers (Druck PTX 610-OI) have a resonance frequency well over 10 kHz.

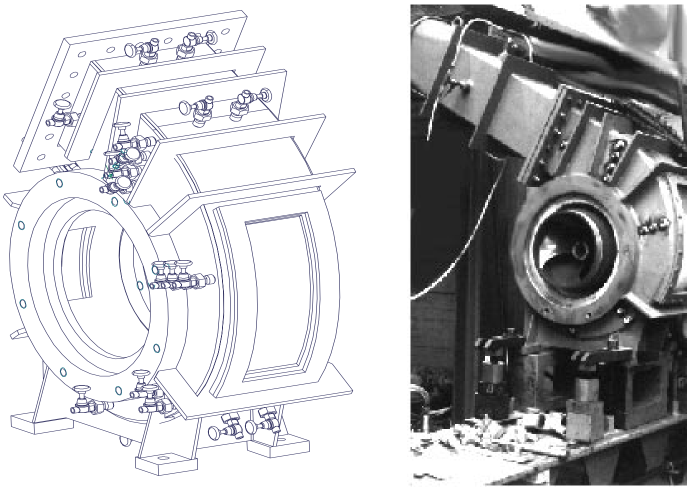

Fig. 6.21: Schematic drawing (left) and photograph of the experimental setup clearly showing the volute shape and some of the pressure taps. In the photograph the impeller can be seen as the inlet pipe is removed.

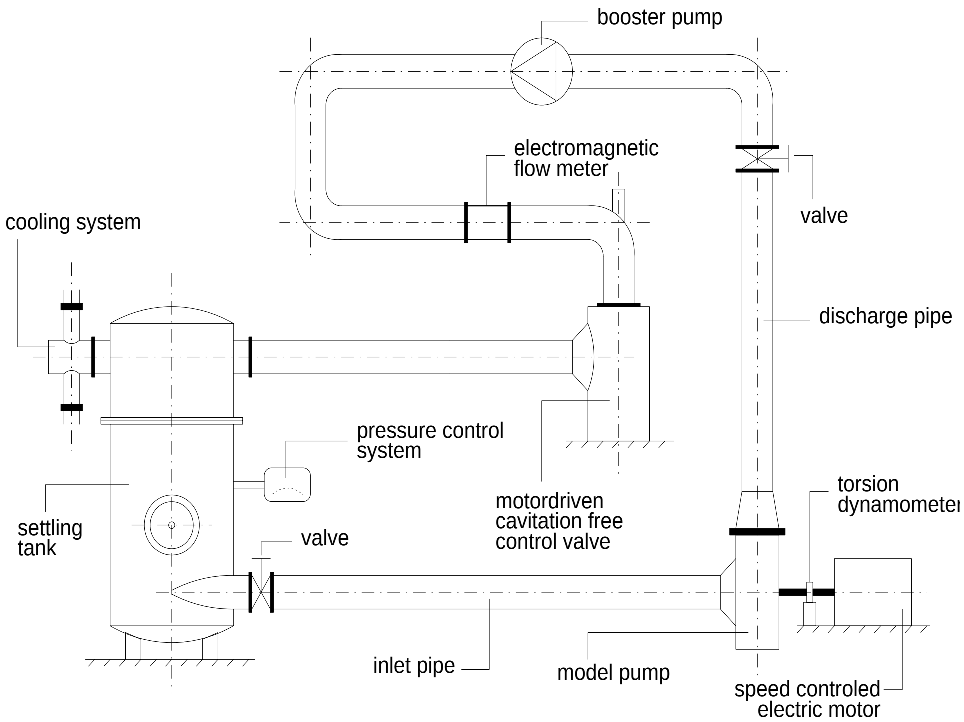

Fig. 6.22: Schematic representation of the test facility at BW/IP Stork Engineered Pumps (The Netherlands). The model pump is mounted in a closed circuit.

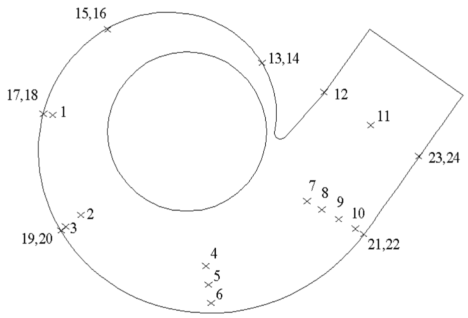

Fig. 6.23: Schematic view of pressure taps in the volute. Taps 13 to 24 are at the side of the volute (even numbers near the shroud, odd numbers near the hub). All other taps are at the shroud surface of the volute.

Unsteady pressure signals are composed of a periodic part, related to the rotational frequency, and a random part resulting from turbulence, vortex shedding etc. The periodic signal is obtained from the instantaneous one by using phase locked sampling and ensemble averaging (Lakshminarayana, 1981). The synchronization pulse is generated by a ring-shaped device attached to the shaft, an infra-red light source and a photocell. All three signals (synchronization, inlet- and volute pressure) are sampled simultaneously by means of a 16 bit A/D converter (APB 200 / D-TAC of Difa Measuring Systems). The revolution frequency pulse is also used for the internal triggering of data acquisition. Each measurement 4096 samples are taken, covering over 168 impeller blade passages. It corresponds to 24 data samples for each impeller passage. Ensemble averaging is done after completion of the measurements using specially developed software.

Nominal operating conditions are at flow coefficient $\phi = 0.056$, head coefficient $\psi = 0.082$, and Reynolds number $Re = 1.3 \cdot 10^7$ (eq. 1.3).

6.5.2 Computational aspects

In fig. 6.24 the computational mesh is shown. Impeller and stator parts are presented separately. The domain is divided into 8 blocks in the rotor and 8 blocks in the stator, with a total number of 80,000 nodes and 380,000 tetrahedral elements. Each impeller pitch revolution is divided into 30 time steps. Computations are performed on a Silicon Graphics Power Challenge operating at a speed of 60 Mflops, and take less than seven hours to complete for each separate flow rate. Maximum internal memory required is 50 Mb.

6.5.3 Results and comparison

Overall performance characteristics

In this section computed values for the delivered head and the efficiency as a function of flow rate are compared with values from experiments. Fig. 6.25 gives the result for

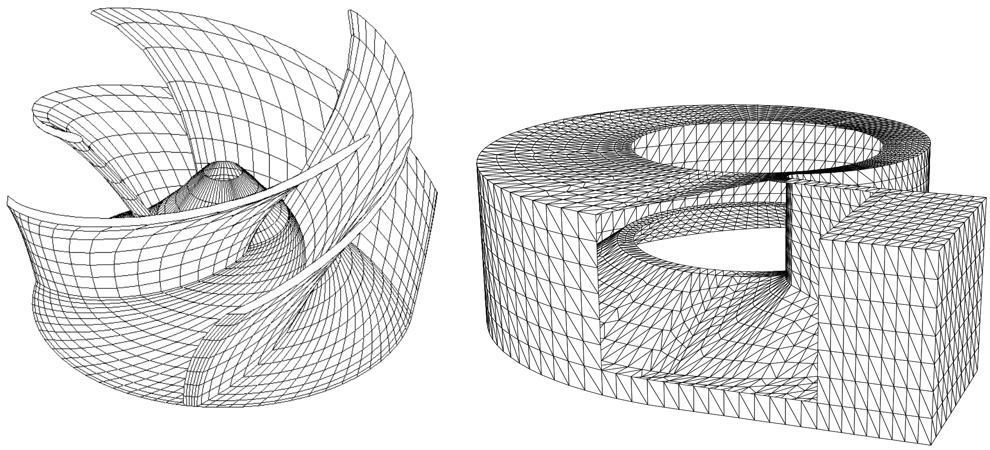

Fig. 6.24: Outer boundaries of meshes for the BW/IP Stork mixed-flow pump, showing the impeller and volute geometry separately. Notice the blade thickness and blunt trailing edge. The shroud and part of the volute wall is removed offering a better view. Depicted are coarse versions of the computational meshes.

the inviscid computations. It clearly shows that inviscid computations overpredict the pump's head considerably. It should be noted that the influence of leakage flow is not considered. In the following, this contribution as well as the hydraulic losses in the impeller and volute will be quantified.

Leakage flow - In order to compute the leakage flow rate the model as discussed in section 5.5 is used. Since surface roughness is known (7 µm for the cavity, 2 µm for the seal), the friction factors of Moody are used. The ratios Q_{leak}/Q are found to be quite high, varying from 1.7 percent at high flow rate (Q/Q_{BEP} = 1.4) up to 6.4 percent at the lowest flow rate considered (Q/Q_{BEP} = 0.6). This is a result of the rather large seal clearance of 0.87 mm. For mixed-flow impellers the radial extent of the

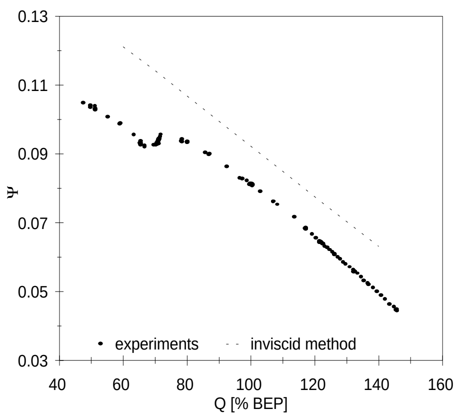

Fig. 6.25: Comparison of experimental head values with results of inviscid computations.

leakage area is limited. As a consequence the wall shear forces are dominant over the ‘free-vortex mechanism’. This leads to a circumferential velocity which tends to be equal to half the rotational speed of the impeller upon leaving the seal. Only for very low flow rates (i.e. large leakage flow rates) circumferential velocities are seen to have somewhat higher values.

Boundary layer - A boundary layer analysis has been performed with the method as described in section 5.7. Three streamlines along the pressure and suction sides of the blade were considered; near the hub, at half span, and near the shroud of the impeller. This was done for several flow rates Q/Q_{BEP} in the range 0.6 to 1.4. A free-stream turbulence intensity of 5 percent was assumed. For most of the streamlines and flow rates considered, transition occurred shortly after the leading edge and no separation, either laminar or turbulent, was predicted. There were however some exceptions. At the pressure side, near the hub, the method predicts separation of the turbulent boundary layer for all of the flow rates considered. For lowest flow rate it occurs at 50% blade length, a position which moves towards the leading edge for larger flow rates, down to 15% blade length at highest flow rate. It should however be noted that this method does not account for the repressing influence of the main flow trough the impeller channels on boundary layer growth. As can be seen in fig. 6.26 turbulent separation occurs in a region were separation is likely to be avoided or at least delayed. An important factor in this respect is the subsequent strong acceleration of the fluid over the second half of the pressure side. In this region the acceleration parameter K which is defined as

$$K = \frac{\nu}{U_e^2}(\partial U_e / \partial s) \tag{6.3}$$

is well over 3.10^{-6}, a value which indicates relaminarization of a turbulent boundary layer (Mayle, 1991). In eq. (6.3), U_e denotes the velocity outside of the boundary layer, ν the viscosity and s the local coordinate along the surface. So, separation bubbles are believed to be small, if at all present, and the extra losses will likewise be small.

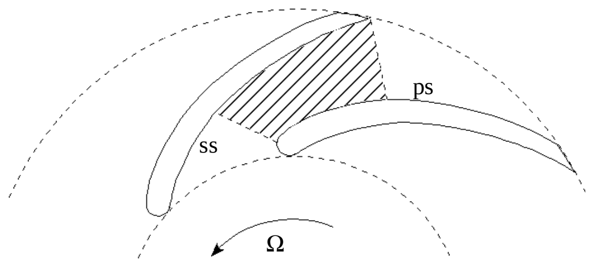

Fig. 6.26: Impeller passage showing region where boundary layer growth is counter-acted by the main flow, due to the presence of adjacent blades (ss = suction side, ps = pressure side).

Some indication of turbulent separation was also given at the pressure side, near the shroud at two third of the blade length, but only for flow rates as low as Q/Q_{BEP} = 0.6. Acceleration along the shroud in this region is not sufficient to prevent separation. If it occurs it will most probably lead to a strongly increased loss over this part of the blade.

The most important phenomenon is laminar separation near the hub at the suction side, immediately after the leading edge, for flow rates below Q/Q_{BEP} = 0.8. This leads to a situation which is generally indicated as stall. Even if the separated flow manages to reattach beyond this point, large losses will inevitably be the result. See also Van Os (1997) for a discussion on boundary layer separation.

Hydraulic losses - In fig. 6.27 several types of hydraulic losses are plotted against flow rate. The models as presented in section 5.4 were used for this purpose. Obviously, boundary layer dissipation in the impeller region of the pump is the most important source of hydraulic loss, followed by boundary layer dissipation loss in the volute. The energy dissipation coefficient c_D (defined in subsection 5.4.1) was taken 0.0038, a constant value for all surfaces and for all flow rates. Although losses along the impeller blades will possibly be much larger at $Q/Q_{BEP} \leq 0.8$ due to boundary layer separation, this is believed to be an acceptable first estimate.

The flow experiences a sudden expansion in radial direction as it leaves the impeller and enters the (much wider) volute. A fraction of the kinetic energy of the radial velocity is dissipated in the process.

The total hydraulic loss is a convex function of flow rate Q, with a minimum at Q=0.8Q_{BEP}.

Euler head - In fig. 6.28 the influence of leakage flow on the (inviscid) theoretical head H_E is shown. The head reduction is caused by the higher flow rate through the impeller, and by the angular momentum of the leakage flow. The relative contribution of the former ranges from 30 percent at lowest flow rate, to 50 percent at highest flow rate considered.

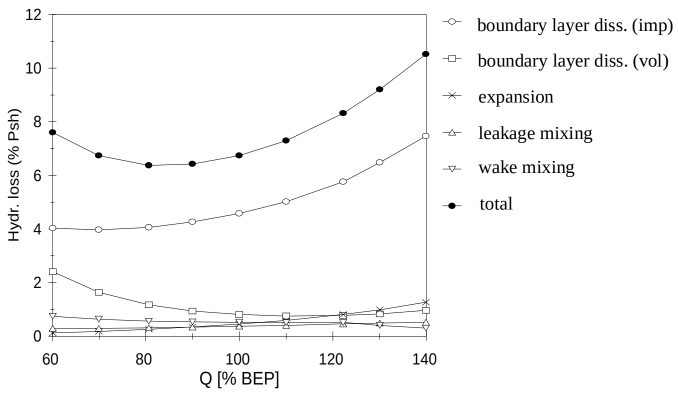

Fig. 6.27: Various types of hydraulic losses plotted against flow rate Q. Loss is presented in percentage of shaft power P_{sh}.

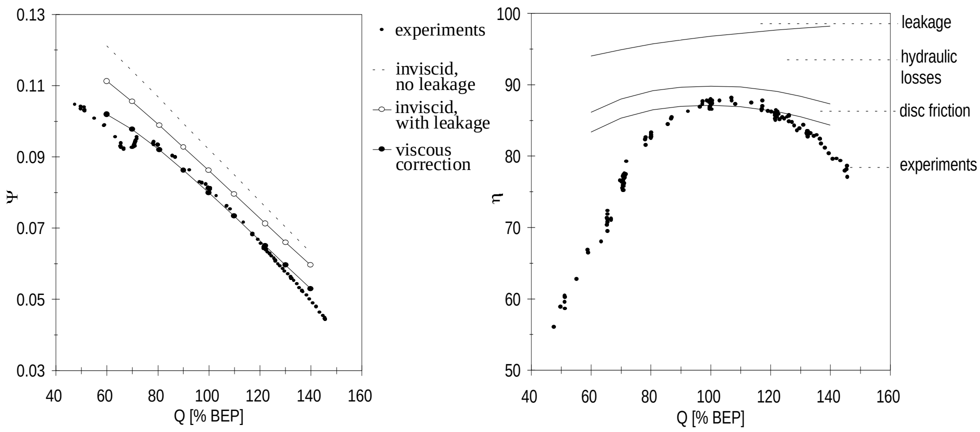

Fig. 6.28: Head coefficient ψ and efficiency η, as a function of flow rate Q. Left figure shows influence of leakage flow and hydraulic losses on head coefficient ψ. Right figure gives computed efficiency and losses.

Disc friction losses - The empirical expressions of Daily and Nece for disc friction without through-flow are used for the cavity below the hub surface. For the leakage area between the shroud and the pump housing the more advanced leakage flow method (discussed in section 5.5) is used. The amount of disc friction varies with flow rate due to a change in circumferential velocity of the incoming fluid. Typical for impellers with backswept blades, the circumferential velocity is below half the value of the rotor speed at cavity entrance. As a consequence disc friction is larger than predicted by the model of Daily and Nece. On the average 3 percent of the shaft power is lost by disc friction at the impeller outer walls.

Shaft power, head and efficiency - In fig. 6.28 the computed head and efficiency are given and compared with experiments. The effects of leakage, hydraulic losses and disc friction are indicated. The computed head characteristic appears to be in very good agreement with measurements, although deviations tend to increase at lowest and highest flow rate considered. As was discussed in this section, the large deviation at flow rates below $Q/Q_{\mathrm{BEP}} = 0.8$ might be attributed to severe losses due to laminar separation (i.e. stall) near the leading edge, at the suction side of the blades. The comparison between computed and measured values for the efficiency shows a much larger deviation at off-design conditions. At optimum design point, the efficiency is slightly too low. The question arises whether it is the computation of the viscous losses which is incorrect or the inviscid computations to begin with, or possibly both.

For this question to be answered, a comparison of in- and output power is made in fig. 6.29. Experimental values for the shaft power $P_{sh}(exp)$, corrected for mechanical losses, and the net power $P_{net}(exp)$, based on delivered head, are plotted. Computed values for P_E, the power which is transferred by the impeller to the internal fluid, and the shaft power $P_{sh}(calc)$ are given as well, the difference being the disc friction losses at the external surfaces of the impeller. The net power $P_{net}(calc)$ is computed by correcting P_E for leakage flow and hydraulic losses (eqs. 5.15 and 5.16), the influ-

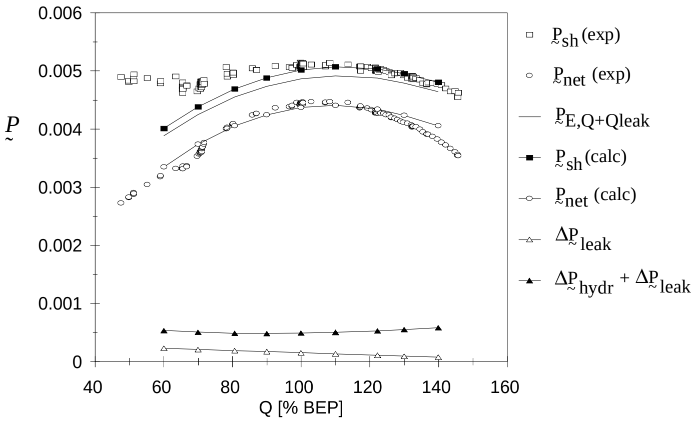

Fig. 6.29: Experimental and computed values for shaft power P_{sh}, and net power P_{net}. Values made dimensionless according to eq. (1.3d)

ence of which is depicted in the lower half of the figure. It appears that for flow rates below best efficiency point the shaft power is underestimated considerably. A likely explanation is provided by inlet recirculation. According to Tuzson (1997) this causes a power loss at flow rates below correct inlet incidence. Like disc friction it considerably affects the required shaft power and efficiency, but leaves the head unaffected. At flow rates larger than Q_{BEP} the shaft power is predicted very well. Hydraulic loss is in the correct range, although at highest flow rate, it appears to be slightly underestimated.

Displacement effect - A boundary layer analysis shows that the displacement thickness of boundary layers along the volute outer wall can become quite substantial. To study the effect of the displacement on quantities like head and efficiency, an iterative procedure of (1) computing the inviscid main flow, (2) a boundary layer analysis at the volute wall and (3) the adjustment of the stator wall according to the displacement thickness was performed until convergence was reached. The implication on the geometry for three flow rates is depicted in fig. 6.30. The volute geometry and computational mesh after subtraction of the boundary layer displacement thickness for low flow rate (80% Q_{BEP}) i shown in fig. 6.31. The influence of these rather extreme displacements on the computed head and efficiency is however surprisingly small and can hardly be distinguished.

Detailed pressure measurements in the volute

The second type of validation concerns values of the static pressure difference p_d (eq. 6.1) between the inlet of the pump and different locations along the volute wall. Time averaged pressure values are shown in fig. 6.32. In computing the pressure account has been given to the leakage flow and the hydraulic losses occurring in the impeller region of the pump, as given in eq. (6.2). Differences between computed and measured values to a large extent reflect the losses in the volute. In most cases these dif-

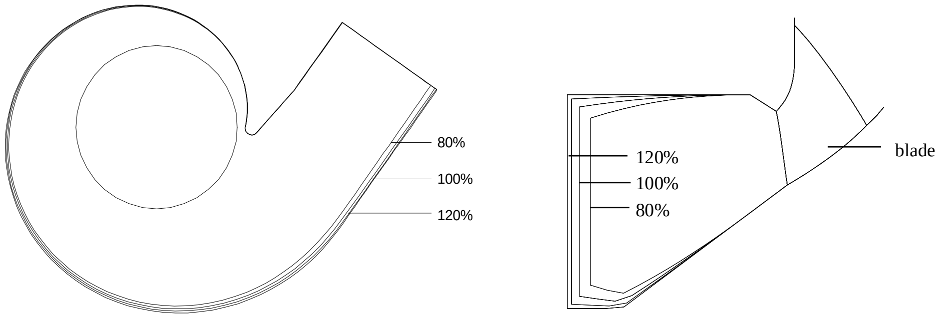

Fig. 6.30: Two-dimensional views of volute shape after boundary layer displacement correction, for three different flow rates (showing percentages of nominal flow rate).

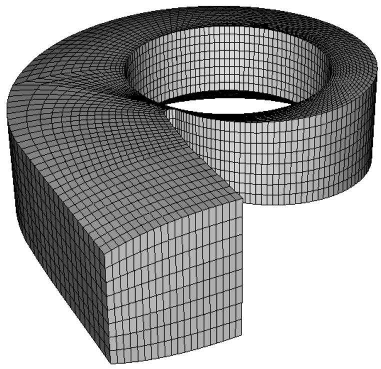

Fig. 6.31: Computational grid of volute geometry after subtraction of boundary layer displacement thickness (Q/Q_{BEP} = 0.8).

ferences tend to increase with distance from the tongue, as the losses in the volute gradually build up along streamlines.

Results of time-dependent pressure measurements prove that it is very difficult to experimentally obtain the periodic signal related to the blade passing frequency. Spectral analysis very often shows that large contributions exist at revolution frequency. This may be due to asymmetry of the impeller, or a slight eccentric motion of the shaft. Higher frequencies, like twice the blade passing frequency, are also frequently encountered. These phenomena tend to confuse comparison of amplitude and phase of the signals to results of computations. In fig. 6.33 some examples of results after ensemble averaging are shown. Comparison of experimental (pitch averaged) and computed time-dependent pressure values for nominal flow rate is shown in fig. 6.34. In order to reduce the amount of data, only the amplitude of the time fluctuations and the phase of the signals are considered. It can be concluded that the computed values show a reasonable agreement with experiments in a large region of the volute, not too close to the tongue. Results at off-design conditions show a similar behaviour, although deviations are larger.

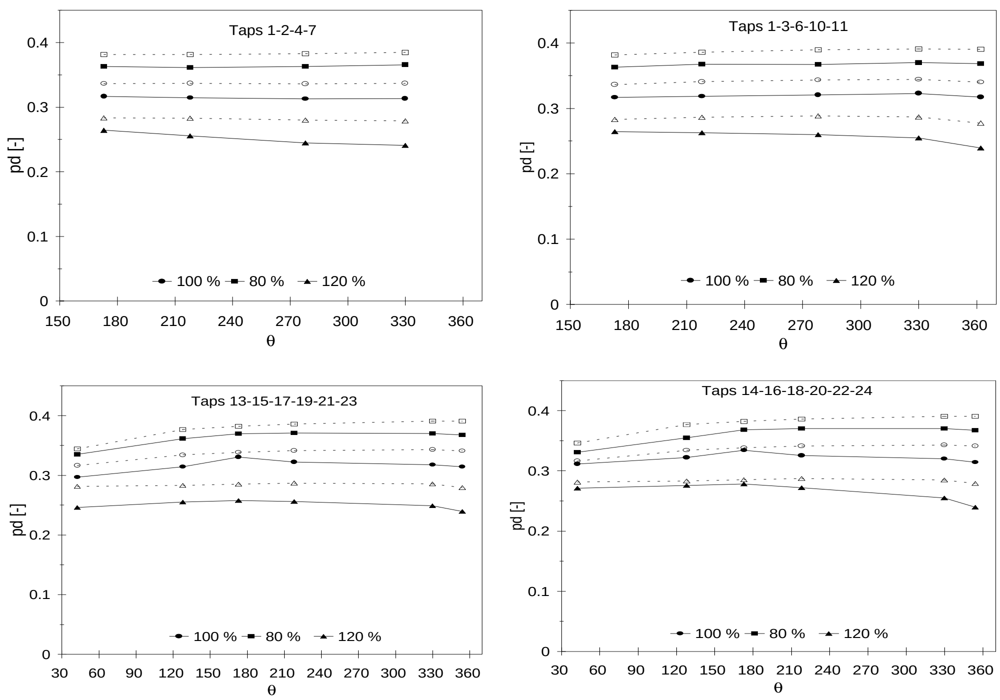

Fig. 6.32: Time averaged static pressure difference between inlet and volute pressure taps (see fig. 6.23) as a function of angular distance from the tongue. Measurements are indicated with solid lines and markers, computations with dashed lines and open markers.

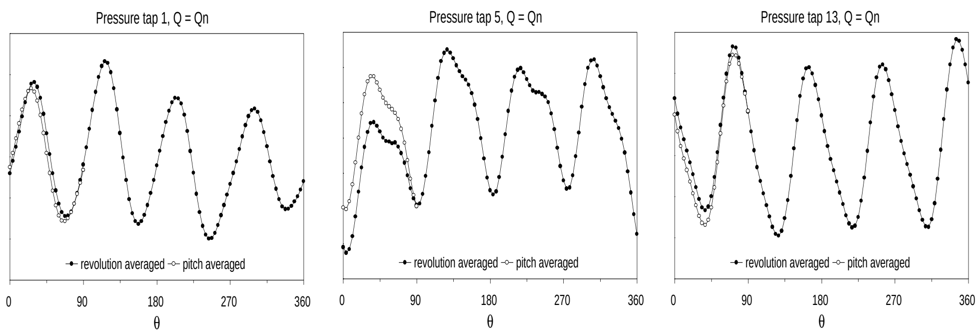

Fig. 6.33: Ensemble averaged signals of pressure differences for three pressure taps, at nominal flow rate. Non-periodicity is illustrated by comparing ensemble averaging over a pitch with averaging over one impeller revolution.

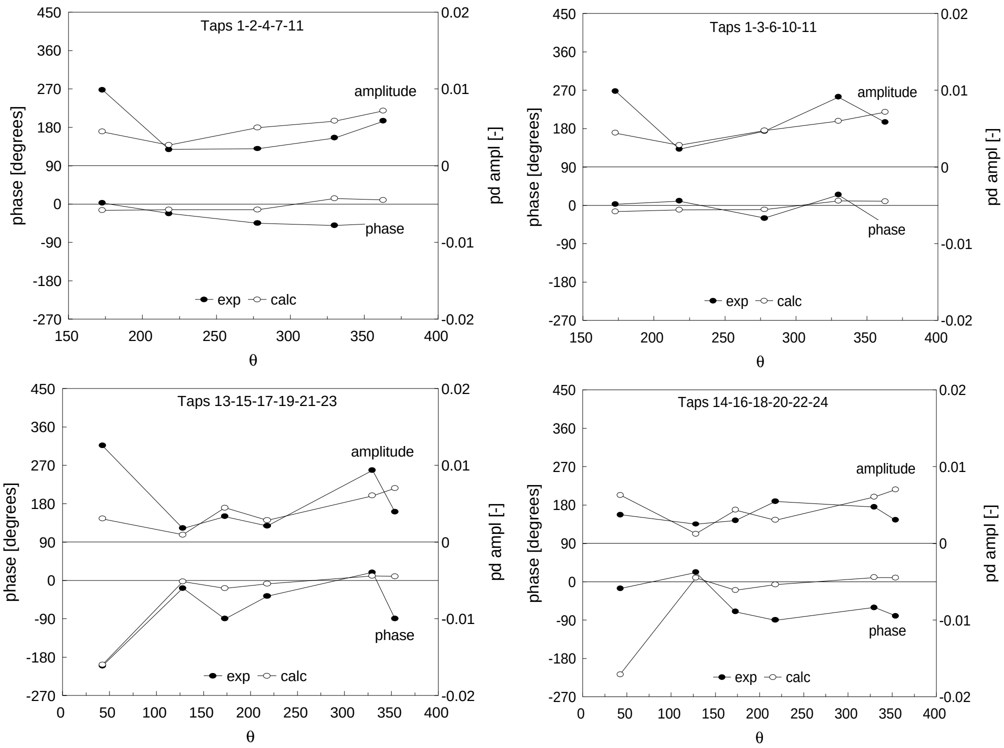

Fig. 6.34: Time-dependent pressure values, phase and amplitude of variation, for optimum flow rate.

6.6 Radial impeller with volute casing

At the California Institute of Technology (CalTech), experiments have been performed on a common type of radial impeller within an unvaned volute to investigate its rotordynamic behaviour. The interaction between the impeller and the volute at off-design conditions, is found to be a very important factor. Although a rotordynamic analysis is not the objective of this thesis (but see Aarts, 1997), some of the experimental results are of particular interest to the validation currently undertaken. It concerns measurements of the head as a function of flow rate for two different seal clearances. Unfortunately, no data on the efficiency of the centrifugal pump are available.

In section 6.6.1 the experimental set-up is briefly described. Some computational aspects are given in section 6.6.2. Results of computations are finally compared with measured values in section 6.6.3.

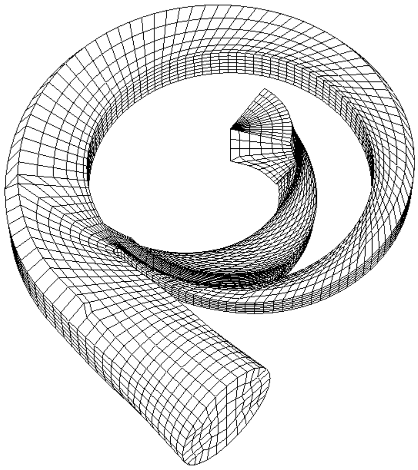

Fig. 6.35: Computational mesh of CalTech pump, showing volute and one of the five impeller channels.

6.6.1 Experimental set-up

At the Rotor Force Test Facility of CalTech extensive measurements have been performed on a common type of centrifugal pump. The impeller, designated impeller X (pronounced as “ten”), is a design of BW/IP International with five blades, a diameter of 162 mm and a specific speed n_ω of 0.57. Blade span at impeller discharge is 16 mm and blade thickness varies from 6 mm at the hub of the impeller up to 8 mm at the shroud. Blade length is 122 mm measured at mid-span. The impeller is mounted in a matching volute (designated volute A) with a logarithmic spiral casing of angle 86°. The design flow coefficient Φ is 0.014 in which case radial forces on the impeller are virtually zero. Rotational speed varies from 600 rpm up to 2000 rpm. Reynolds numbers vary between $1.6 \cdot 10^5$ and $5.5 \cdot 10^6$ (eq. 1.3c). The shaft of the impeller is artificially set into an orbital motion of radius 1.26 mm and a whirl speed of 3 rpm to measure the rotordynamic forces. Tests were conducted at several seal clearances, two of which are presented in Chamieh et al. (1982). One of the conclusions was that the influence of the whirling motion of the impeller on the head characteristic was negligible.

6.6.2 Computational aspects

The geometry for the volute is taken from Adkins (1985), while for the impeller geometry the original technical drawings of BW/IP International are used. The computational domain is divided into 22 blocks in total, of which 10 in the impeller region. Total number of nodes is approximately 40,000 forming 180,000 linear tetrahedral elements (fig. 6.35). Rotational speed is set to 1000 rpm. Each impeller pitch revolution is divided into 15 time steps. The whirling motion of the impeller is not simulated. Computations are performed on a Silicon Graphics Power Challenge operating at a speed of 60 Mflops, and take less than 1.5 hours for each separate flow rate. Maximum internal memory required is 22 Mb.

6.6.3 Results and comparison

A viscous analysis has been carried out which is very similar to the one for the mixed-flow pump (section 6.5), the only difference being the neglect of expansion losses. As the width of the impeller discharge and volute entrance are virtually equal, expansion losses are assumed to be zero. In the following several aspects of the viscous analysis will be discussed.

Leakage flow - The leakage flow area is taken from a sketch in Adkins and Brennen (1988) and its exact geometry is not known. A parameter study shows that the length of the radial seal and, most of all, the clearance itself are the only parameters which have a considerable influence on leakage flow characteristics. Rotor and stator walls are assumed perfectly smooth, as no information on surface roughness is available. Two seal clearances are considered: 0.14 mm and 0.79 mm (Chamieh et al., 1982). The model of section 5.5 with friction factors of Moody was used to compute the characteristics of the leakage flow. For small seal clearance the fraction of the leakage flow rate to the actual flow rate has a moderate value of almost 3 percent, ranging from over 7 percent at lowest flow rate considered (Q/Q_{BEP} = 0.4) to 2 percent at Q/Q_{BEP} = 1.3. The circumferential velocity of the leakage flow upon re-entrance in the impeller inlet is roughly 90 percent of the rotor speed, for all flow rates. This rather high value is a result of the larger radial extent of the leakage flow area when compared to typical mixed flow configurations. At large seal clearance huge leakage flow rates are computed ranging from 11 up to 30 percent of the actual flow rate, with a value of 15 percent at design conditions. As a consequence circumferential velocities tend to be very high as well (1.5 to 1.75 times the rotor speed).

Boundary layers - A boundary layer analysis (section 5.7) has been performed for the impeller blades, based on relative velocities along a streamline at mid-span. Turbulence intensity of the free-stream was assumed to be 5 percent. For all flow rates considered, laminar separation occurred at the pressure side of the blade, roughly 1 cm behind the leading edge, followed by turbulent separation a few centimeters later. Whether or not separation really occurs is a question which can not be answered conclusively (see section 6.5.3 for a discussion). As was the case for the mixed-flow impeller, a zone of strong acceleration (strong enough to force a turbulent boundary layer to a laminar state) is present at the second half of the pressure side. For most of the flow rates transition of the laminar boundary layer occurs at the suction side roughly 1 cm behind the leading edge. No subsequent separation is predicted. Only for very low flow rates (Q/Q_{BEP} < 0.6) laminar separation was predicted at the suction side immediately after the leading edge (stall).

Hydraulic losses - Various contributions to the hydraulic loss are depicted in fig 6.36 for both seal clearances. The energy dissipation coefficient c_D is taken 0.0038 for all surfaces and for all flow rates. Unlike the situation for the mixed-flow pump, hydraulic loss is dominated at low flow rates by dissipation in the boundary layers along the volute wall. For large seal clearances the mixing loss which occurs when the leakage flow enters the impeller inlet is quite substantial, while for small clearance its contribution can be neglected. In order to compute the wake mixing loss, the value for the momentum thickness at the pressure side trailing edge is assumed to be small (0.1

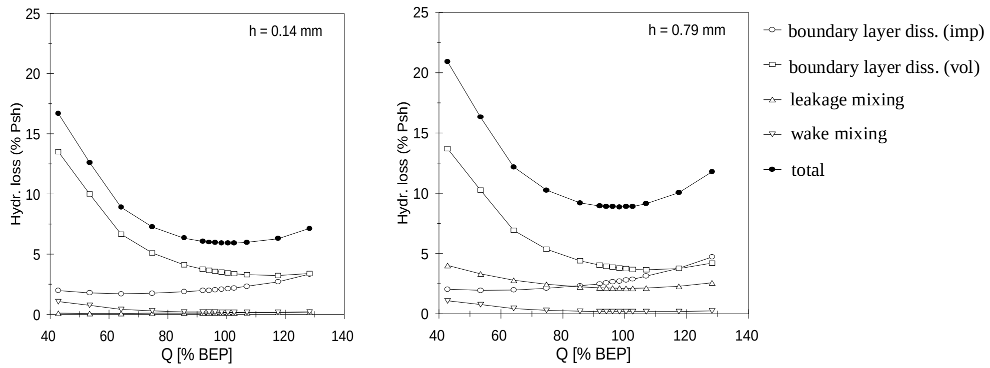

Fig. 6.36: Total hydraulic loss and its components, for two seal clearances *h*.

mm) as opposed to the suction surface (computed values: 0.2 to 0.4 mm).

Euler head - In fig. 6.37 the head coefficient which results from inviscid computations without leakage is given, as well as the influence of leakage flow and hydraulic losses, for both seal clearances. It appears that the estimated head value for the small seal clearance is far too high when compared to measurements. However, the result for the large seal clearance is in a very good agreement indeed. The reason for this difference is unknown. The leakage flow is not that sensitive to the magnitude of the clearance for such a large effect to be explained by mis-measurement; to fit the computed data to the experiments a clearance which is nearly three times as high must be assumed! On the other hand, it seems impossible to change the result for the small clearance by altering geometric quantities (or surface roughness) of the leakage area, without changing the results for the large clearance as well. Another explanation could be an over-prediction of the influence of the leakage flow, especially for large clearances, combined with an underestimation of the hydraulic losses.

Disc friction losses - The leakage flow model which was mentioned earlier also provides the wall shear stresses at the rotor surface of the leakage flow area. Thus, the

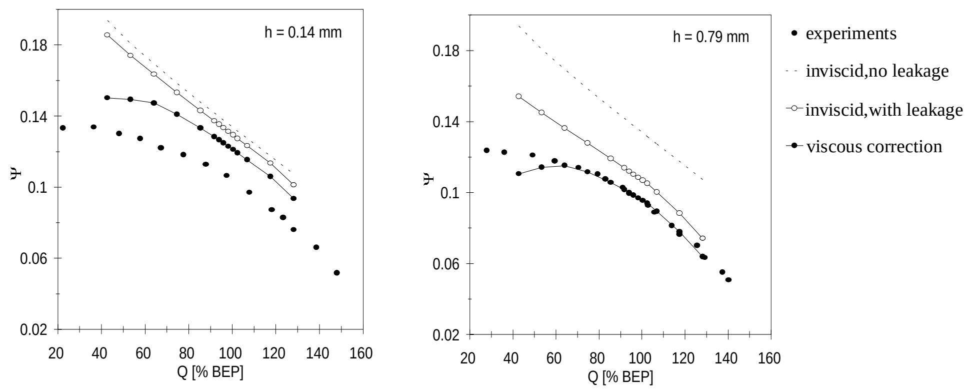

Fig. 6.37: Influence of leakage flow and hydraulic losses on head coefficient Ψ, for two seal clearances *h*.

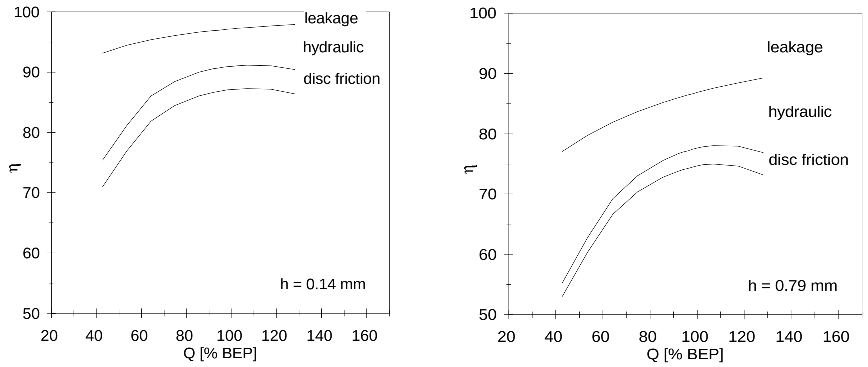

Fig. 6.38: Effect of leakage flow, hydraulic losses and disc friction on efficiency η.

loss in shaft power as a result of disc friction can be determined. For the small seal clearance, the loss is roughly 4.3 percent of the shaft power. For the large clearance this value reduces to 3.7 percent at optimum flow rate. In fig. 6.38 the effect of disc friction on efficiency is given, as well as the influence of leakage and hydraulic losses. Clearly, disc friction plays a minor role compared to leakage and hydraulic losses. Unfortunately, computed values can not be compared to measurements, as measured values for the efficiency are not available.

6.7 Conclusions

6.7.1 Summary

As a final section to this chapter, the results for validation are summarized for the six test cases considered.

Free radial impeller

The agreement between measured and computed values for the static pressure distribution at the impeller blades is reasonable. Computed circumferential and radial velocity distributions across the impeller passage near the trailing edge are in good agreement with measurements, except for areas with strong secondary flow.

Radial impeller with vaned diffuser

Measurements of the unsteady velocity in the region between the impeller and stator vanes are available. Computations show a qualitative agreement which is reasonably good. The magnitude of the velocity fluctuations is predicted well. The overall impression is that computed relative velocities are too small, possibly caused by boundary layer separation and tip clearance effects.

Radial impeller with spiral volute

At flow rates near best efficiency point, the computed head characteristic agrees well with measurements. Standard models for boundary layer dissipation and wake mixing are used to quantify the viscous losses. At high flow rate, the head is overpredicted, possibly because of boundary layer separation in the volute.

Computed circumferential velocities in the volute are in good agreement in regions not too close to the tongue. At off-design flow conditions, this agreement deteriorates. Radial velocities deviate considerably. This can be explained by strong secondary flow.

Static pressure values in the volute are computed by correcting the results (obtained with the inviscid model) for hydraulic losses occurring in the impeller. Agreement with experimental values is good, although large deviations exist in the immediate vicinity of the volute tongue.

Free mixed-flow impeller

The relative velocity in a region close to the impeller blade trailing edge was measured, in order to validate the flow direction which was assumed in the computational model. Although computations show a good agreement with measurements, the current analysis is not sufficient to conclusively decide between the two flow directions which are possible from a theoretical consideration.

Mixed-flow impeller with volute casing

For flow rates in the range 80% to 120% Q_{BEP}, the computed head characteristic is in excellent agreement with measurements. Leakage loss is found to be considerable, due to a rather large seal clearance. Shaft power is predicted well at design flow rate and higher. For low flow rate, the shaft power is underpredicted. Hydraulic losses at lowest and highest flow rates considered are probably estimated too small. Efficiency is predicted correctly at optimum flow rate. At off-design conditions, large deviations from measurements are observed, due to the underprediction of hydraulic losses and, at low flow rate, the shaft power. It appears that a slight deviation in either the shaft power or the hydraulic losses leads to large deviations in efficiency.

Boundary layer displacement in the volute, which may be quite substantial at low flow rate, does not lead to a considerable change in computed head and efficiency characteristics.

Computed time averaged static pressure values in the volute are in good agreement with measurements. Computations of the time-dependent static pressure variation shows a reasonable qualitative agreement in regions not too close to the tongue.

Radial impeller with volute casing

Head characteristics for this test case are available from measurements for two values of the seal clearance gap. For the large seal clearance, the computed head characteristic agrees very well with measurements for a wide range in flow rate (60% to 130% Q_{BEP}). However, head values for the small seal clearance are overpredicted considerably. A full analysis of this test case is not possible because measurements of shaft power are not available.

6.7.2 Evaluation of viscous loss models

For the radial and mixed-flow pump operating at best efficiency point, the power loss resulting from hydraulic losses, leakage flow and disc friction can be compared with values given by e.g. Stepanoff (1964) for pumps with similar specific speed. For the radial pump, the small seal clearance is taken. Results are depicted in tables 6.1 and 6.2. Hydraulic losses are divided into losses in the impeller and losses in the volute. Leakage losses include mixing loss at the impeller entrance. Wake mixing loss is attributed to impeller hydraulic losses, expansion losses to volute hydraulic losses. Several conclusions can be drawn referring to the data as given by Stepanoff. It appears that hydraulic losses are well estimated for the mixed-flow pump, although the distribution over impeller and volute is quite different. For the radial impeller the computed hydraulic loss in the volute is rather high. Loss attributed to leakage is well estimated for the radial pump with small seal clearance. For the mixed-flow pump, the seal clearance is quite large, resulting in a relatively high loss. Loss due to disc friction is predicted very well; the leakage flow model correctly predicts a larger relative loss in shaft power for radial pumps, when compared to pumps of higher specific speed.

Radial pump	Computed	Stepanoff
Impeller loss	2.4	2.2
Volute loss	3.6	2.2
Leakage loss	3.0	2.5
Disc friction loss	4.0	4.5

Table 6.1: Comparison for relative power losses (in percentage shaft power) for a radial pump with specific speed $n_\omega = 0.57$.

Mixed-flow pump	Computed	Stepanoff
Impeller loss	4.7	2.2
Volute loss	1.0	3.5
Leakage loss	3.6	1.0
Disc friction loss	2.7	2.2

Table 6.2: Comparison for relative power losses (in percentage shaft power) for a mixed-flow pump with specific speed $n_\omega = 1.6$.

The power loss as a function of flow rate can be found in fig. 6.28 (right) for the mixed-flow pump, and in fig. 6.38 (left) for the radial pump with small seal clearance. In both cases, leakage loss decreases with flow rate, in agreement with data provided by Stepanoff. Hydraulic losses show a parabolic-like dependency on flow rate with a minimum value near best efficiency point (refer to fig. 6.27 and 6.36, left). This corresponds to what is given by almost all textbooks on this subject (e.g. Pfleiderer and Petermann, 1972). According to Stepanoff, disc friction decreases with flow rate. Computations show a value which is roughly constant for all flow rates, or even tends to increase with flow rate. The first reason is that a larger leakage flow rate (at low capacity) will lead to a lower friction at the impeller external shroud wall. Secondly, a lower capacity results in a higher circumferential velocity at the impeller discharge. This in turn leads to a larger overall circumferential velocity within the clearance region, resulting in a lower disc friction at the impeller surface.

Nomenclature

D	Impeller diameter
H	Head
P	Power
$\tilde{P}$	Dimensionless power
Q	Flow rate
Re	Reynolds number: $\Omega D^2/\nu$
T_i	Blade passing period
U	Blade tip speed
b	Blade coordinate in axial direction
b_{max}	Blade span
n_ω	Specific speed: $\Omega Q^{1/2}/(gH)^{3/4}$
r	Radius
s	Local coordinate
t	Time
v	Absolute velocity
w	Relative velocity

Greek symbols

Φ	Flow coefficient: $Q/(\Omega D^3)$
Ω	Angular velocity of impeller
Ψ	Head coefficient: $gH/(\Omega D)^2$
η	Efficiency
ν	Kinematic viscosity
ρ	Fluid density

θ Angular coordinate
Θ Angular width of impeller passage

Subscripts
BEP Best efficiency point
2 Trailing edge

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Acknowledgement

Much of the work presented in this thesis has been done with the help of others. I would like to thank everyone who contributed. There are, however, some people that I feel particularly grateful to.

First of all, I would like to thank Bert Brouwers for the initiation of this research. His expertise proved to be of high value in the completion of this thesis. Ben Jonker is thanked for the confidence he has put in me from the beginning, and for his enthusiasm which was a source of continuous inspiration. I also would like to express my deep appreciation to Niels Kruyt for his guidance in every step of this work. The many discussions we had were always open and unbiased. I strongly felt this to be the right atmosphere to achieve progress, although we never seemed to fully grasp the essence of the unsteady Bernoulli equation.

I gratefully appreciate the co-operation with BW/IP Stork Engineered Pumps. By providing the funds and the opportunity to perform experiments on one of their industrial pumps, I was allowed to carry out a major part of the validation of the method. In particular, I should mention Johan op de Woerd and Ronald Dijkers. I appreciate the many discussions we have had on the practical value of this work to pump engineers.

Marijn van Os has been my room-mate for almost the entire four-year period. I thank him for his companionship. Through our frequent conversations I learned to know him as an open-hearted and straightforward person with a great sense of justice. A special word of thanks is directed to Ronald Aarts, to whom an appeal is never made in vain. Graduate students Paul Braam, Peter Duineveld, Jasper Kelder, Wim Super and Jan Wierenga are thanked for their assistance.

A list of people who played an important role would not be complete without the name of Hans Kuerten, a highly esteemed colleague and dear friend.

Finally, I would like to thank Annemarie for her love and understanding.

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